Syllabus

Science (Code No. 086)
Class - X
(Issued by CBSE Dated on 28th March, 2025)

Time: 03 Hours **Marks: 80**

Unit No.	Unit	Marks
I	Chemical Substances- Nature and Behaviour	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	05
	Total	**80**
	Internal Assessment	**20**
	Grand Total	**100**

Theme : Materials

Unit I : Chemical Substances - Nature and Behaviour

Chemical Reactions and Equations: Chemical reactions, Chemical equation, Balanced chemical equation, types of chemical reactions: combination, decomposition, displacement, double displacement, precipitation, endothermic exothermic reactions, oxidation and reduction.

Acids, Bases and Salts: Acids and Bases – definitions in terms of furnishing of H^+ and OH^- ions, identification using indicators, chemical properties, examples and uses, neutralization, concept of pH scale (Definition relating to logarithm not required), importance of pH in everyday life; preparation and uses of Sodium Hydroxide, Bleaching powder, Baking soda, Washing soda and Plaster of Paris.

Metals and Non-metals: Properties of metals and non-metals; Reactivity series; Formation and properties of ionic compounds; Basic metallurgical processes; Corrosion and its prevention.

Carbon and its Compounds: Covalent bonds – formation and properties of covalent compounds, Versatile nature of carbon, Hydrocarbons – saturated and unsaturated Homologous series. Nomenclature of alkanes, alkenes, alkyne and carbon compounds containing functional groups (halogens, alcohol, ketones, aldehydes). Chemical properties of carbon compounds (combustion, oxidation, addition and substitution reaction). Ethanol and Ethanoic acid (only properties and uses), soaps and detergents.

Theme : The World of the Living

Unit II : World of Living

Life processes: 'Living Being'. Basic concept of nutrition, respiration, transport and excretion in plants and animals.

Control and co-ordination in animals and plants: Tropic movements in plants; Introduction of plant hormones; Control and co-ordination in animals: Nervous system; Voluntary, involuntary and reflex action; Chemical co-ordination: animal hormones.

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Reproduction: Reproduction in animals and plants (asexual and sexual) reproductive health - need and methods of family planning. Safe sex vs HIV/AIDS. Child bearing and women's health.

Heredity and Evolution: Heredity; Mendel's contribution- Laws for inheritance of traits: Sex determination; brief introduction.

Theme : Natural Phenomena

Unit III : Natural Phenomena

Reflection of light by curved surfaces; Images formed by spherical mirrors, centre of curvature, principal axis, principal focus, focal length, mirror formula (Derivation not required), magnification.

Refraction; Laws of refraction, refractive index.

Refraction of light by spherical lens; Image formed by spherical lenses; Lens formula (Derivation not required); Magnification. Power of a lens.

Functioning of a lens in human eye, defects of vision and their corrections, applications of spherical mirrors and lenses.

Refraction of light through a prism, dispersion of light, scattering of light, applications in daily life (excluding colour of the sun at sunrise and sunset).

Theme : How Things Work

Unit IV : Effects of Current

Electric current, potential difference and electric current. Ohm's law; Resistance, Resistivity, Factors on which the resistance of a conductor depends. Series combination of resistors, parallel combination of resistors and its applications in daily life. Heating effect of electric current and its applications in daily life. Electric power, Interrelation between P, V, I and R.

Magnetic effects of current: Magnetic field, field lines, field due to a current carrying conductor, field due to current carrying coil or solenoid; Force on current carrying conductor, Fleming's Left Hand Rule, Direct current. Alternating current: frequency of AC. Advantage of AC over DC. Domestic electric circuits.

Theme : Natural Resources

Unit V : Natural Resources

Our environment: Eco-system, Environmental problems, Ozone depletion, waste production and their solutions; Biodegradable and non-biodegradable substances.

Note for the Teachers:

The NCERT text books present information in boxes across the book. These help students to get conceptual clarity. However, the information in these boxes would not be assessed in the year-end examination.

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Practicals

Practical should be conducted alongside the concepts taught in theory classes.

List of Experiments

1. A. Finding the pH of the following samples by using pH paper/universal indicator: **Unit-I**

a) Dilute Hydrochloric Acid

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b) Dilute NaOH solution
c) Dilute Ethanoic Acid solution
d) Lemon juice
e) Water
f) Dilute Hydrogen Carbonate solution

B. Studying the properties of acids and bases (HCl & NaOH) on the basis of their reaction with: **Unit-I**

a) Litmus solution (Blue/Red)
b) Zinc metal
c) Solid sodium carbonate

2. Performing and observing the following reactions and classifying them into: **Unit-I**

a) Combination reaction
b) Decomposition reaction
c) Displacement reaction
d) Double displacement reaction

- Action of water on quicklime
- Action of heat on ferrous sulphate crystals
- Iron nails kept in copper sulphate solution
- Reaction between sodium sulphate and barium chloride solutions

3. Observing the action of Zn, Fe, Cu and Al metals on the following salt solutions: **Unit-I**

a) $ZnSO_4$ (aq)
b) $FeSO_4$ (aq)
c) $CuSO_4$ (aq)
d) $Al_2 (SO_4)_3$ (aq)

Arranging Zn, Fe, Cu and Al (metals) in the decreasing order of reactivity based on the above result.

4. Studying the dependence of potential difference (V) across a resistor on the current (I) passing through it and determine its resistance. Also plotting a graph between V and I. **Unit-IV**

5. Determination of the equivalent resistance of two resistors when connected in series and parallel. **Unit-IV**

6. Preparing a temporary mount of a leaf peel to show stomata. **Unit- II**

7. Experimentally show that carbon dioxide is given out during respiration. **Unit-II**

8. Study of the following properties of acetic acid (ethanoic acid): **Unit- I**

a) Odour
b) solubility in water
c) effect on litmus
d) reaction with Sodium Hydrogen Carbonate

9. Study of the comparative cleaning capacity of a sample of soap in soft and hard water. **Unit- I**

10. Determination of the focal length of: **Unit-III**

a) Concave mirror
b) Convex lens by obtaining the image of a distant object.

11. Tracing the path of a ray of light passing through a rectangular glass slab for different angles of incidence. Measure the angle of incidence, angle of refraction, angle of emergence and interpret the result. **Unit - III**

12. Studying (a) binary fission in *Amoeba*, and (b) budding in yeast and Hydra with the help of prepared slides. **Unit-II**

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13. Tracing the path of the rays of light through a glass prism. **Unit-III**

14. Identification of the different parts of an embryo of a dicot seed (pea, gram or red kidney bean).

Unit-II

PRESCRIBED BOOKS:

- Science-Textbook for class IX-NCERT Publication
- Science-Text book for class X- NCERT Publication
- Assessment of Practical Skills in Science-Class IX - CBSE Publication
- Assessment of Practical Skills in Science- Class X- CBSE Publication
- Laboratory Manual-Science-Class IX, NCERT Publication
- Laboratory Manual-Science-Class X, NCERT Publication
- Exemplar Problems Class IX – NCERT Publication
- Exemplar Problems Class X – NCERT Publication
- Reading Material – Science – Class IX – CBSE

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Question Paper Design (Class X)

Theory (80 marks)

Competencies	Total
Demonstrate Knowledge and Understanding	50%
Application of Knowledge/Concepts	30%
Formulate, Analyse, Evaluate and Create	20%

Note:
- Typology of Questions: VSA including objective type questions, Assertion – Reasoning type questions; SA; LA; Source-based/ Case-based/ Passage-based/ Integrated assessment questions.
- An internal choice of approximately 33% would be provided.

Internal Assessment (20 Marks)

- **Periodic Assessment -** 05 marks + 05 marks
- **Subject Enrichment** (Practical Work) - 05 marks
- **Portfolio** - 05 marks

Suggestive verbs for various competencies

- **Demonstrate Knowledge and Understanding**
  State, name, list, identify, define, suggest, describe, outline, summarize, etc.
- **Application of Knowledge/Concepts**
  Calculate, illustrate, show, adapt, explain, distinguish, etc.
- **Formulate, Analyse, Evaluate and Create**
  Interpret, analyze, compare, contrast, examine, evaluate, discuss, construct, etc.

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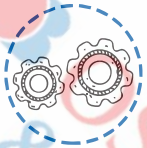
MIND MAPS
Learning Made Simple



anytime, as frequently as you like till it becomes a habit!

When?

Presenting words and concepts as pictures!!

What?

MIND MAP
AN INTERACTIVE MAGICAL TOOL

Why?

- To Unlock the imagination and come up with ideas
- To Remember facts and figures easily
- To Make clearer and better notes
- To Concentrate and save time
- To Plan with ease and ace exams

Result

Learning made simple
'a winning combination'

How?

With a blank sheet of paper coloured pens and your creative imagination!

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What are Associations?

It's a technique connecting the core concept at the Centre to related concepts or ideas. Associations spreading out straight from the core concept are the First Level of Association. Then we have a Second Level of Association emitting from the first level and the chronology continues. The thickest line is the First Level of Association and the lines keep getting thinner as we move to the subsequent levels of association. This is exactly how the brain functions, therefore these Mind Maps. Associations are one powerful memory aid connecting seemingly unrelated concepts, hence strengthening memory.

Chemical Reactions and Equations

1. Chemical Equations

- **Writing a Chemical Equation**
  - Word Equations: Represent reactions using words.
    - Example: Magnesium + Oxygen → Magnesium Oxide
  - Chemical Equations: Use symbols and formulas.
    - Example: $Mg + O_2 \rightarrow MgO$
- **Balanced Chemical Equations**
  - Law of Conservation of Mass: Mass remains constant in a reaction
  - Balancing Steps:
    1. Draw Boxes: Around each formula.
    2. List Atoms: Count elements on both sides.
    3. Choose Element with Maximum Atoms: Start balancing from the most complex compound.
    4. Balance Oxygen first: Example: $Fe + H_2O \rightarrow Fe_2O_3 + H_2$ $Fe + 4H_2O \rightarrow Fe_2O_3 + 4H_2$
    5. Balance Hydrogen: Adjust coefficients to balance H atoms.
    6. Balance Iron: Adjust Fe atoms to balance the equation.
    7. Include Physical States: Indicate (s), (l), (g) and (aq).
    8. Final Balanced Equation: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$.

2. Types of Chemical Reactions

- **Combination Reaction**
  - Definition: Two/more reactants form one product.
  - Example: $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$
- **Decomposition Reaction**
  - Definition: One compound breaks into simpler substances.
  - Example: $2FeSO_4(s) \xrightarrow{\Delta} Fe_2O_3(s) + SO_2(g) + SO_3(g)$
  - Types:
    1. Thermal Decomposition: Uses Heat
       Example: $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
    2. Electrolytic Decomposition: Uses electricity
       Example: $2H_2O(l) \xrightarrow{\text{electric current}} 2H_{2(g)} + O_{2(g)}$
    3. Photodecomposition: Uses light
       Example: $2AgCl \xrightarrow{\text{light}} 2Ag + Cl_2$
- **Displacement Reaction**
  - Definition: A more reactive element displaces a less reactive one.
  - Example: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$
- **Double Displacement Reaction**
  - Definition: Exchange of ions between reactants.
  - Example: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$
  - Precipitation: Formation of an insoluble substance (e.g., $BaSO_4$).
- **Oxidation and Reduction**
  - Oxidation: Gain of oxygen/loss of hydrogen.
  - Reduction: Loss of oxygen/gain of hydrogen.
  - Examples:
    - $2Cu + O_2 \xrightarrow{\Delta} 2CuO$ (oxidation)
    - $CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$ (reduction)
  - Redox Reaction: Oxiation and reduction occurring simultaneously
    - $ZnO + C \rightarrow Zn + CO$ (ZnO reduced, C oxidised)

3. Have you observed the effects of oxidation reactions in everyday life?

- **Corrosion**
  - Definition: Metal deterioration due to environmental factors.
  - Examples:
    - Rusting of iron (reddish-brown coat).
    - Black coat on silver green coat on copper.
  - Impact: Damages bridges, ships, and iron objects.
- **Rancidity**
  - Definition: Oxidation of fats/oils causing bad taste/smell.
  - Prevention:
    - Use antioxidants.
    - Store in airtight containers.
    - Flush chips bags with nitrogen gas.

4. Energy Changes in a Reaction

- Exothermic
  - Definition: Heat evolves in a reaction
  - Example: Respiration : $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l)$ + Energy
- Endothermic
  - Definition: Heat is absorbed in a reaction
  - Example: $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$

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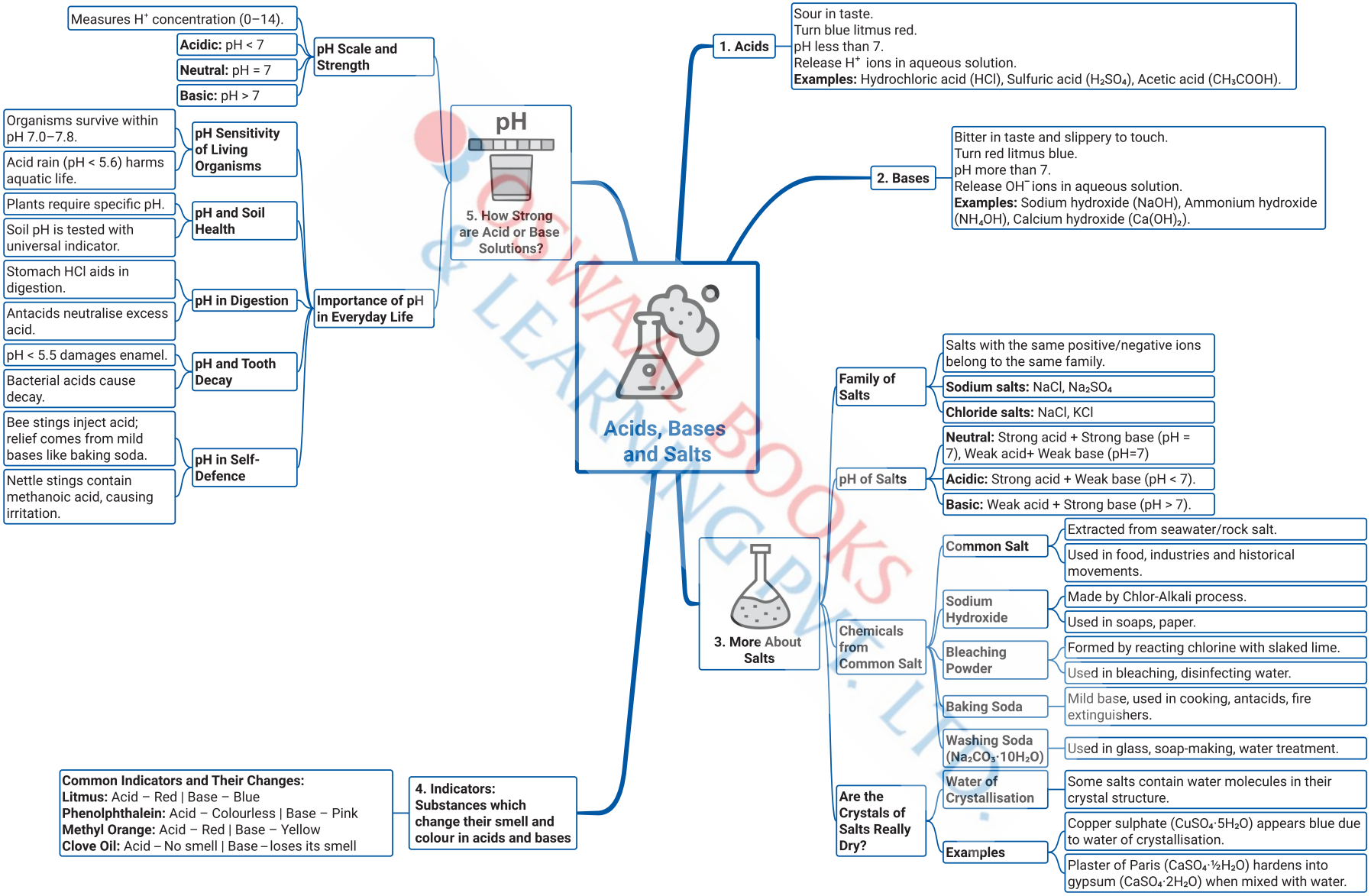

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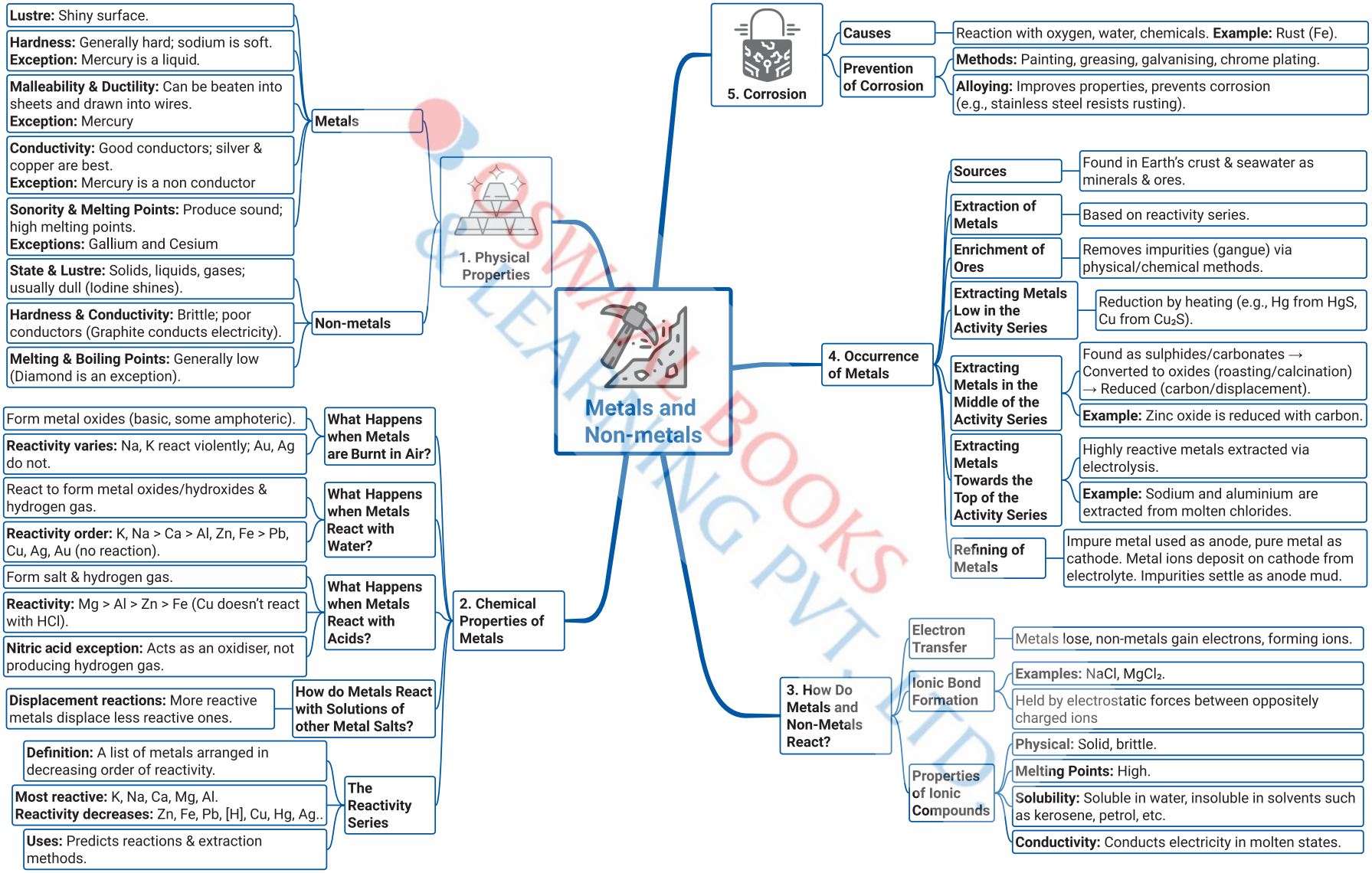

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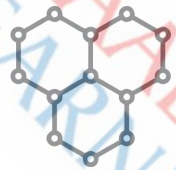
Carbon and its Compounds



1. Bonding in Carbon – The Covalent Bond

- **Nature of Carbon Compounds**: Low melting/boiling points, poor electrical conductivity due to absence of ions.
- **Electronic Configuration of Carbon**: **Atomic number:** 6. Need to share 4 electrons to attain stability.
- **Formation of Covalent Bonds**: Carbon shares electrons to form stable covalent bonds without ions.
- **Types of Covalent Bonds**: **Single:** H_2, Double: O_2, Triple: N_2.
- **Methane – A Simple Carbon Compound**: CH_4 is forms by sharing electrons with four H atoms.

2. Versatile Nature of Carbon

- **Unique Bonding Properties**
  - **Catenation:** Forms chains, branches, rings, enables vast compound diversity.
  - **Multiple Bonds:** Single, double, triple bonds.
- **Saturated and Unsaturated Carbon Compounds**
  - **Saturated:** Single bonds (e.g., Ethane - C_2H_6).
  - **Unsaturated:** Double/triple bonds (e.g., Ethene - C_2H_4, Ethyne - C_2H_2).
- **Chains, Branches and Rings**: Forms straight chains (Butane - C_4H_{10}), branched chains and ring chains (Cyclohexane - C_6H_{12}).
- **Will You Be My Friend? – Functional Groups**
  - Carbon forms bonds with elements like oxygen, nitrogen, sulphur, etc.
  - Functional groups determine chemical properties.
  - **Examples:** Alcohol, Aldehyde.
- **Homologous Series**
  - A series of compounds with the same functional group but different chain lengths.
  - Each successive compound differs by a - CH_2- unit.
  - **Examples:** Alkanes, Alkenes, Alkynes.
- **Nomenclature of Carbon Compounds**: Is based on the basic carbon chain, with prefixes or suffixes indicating functional groups, or vowel-starting suffixes.

3. Chemical Properties of Carbon Compounds

- **Combustion**
  - **Complete:** Produces CO_2 + heat/light.
  - **Incomplete:** Produces CO/soot.
  - **Blue flame:** Complete combustion. **Yellow flame:** Incomplete combustion.
- **Oxidation**
  - **Definition:** Oxygen addition.
  - **Examples:** Alcohols oxidise to carboxylic acids.
  - **Oxidising Agents:** $KMnO_4$, $K_2Cr_2O_7$.
- **Addition Reaction**
  - Unsaturated hydrocarbons add hydrogen (in the presence of palladium or nickel catalyst) to give saturated hydrocarbons.
  - **Application:** Hydrogenation of vegetable oils.
- **Substitution Reaction**
  - **Definition:** Atom/group replacement.
  - **Example:** $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (in the presence of sunlight).

4. Some Important Carbon Compounds – Ethanol and Ethanoic Acid

- **Properties of Ethanol**
  - Liquid, soluble in water, found in alcohol, medicine.
  - Reacts with Na: Produces hydrogen + sodium ethoxide.
  - **Dehydration:** Converts to ethene with concentrated H_2SO_4.
- **Properties of Ethanoic Acid**
  - **Common Name:** Acetic acid (vinegar component).
  - **Esterification:** Reacts with alcohols in presence of concentrated sulphuric acid to form esters (used in perfumes).
  - Reacts with bases (forms sodium ethanoate and water) & carbonates and hydrogencarbonates (produces salt, carbon dioxide and water).

5. Soaps and Detergents

- **Soap Structure and Function**
  - **Composition:** Sodium/potassium salts of long-chain carboxylic acids.
  - **Action:** Forms micelles to trap dirt.
  - **Hard Water Issue:** Forms insoluble precipitate with calcium and magnesium ions in hard water.
- **Detergents vs Soaps**
  - **Detergents:** Sulphonic acid salts, effective in hard water.
  - **Comparison:** Soaps form scum; detergents do not.

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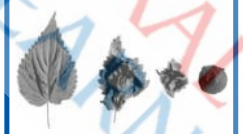
Life Processes

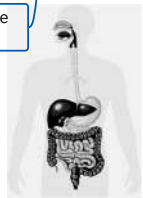


1. What Are Life Processes?

- **Maintenance in Living Organisms:** Prevents breakdown, active at rest.
- **Energy Requirement:** Needs external food sources.
- **Molecular Movements:** Essential for growth, powered by energy.
- **Specialised Systems:** Tissues manage nutrition, respiration, excretion. Transport systems distribute nutrients, remove waste.

2. Nutrition

- **Energy and Material Needs:** Food fuels movement, growth, maintenance.
- **Autotrophic Nutrition**
  - **Photosynthesis:** Light absorption → Energy conversion → CO_2 to carbohydrates.
  - **Stomata:** Gaseous exchange regulated by guard cells.
  - **Nutrient Uptake:** Water and soil, minerals (e.g., nitrogen for proteins).
- **Heterotrophic Nutrition**
  - Absorb decomposed matter (fungi), ingest food (humans), parasitic (tapeworms).
- **Nutrition in Organisms**
  - **Unicellular:** Absorption, photosynthesis, ingestion (Amoeba).
  - **Multicellular:** Specialised digestion (humans).
- **Nutrition in Human Beings**
  - Alimentary Canal
    - Tube (mouth to anus) for digestion and absorption.
  - Digestive Process
    - **Ingestion:** Chewing, salivary amylase.
    - **Stomach:** HCl (acidic), pepsin (proteins), mucus (lining protection).
    - **Small Intestine:** Bile (fat emulsification), pancreatic & intestinal enzymes (nutrient conversion).
    - **Absorption:** Villi absorb nutrients, large intestine absorbs water, excretes waste.



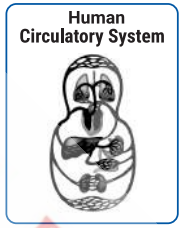
3. Respiration

- Cellular Respiration
  - **Aerobic:** $O_2 \rightarrow CO_2 + H_2O$ + energy
  - **Anaerobic:** No O_2 (e.g., lactic acid in muscles)
- ATP – The Energy Currency
  - Stores energy for cell activities.
- Gas Exchange
  - **Plants:** Stomata (CO_2 in, O_2 out).
  - **Animals:** Specialised respiratory organs.

Human Respiratory System

- **Pathway:** Nostrils → Throat → Lungs → Alveoli (gas exchange)
- **Breathing:** Inhale (O_2 in), Exhale (CO_2 out)
- **Oxygen Transport:** Haemoglobin carries O_2, CO_2 dissolves in blood

4. Transportation



Human Circulatory System

- Transportation in Human Beings
  - Blood Composition
    - Plasma (nutrients/wastes), RBCs (O_2 transport).
  - The Heart
    - Muscular pump, four chambers.
  - Double Circulation
    - Blood passes heart twice per cycle.
  - Blood Pressure
    - **Systolic:** 120 mm Hg, Diastolic: 80 mm Hg.
  - Blood Vessels
    - Arteries except pulmonary artery (oxygenated, high pressure), Veins except pulmonary vein (deoxygenated, low pressure Capillaries (exchange of substances).
  - Platelets and Clotting
    - Prevents blood loss.
  - Lymph
    - Transports fats, fluids.
- Transportation in Plants
  - Water and Mineral Absorption
    - Active root uptake.
  - Water Transport (Xylem)
    - Moves water via transpiration pull.
  - Food Transport (Phloem)
    - Translocates food via osmotic pressure.

5. Excretion

- Excretion in Human Beings
  - Excretory System
    - **Organs:** Kidneys, ureters, bladder, urethra.
  - Urine Formation
    - Nephron filtration → Urine excretion.
  - Artificial Kidney (Dialysis)
    - Filters blood in kidney failure.
- Excretion in Plants
  - O_2, H_2O via stomata; waste in vacuoles or shed with leaves.

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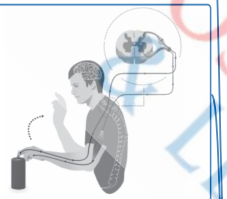
Control and Coordination in Animals and Plants

1. Animals – Nervous System

Nervous Control in Animals
- Controlled by nervous and muscular tissues.
- Receptors detect stimuli (e.g., gustatory for taste, olfactory for smell).
- Signals trigger electrical impulses, transmitted via neurons across synapses.

Reflex Action
- Automatic, rapid responses (e.g., withdrawing from heat).
- **Reflex arc:** stimulus → receptors → sensory neurons → spinal cord → motor neurons → response.
- Bypasses the brain for quicker reactions.



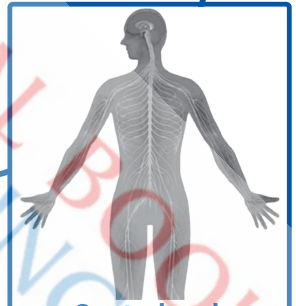
Brain as the Coordinating Centre
- Central nervous system (brain + spinal cord) integrates body functions.
- Controls voluntary (e.g., writing) and involuntary (e.g., heartbeat) actions.

Structure of the Human Brain
- **Fore-brain:** Thinking, sensory interpretation, voluntary movements.
- **Mid-brain:** Reflexes involving sight and hearing.
- **Hind-brain:** Involuntary actions (breathing, digestion), balance, posture (cerebellum), vital functions (medulla).

Human Brain

- **Brain:** Enclosed in the skull, cushioned by cerebrospinal fluid.
- **Spinal Cord:** Protected by the vertebral column.

Muscular Response to Nerve Stimulus
- Nervous tissue processes and transmits information.
- Muscle cells respond by contracting via protein-triggered shape changes.



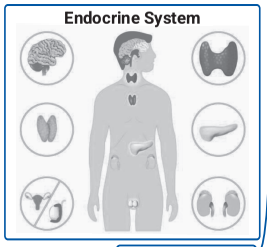
2. Coordination in Plants

Immediate Response to Stimulus
- Some plants respond to touch via electrochemical signals.
- Movement due to water content changes in cells.

Movement Due to Growth
- **Tendril Movement:** Unequal growth helps climbing plants cling to support.
- **Tropic Movements:** Shoots bend towards light (phot otropism), roots towards gravity (geotropism), roots towards water (hydrotropism), pollen tubes toward ovule (chemotropism).

Plant Hormones
- Auxins (cell elongation) gibberellins (stem growth),
- cytokinins (cell division), abscisic acid (inhibits growth)

3. Hormones in Animals



Endocrine System

Chemical Communication
- Hormones act as messengers, ensuring coordinated body functions.
- **Adrenaline:** Prepares for "fight or flight way," increasing heart rate and breathing.

Growth Regulation
- **Thyroxin:** Regulates metabolism; iodine deficiency causes goitre.
- **Growth Hormone:** Controls body growth; imbalance leads to dwarfism/gigantism.
- **Puberty/Sex Hormones:** Testosterone (males), Oestrogen (females).

Insulin and Blood Sugar
- **Insulin:** Regulates blood sugar; deficiency leads to diabetes.
- **Feedback Mechanism:** Regulation of hormone secretion (e.g., insulin release lowers high blood sugar).

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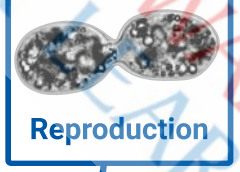
Reproduction



1. Importance of Variations

- **Similar Body Designs:** DNA-based blueprints ensure similar body designs.
- **DNA Copying Process:** DNA duplicates via chemical reactions, forming two copies that separate into new cells.
- **Inaccuracy in DNA Copying:** Imperfect copying causes variations, some harmful, others driving evolution.
- **Importance of DNA Consistency:** Preserves body designs suited for specific environments.
- **Importance of Variation:** Helps species survive environmental changes.

2. Modes of Asexual Reproduction

Fission

- **Binary Fission:** *Amoeba* splits into two; *Leishmania* divides directionally.
- **Multiple Fission:** *Plasmodium* produces several daughter cells.

Fragmentation

- *Spirogyra* fragments grow into new individuals.

Regeneration

- *Planaria* regrows from body parts using specialised cells.

Budding

- *Hydra* develops buds that detach upon maturity.

Vegetative Propagation

- **Plant Parts:** Roots, stems, leaves form new plants.
- **Agricultural Use:** Layering & grafting in crops like sugarcane and roses.
- **Genetic Similarity:** Offspring are identical to parents.

Tissue Culture

- Production or propagation of new plants from isolated plant cells or small pieces of plant tissue in a nutrient medium.

Spore Formation

- ***Rhizopus* Spores:** Sporangia produce spores.
- **Protection & Growth:** Thick-walled spores grow in moist environments.

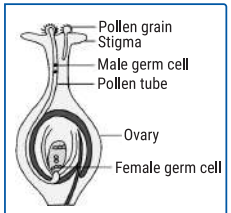
3. Sexual Reproduction

Importance of Sexual Reproduction

- Genetic Variation: Increases diversity.
- DNA Combination: Enhances ability to adapt

Sexual Reproduction in Flowering Plants

- Flower Structure: Stamens (male) & pistil (female).



Pollen grain germination

- **Pollination:** Self-/cross-pollination enables fertilisation. Male gametes are motile, female gametes are non-motile, store energy.
- **Germination:** Seeds grow under favourable conditions.

Reproduction in Human Beings

- Humans reproduce sexually via germ-cell fusion.

Puberty

- Puberty is the stage of sexual maturation triggered by hormones that makes males and females capable of reproduction.
- **Changes during puberty :** Breast development in girls, voice deepening and facial hair growth in boys and the development of pubic and underarm hair in both sexes.

Male Reproductive System

Structure and Function

- **Testes:** Produce sperm & testosterone.
- Sperm moves via vas deferens & urethra.
- **Gland Secretions:** Seminal vesicles & prostate nourish sperm.

Female Reproductive System

Structure and Function

- **Ovaries:** Produce eggs & hormones.
- **Oviduct or Fallopian Tubes:** Site for fertilisation
- **Uterus:** Supports embryo development.
- **Vagina:** Site for sperm entry & menstrual discharge.

Fertilisation and Development

- **Egg Release:** Monthly ovulation.
- **Zygote Formation:** Sperm-egg fusion forms zygote.
- Embryo is implanted in inner lining of uterus where it is supplied with nutrients and oxygen through placenta.

Menstruation

Menstrual Cycle

- Unfertilised egg disintegration followed by the shedding of uterine lining called menstruation (lasts 2-8 days)
- **Cycle Duration:** Average 28 days

Reproductive Health

Sexually Transmitted Infections (STIs)

- **Common STIs:** Gonorrhoea, syphilis (bacterial); HIV-AIDS (viral).
- **Prevention:** Condom use reduces risk.

Contraceptive Methods

- Condoms block sperm. Copper-T prevents implantation, oral pills inhibit ovulation.
- Vasectomy in males and tubectomy in females prevent transfer of gametes.

Population Control and Ethics

- **Family Planning:** Prevents unplanned pregnancies.
- **Impact of Population:** Strains resources & lowers living standards.
- **Sex-selective abortion:** illegal & unethical.

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Heredity

Inheritance ensures stability and variation. Asexual reproduction causes minor differences, while sexual reproduction enhances diversity, with environmental factors selecting advantageous traits.

1. Accumulation of Variation During Reproduction

2. Heredity and Evolution

Heredity transmits traits across generations, ensuring both similarity and variation in offspring.

Inherited Traits

Similarities and Differences: Offspring inherit traits from parents but also exhibit unique characteristics, contributing to diversity.

Rules for the Inheritance of Traits – Mendel's Contributions

Gregor Mendel's Work

Mendel's pea plant experiments led to the discovery of dominant and recessive traits through systematic analysis.

Mendel's Experiments

Crossbreeding pea plants revealed that dominant traits appear in F_1, while recessive traits reappear in F_2 when both inherited factors are recessive.

Independent Inheritance

Traits are inherited independently, leading to diverse genetic combinations in offspring.

Expression of traits in organisms

Role of DNA and Genes

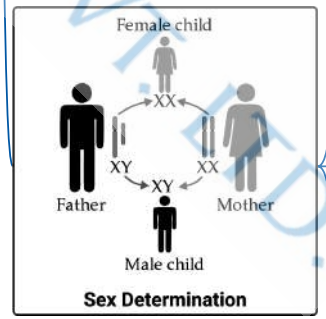
DNA encodes proteins that control traits. Genes are DNA sections influencing characteristics like plant height via enzyme activity.

Genetic Contribution in Sexual Reproduction

Offspring inherit two gene sets, one from each parent, explaining trait inheritance.

Chromosomes and Gene Sets

Genes are independent DNA pieces located on chromosome. Each germ cell carries one chromosome from each pair, ensuring a complete gene set in offspring.



Sex Determination

Environmental and Genetic Factors

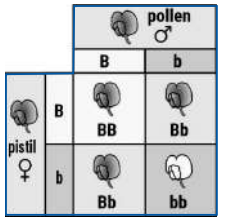
Some species determine sex environmentally, while in humans, it is genetically determined.

Human Sex Chromosomes

Of 23 chromosome pairs, 22 are identical in both sexes. The 23rd pair determines sex: XX in women, XY in men.

Inheritance of Sex Chromosomes

Children inherit an X from their mother. The father's X results in a girl, while a Y results in a boy.



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Light - Reflection and Refraction

1. Reflection of Light

- **Definition**: Bouncing of light when it strikes a polished surface.
- **Laws of Reflection**:
  - Angle of incidence = Angle of reflection.
  - Incident ray, reflected ray and normal lies in the same plane.

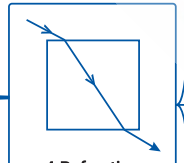
2. Spherical Mirrors

- **Types**:
  - **Concave Mirror:** Reflecting surface curved inward.
  - **Convex Mirror:** Reflecting surface curved outward.
- **Image Formation by Spherical Mirrors**:
  - Concave Mirror: Forms real/inverted or virtual/erect images depending on object position. Real image may be magnified or diminished. Virtual image is always erect and enlarged.
  - Convex Mirror: Always forms virtual, erect, and diminished images.
  - **Ray Diagrams for Image Formation**:
    - **Parallel Ray Rule (Principal):** A ray parallel to the axis passes through (or appears to diverge from) Focus(F).
    - **Focus Ray Rule:** A ray passing through Focus (F) emerges parallel to principal axis
    - **Centre of Curvature Rule:** A ray passing through C reflects back on the same path.
    - **Pole Rule:** A ray striking the pole follows the law of reflection.
- **Uses of Mirrors**:
  - Concave Mirrors: Used in torches, shaving mirrors, solar furnaces and dentists' mirrors.
  - Convex Mirrors: Used in vehicle rear-view mirrors for a wider field of view.
- **Mirror Formula and Magnification**:
  - Mirror Formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
  - Magnification Formula $m = \frac{h'}{h} = -\frac{v}{u}$
  - Magnification is positive for virtual images, and negative for real images.

3. Sign Conventions for Mirrors (New Cartesian Sign Convention)

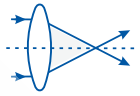
- **Distance Measurement**:
  - **Parallel to Axis:** Measured from the pole (P); in the direction of incident light is positive, opposite to the direction of incident light is negative.
  - **Perpendicular to Axis:** Above principal axis is positive, below is negative.
- **Focal Length and Image Properties**:
  - **Concave Mirror:** Focal length (f) and radius of curvature (R) are negative.
  - **Convex Mirror:** Focal length (f) and radius of curvature (R) are positive.

4.Refraction of Light



- **Definition**: Bending of light when passing from one medium to another due to a change in speed.
- **Laws of Refraction (Snell's Law)**:
  - (constant for a given medium pair) and a definite colour. $\frac{\sin i}{\sin r} = n$
  - Incident ray, refracted ray and normal, all in the same plane.
- **Refractive Index**: Absolute Refractive Index, $n = \frac{\text{Speed of light in vacuum } (c)}{\text{Speed of light in medium } (v)}$

5. Refraction by Lenses



- **Lens**: A transparent material bound by curved surface that refracts light.
- **Types of Lenses**:
  - Convex Lens (Converging Lens): Thicker at the centre; converges light to a focal point.
  - Concave Lens (Diverging Lens): Thinner at the centre; diverges light as if originating from a focal point.
- **Image Formation by Lenses**:
  - Convex Lens: Forms real/inverted or virtual/erect images depending on object position. Real image may be erect and magnified or diminished. Virtual image always magnified.
  - Concave Lens: Always forms virtual, diminished images.
- **Lens Formula and Magnification**:
  - Lens Formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
  - Magnification Formula $m = \frac{h'}{h} = \frac{v}{u}$
  - Magnification is positive for virtual images ,and negative for real images.

6. Power of a Lens

- **Definition**: The ability of a lens to converge or diverge light.
- Formula $P = \frac{1}{f \text{ (in metres)}}$
- Power of Combined Lenses $P_{\text{total}} = P_1 + P_2 + P_3 + \ldots$

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The Human Eye and the Colourful World

1. The Human Eye

Structure of the Human Eye

- **Cornea:** Transparent layer that refracts most of the incoming light.
- **Iris:** Coloured part that controls the pupil size to regulate light entry.
- **Pupil:** Small opening through which light enters the eye.
- **Lens:** A flexible, transparent convex lens that focuses light on the retina.
- **Retina:** A light-sensitive membrane with light sensitive rods and cone cells.
- **Optic Nerve:** Carries electrical signals from the retina to the brain.

Power of Accommodation

- **Focal Length Adjustment:** The eye lens changes its focal length using ciliary muscles.
- **Near Point:** Closest point of clear vision for a normal eye (25 cm).
- **Far Point:** Farthest point visible to a normal eye (infinity).
- **Cataract:** Clouding of the eye lens causing vision impairment.

2. Defects of Vision and Their Correction

Myopia (Short-sightedness)

- **Symptoms:** Near objects are clear, distant objects appear blurred.
- **Cause:** Either excessive curvature of the lens or an elongated eyeball.
- **Correction:** Using a concave lens to diverge light rays before they enter the eye.

Hypermetropia (Long-sightedness)

- **Symptoms:** Distant objects are clear, nearby objects appear blurred.
- **Cause:** The eye lens is too flat or the eyeball is too short.
- **Correction:** Using a convex lens to converge light rays before they enter the eye.

Presbyopia

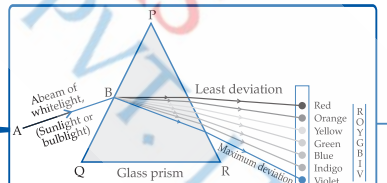
- **Symptoms:** Difficulty in seeing both near and distant objects.
- **Cause:** Weakening of ciliary muscles and loss of lens flexibility.
- **Correction:** Using bi-focal lenses (concave for distance, convex for near vision).

3. Refraction of Light Through a Prism

Refraction in a Triangular Prism

- **Prism Effect:** Light bends twice at inclined surfaces.
- **Angle of Deviation:** The angle between the emergent and incident rays.

4. Dispersion of White Light



Spectrum Formation

- **Definition:** Splitting of white light into seven colours (VIBGYOR).
- **Cause:** Different wavelengths bend at different angles.
- **Red Light:** Bends the least, Violet Light: Bends the most.
- **Rainbow Formation:** Caused by dispersion, refraction and internal reflection inside raindrops.

5. Atmospheric Refraction

Twinkling of Stars

- **Cause:** Continuous refraction due to varying air refractive index.
- **Effect:** Apparent position of stars fluctuates, making them appear to twinkle.

No Twinkling of Planets

- **Reason:** Planets appear as extended sources of light.
- **Effect:** Light variations cancel out, making them appear steady.

Advance Sunrise and Delayed Sunset

- **Cause:** Refraction bends light, making a virtual image of Sun visible before actual sunrise.
- **Time Difference:** The Sun appears 2 minutes early and 2 minutes late.

6. Scattering of Light

- **Definition:** It is the phenomenon where light rays deviate from their original path due to interactions with particles in the medium.

Applications in Daily Life

Tyndall Effect

- **Examples 1:** Sunlight passing through a dense forest canopy.
- **Example 2:** In colloid solutions like milk or soap water, light is scattered, making the path of light visible.

Why is the Sky Blue?

- **Cause:** Shorter wavelengths (blue) scatter more than red.
- **Effect:** The scattered blue light reaches our eyes.

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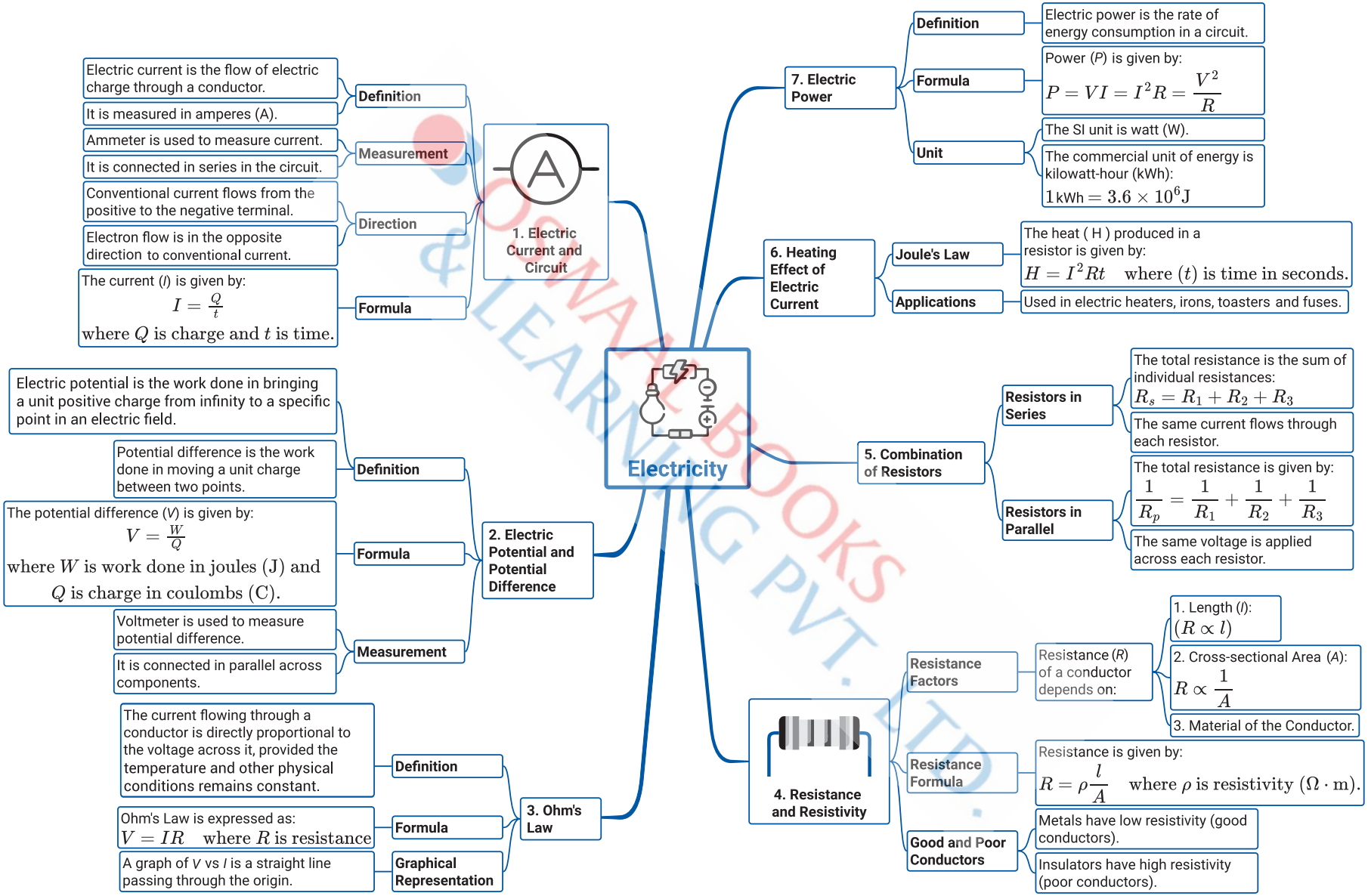

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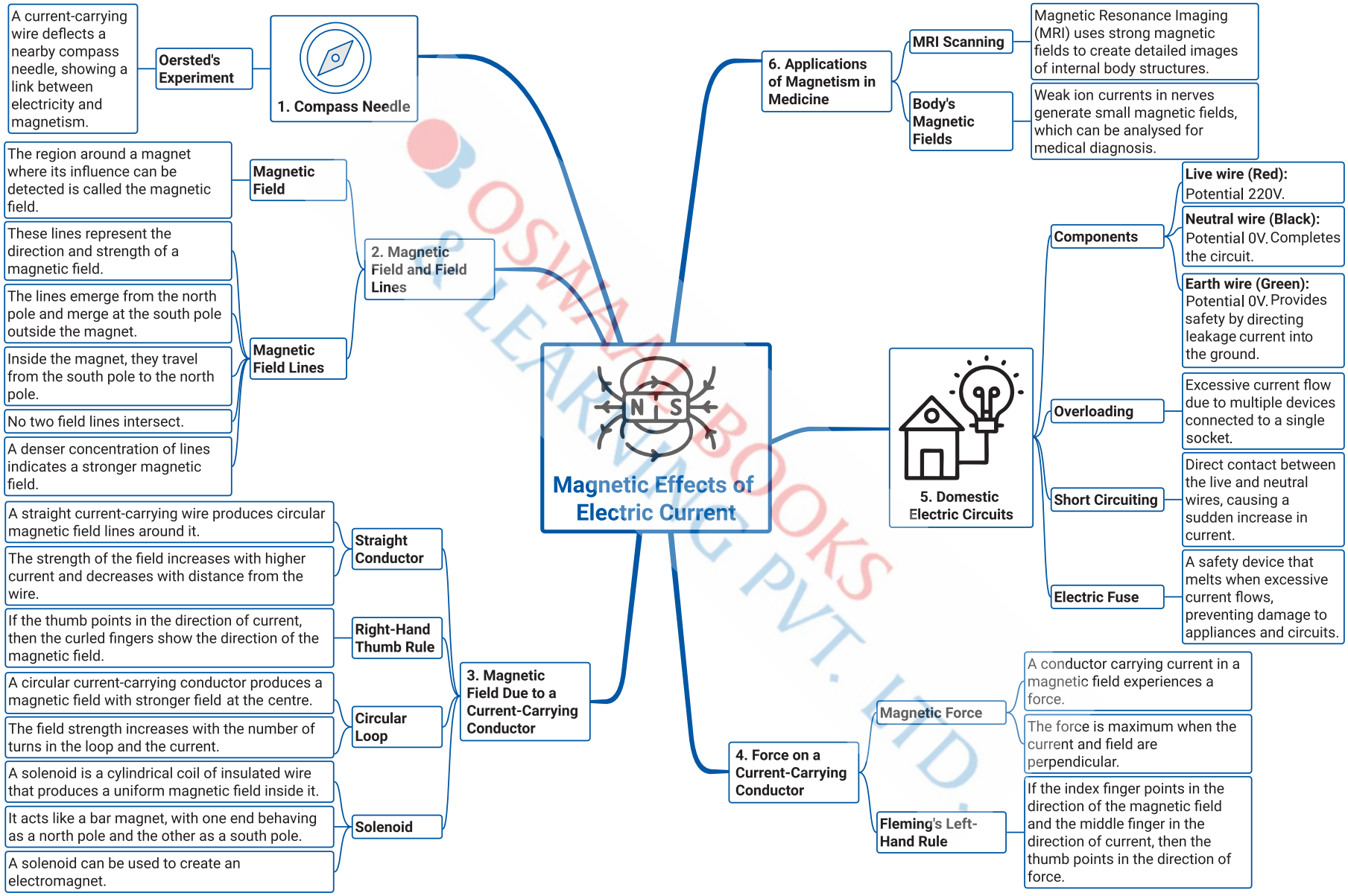

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Our Environment

1. Ecosystem and its Components

Biotic and Components
- **Biotic:** Living organisms (plants, animals, microorganisms) interact with their surroundings.
- **Abiotic:** Non-living factors (temperature, rainfall, wind, soil, minerals) influence ecosystems.

Types of Ecosystems
- **Natural:** Forests, ponds, lakes.
- **Artificial:** Gardens, crop fields.

Role of Organisms in an Ecosystem
- **Producers:** Green plants/bacteria make food via photosynthesis.
- **Consumers:** Depend on producers for food and nutrients.
- **Decomposers:** Recycle nutrients by breaking down dead and decaying matter.

Food Chains and Webs

Food Chains
Series of organisms feeding on one another.

Trophic Levels:
- **First:** Producers (autotrophs capture solar energy).
- **Second:** Primary consumers (herbivores).
- **Third:** Secondary consumers (small carnivores).
- **Fourth:** Tertiary consumers (large carnivores).

Energy Flow in Ecosystems
- Energy flows unidirectionally from producers to consumers.
- Only 10% of energy transfers to the next level.
- More organisms exist at lower trophic levels.

Food Web
- Interconnected food chains for a realistic ecosystem model.

Biological Magnification
- Pesticides and pollutants enter the food chain through polluted water.
- Non-biodegradable chemicals accumulate at higher trophic levels comprises.

2. Impact of Human Activiites on the Environment

Ozone Layer and its depletion

Importance of Ozone
- Ozone (O_3) absorbs harmful UV radiation, preventing skin cancer and ecological damage.

Causes of Ozone Depletion
- CFCs from refrigerators and fire extinguishers degrade ozone.
- Industrial emissions accelerate depletion.

Global Efforts to Protect Ozone
- The United Nations Environment Programme (UNEP) 1987: Agreement to reduce CFCs.
- Mandatory shift to CFC-free products.

Managing the Garbage

Types of Waste
- **Biodegradable:** Breaks down naturally (vegetable and fruits peel, paper, etc.).
- **Non-Biodegradable:** Does not break down naturally (plastics, metals, etc.).

Environmental Impact of Waste
- Excess waste leads to pollution.
- Disposable culture increases non-biodegradable waste.

Sustainable Waste Management
- 3Rs: Reduce, Reuse, Recycle.
- Promote biodegradable materials and eco-friendly packaging.
- Proper waste segregation and disposal.

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Chapter-1
Chemical Reactions and Equations

1 Chemical Reactions

Concept covered: • Chemical reaction and examples • Skeletal and balanced chemical equation • Steps to balance a chemical equation.

Chemical Reaction

[Board, 2020]

Chemical Reactions

Scan Me!

- A chemical reaction is a process in which the original substance(s) loses its nature and identity and forms new substance(s) with different properties.
- Breaking of the chemical bonds and formation of new chemical bonds 'results in a' or 'causes' a chemical reaction.
- The substances that take part in a chemical reaction are called Reactants.
- The substances that are formed in a chemical reaction are called Products.
- A chemical reaction can be identified by either of the following observations: [Board, 2024]

S. No.	Characteristics	Examples
1.	Change in state [Board, 2024]	The combustion reaction of candle wax is characterised by a change in state from solid to liquid and gas.
2.	Change in colour	The chemical reaction between citric acid and purple-coloured potassium permanganate solution is characterised by a change in colour from purple to colourless.
3.	Evolution of gas	The chemical reaction between zinc and dilute sulphuric acid is characterised by hydrogen gas. $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2\uparrow(g)$
4.	Change in temperature	The reaction between quicklime and water to form slaked lime is characterised by an increase in temperature. $CaO + H_2O \rightarrow Ca(OH)_2 + Heat$
5.	Formation of a precipitate	When an aqueous solution of sodium sulphate is mixed with the aqueous solution of barium chloride, barium sulphate comes in the form of white precipitate.
		$Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(\downarrow) + 2NaCl(aq)$

Chemical Equations

[Board, 2016]

- A chemical equation is the symbolic representation of a chemical reaction in the form of symbols and formulae.
- It is a way to represent the chemical reaction in a concise and informative way.
- For example,

Magnesium + Oxygen → Magnesium oxide
(Reactants) (Product)

This equation is called word equation.

- The word equation can be converted into chemical equation by writing symbols and formulae of the substance instead of their names $2Mg + O_2 \rightarrow 2MgO$

Writing a Chemical Equation:

(i) The symbols of elements and the formulae of reacting substances (reactants) are written on the left-hand side of the equation, with a plus (+) sign between them.

(ii) The symbols and formulae of the substances formed (products) are written on the right-hand side of the equation, with a plus sign (+) between them.

(iii) An arrow sign (→) is put between the reactants and the products.

(iv) The physical states of the reactants and products are also mentioned in a chemical equation.

Skeletal Chemical Equation: A chemical equation which simply represents the symbols and formulae of reactants and products taking part in the reaction is known as skeletal chemical equation for a reaction.

For example: $Mg + O_2 \rightarrow MgO$ is a skeletal equation.

Balanced Equation: The equation in which atoms of various elements on both sides of a chemical equation are equal in accordance with the Law of Conservation of Mass. [Board, 2024]

For example:

(i) $CO(g) + 2H_2(g) \xrightarrow{340\ atm} CH_3OH(l)$

(ii) $6CO_2(g) + 12H_2O(l) \xrightarrow[chlorophyll]{sunlight} \underset{Glucose}{C_6H_{12}O_6(aq)} + 6O_2(aq) + 6H_2O(l)$

The process of equalising the atoms of the elements on both sides of an equation is called balancing a chemical equation. This is known as hit and trial method. Let us understand this with the help of an example given below:

Example 1

Balancing a chemical equation:

Step 1: Write the chemical equation.

$$Fe + H_2O \rightarrow Fe_3O_4 + H_2$$

Step 2: Count the number of atoms of each element on both sides of the chemical equation.

	Element	No. of atoms at reactant side	No. of atoms at product side
1.	Fe	1	3
2.	H	2	2
3.	O	1	4

Step 3: Equalise the number of atoms of element with the maximum number by trial and error method 'by placing it in front of it.' $Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$

Step 4: Try to equalise all the atoms of elements on both reactant and product sides by adding coefficient in front of it. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

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2 Types of Chemical Reactions

Concept covered: • Combination reaction • Decomposition reaction • Displacement reaction • Double displacement reaction • Redox reaction • Oxidation and reduction reaction • Exothermic and endothermic reaction.

Types of Chemical Reactions

I. Combination Reaction: The reaction in which two or more reactants combine to form a single product. e.g.,

(i) Burning of coal

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

(ii) Formation of water

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

(iii) $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$

(Quicklime) (Slaked lime)

Exothermic Reactions: Reaction in which heat is released along with the formation of products e.g.,

(i) Burning of natural gas.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + Heat$$

(ii) Respiration is also an exothermic reaction.

$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(aq) + 6H_2O(l) + Energy$

(Glucose)

II. Decomposition Reaction: [Board 2024]

The reaction in which a compound splits into two or more simpler substances is called decomposition reaction.

$$A \rightarrow B + C$$

(a) Thermal Decomposition: When decomposition is carried out by heating. e.g.,

(i) $2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$

(Ferrous sulphate) Green colour — (Ferric oxide) Red-brown colour

(ii) $CaCO_3(s) \xrightarrow{Heat} CaO(s) + CO_2(g)$

(Lime stone) (Quicklime)

(b) Electrolytic decomposition: When decomposition is carried out by passing electricity. e.g.,

$$2H_2O(l) \xrightarrow[current]{Electric} 2H_2(g) + O_2(g)$$

(c) Photolytic decomposition: When decomposition is carried out in presence of sunlight. e.g.,

(i) $2AgCl(s) \xrightarrow{Sunlight} 2Ag(s) + Cl_2(g)$

(ii) $2AgBr(s) \xrightarrow{Sunlight} 2Ag(s) + Br_2(g)$

Endothermic Reaction: The reactions which require energy in the form of heat, light or electricity to break reactants are called endothermic reactions. e.g.,

(i) $NH_4Cl\ (s) + H_2O\ (l) \rightarrow NH_4Cl\ (aq) - Heat$

III. Displacement Reaction: The chemical reactions in which more reactive element displaces less reactive element from its salt solution. e.g.,

(i) $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

(Iron) (Copper sulphate) (Ferrous sulphate) (Copper)

(ii) Zinc displaces copper forming zinc sulphate. Zinc is more reactive than copper.

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

(Zinc Sulphate)

IV. Double Displacement Reaction: A reaction in which new compounds are formed by mutual exchange of ions between two compounds. e.g.,

$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

(Sodium sulphate) (Barium chloride) (Barium sulphate) (Sodium chloride)

White precipitate of $BaSO_4$ is formed, so it is also called **precipitation rection.**

Precipitation reaction: Precipitation reactions occur when cations and anions in aqueous solution combine to form an insoluble ionic solid called a precipitate. e.g., $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$ (Precipitate)

V. Oxidation and Reduction:

Loss of electron/s is called **Oxidation** and gain of electron/s is call **Reduction.**

Oxidation: It is a process of gaining oxygen or loss of hydrogen atom during a reaction by an atom, molecule, or ion. When any atom loses electron or increase in oxidation number is observed by another atom, ion or molecule, it is said to be oxidation.

$$2Cu + O_2 \xrightarrow{Heat} 2CuO$$ [Board, 2019]

Reduction: It is a process of gaining hydrogen or loss of oxygen atom during a reaction by an atom, molecule, or ion. When any atom gain electron or decrease in oxidation number is observed by another atom, ion or molecule, it is said to be reduction.

$$CuO + H_2 \longrightarrow Cu + H_2O$$

MNEMONICS

Concept 1: Types of decomposition reaction
Mnemonics: PET
Interpretations:
PET: Photolytic reaction, Electrolytic reaction, Thermal reaction

Concept 2: Oxidation and reduction reaction
Mnemonics: OIL RIG
Interpretations:
OIL RIG: Oxidation Is Loss of electrons, Reduction Is Gain of electrons

Concept 3: Types of chemical reactions
Mnemonics: ROC.D^3
Interpretations:
ROC.D^3: Reduction, Oxidation, Combination, Decomposition, Displacement, Double Displacement

Redox (Oxidation and Reduction) Reaction: In this reaction, CuO is reduced to Cu and H_2 is oxidised to H_2O. In other words, one reactant gets oxidised while the other gets reduced. Such reactions are called oxidation–reduction reactions or redox reactions. [Board 2024]

Oxidation (Cl_2 is Oxidised)

$$MnO_2(s) + 4HCl\ (conc.) \rightarrow Mn(aq) + Cl_2(g) + 2H_2O$$

Reduction (Mn is reduced)

A redox (oxidation-reduction)reaction involves the transfer of electrons between reactants. Oxidation involves the loss of electrons, while reduction involves the gain of electrons. [Board, 2023]

Oxidation and Reduction

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Chapter-2
Acids, Bases and Salts

1 Acids and Bases

Concept covered: • Definition of acids and bases • Properties of acids and bases • Indicators and its type • pH scale and importance of pH in everyday life.

Acids

• Acids are the substances that furnish H^+ ions in aqueous solution. Acids are sour in taste. They turn blue litmus red.

• Its examples include: Sulphuric acid (H_2SO_4), Acetic acid (CH_3COOH), Nitric acid (HNO_3), etc.

• If in an aqueous solution, concentration of acid is low as compared to water, it is called dilute solution and if concentration of acid is high, it is called concentrated solution.

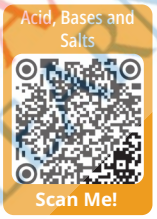
• Acids which dissociate into ions completely are called strong acids, e.g., H_2SO_4, HCl, HNO_3, etc.

• Acids which do not dissociate into ions completely are called weak acids, e.g., citric acid, acetic acid, formic acid, etc.

Note:
Although we talk about 'taste' of acids and bases, it is not advisable to taste any acid or base. Most of them are harmful. Similarly, touching the solutions of strong acids and bases should be avoided. They may harm the skin.

Bases

• Bases are those chemical compounds which are bitter in taste, soapy in touch, turn red litmus blue and give OH^- ions in aqueous solution.



• The examples include sodium hydroxide (NaOH), potassium hydroxide (KOH), etc.

• The substances/bases which ionise completely to furnish OH^- ions are called strong bases, e.g., KOH, NaOH, etc.

• The bases which ionise only partially are called weak bases, e.g., $Mg(OH)_2$, $Cu(OH)_2$, etc.

• Both acids and bases conduct electricity in their aqueous solution due to the presence of free ions.

• Indicators: These are the substances which change their colour/smell in different acids or bases.

• **Identification using indicators:**

There are different type of indicators used to distinguish between acids and bases

1.Natural Indicators: These are the indicators which changes their colour in bases and acids,but extracted naturally.

Example: Turmeric, Litmus, China rose

(i) Litmus Paper: For Acids: Blue litmus paper changes colour to red in acids.

For Bases: Red litmus paper changes colour to blue in bases.

(ii) China Rose: For Acids: It changes colour to red in acids.

For Bases: It changes colour to green in bases.

(iii) Turmeric: For Acids: It remains yellow in acids.

For Bases: It changes colour to red in bases.

2.Synthetic Indicators: These are the indicators which changes their colour in bases and acids, but made synthetically.

Example: Methyl Orange, Phenolphthalein

(i) Phenolphthalein: For Acids: It remains colourless in acids.

For Bases: It changes colour to pink in bases.

(ii) Methyl Orange: For Acids: It changes colour to red in acids

For Bases: It changes colour to yellow in bases.

3.Olfactory Indicators: These are the indicators which changes their smell in bases and acids

Example: Onion ,Clove oil, Vanilla Extract

(i) Onion: For Acids: Its smell retains in acids.

For Bases: It loses its smell in bases.

(ii) Vanilla Extract: For Acids: Its smell retains in acids.

For Bases: It loses its smell in bases.

(iii) Clove Oil: For Acids: Its smell retains in acids.

For Bases: It loses its smell in bases.

• **Chemical Properties of Acids and Bases:** [Board 2018]

1.Reaction of Metals with: [Board 2024]

Acids	Bases
Acid + Metal → Salt + Hydrogen gas	Base + Metal → Salt + Hydrogen gas
e.g., $2HCl + Zn \rightarrow ZnCl_2 + H_2(g)$ Zinc chloride	e.g., $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2(g)$ Sodium zincate

Test for H_2 gas: Hydrogen gas released can be tested by bringing a burning candle near gas bubbles; it bursts with pop sound. [Board 2024]

2.Reaction of Metal Carbonates/Metal Hydrogen Carbonates with:

Acids	Bases
Acid + Metal Carbonate /Metal Hydrogen Carbonate	Base + Metal Carbonate /Metal Hydrogen Carbonate
↓	↓
Salt + $CO_2(g)$ + H_2O	No Reaction

e.g., $2HCl + Na_2CO_3 \rightarrow 2NaCl + CO_2(g) + H_2O$

$HCl + NaHCO_3 \rightarrow NaCl + CO_2(g) + H_2O$

Test for CO_2: CO_2 can be tested by passing it through lime water. Lime water turns milky due to formation of insoluble calcium carbonate (white precipitate).

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

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When excess CO_2 is passed, milkiness disappears due to formation of soluble calcium hydrogen-carbonate.

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

3. Reaction of Acids and Bases with Each Other

When an acid reacts with base, the hydrogen ion of acid combines with hydroxide ion of base and forms water. As these ions combine together, they form water instead of remaining free, thus both neutralise each other.

Acid + Base → Salt + Water [Board 2024]

Since in the reaction between an acid and a base both neutralise each other, it is also known as neutralisation reaction.

Properties of Acids and Bases

Scan Me!

Example: Sodium hydroxide (a strong base) reacts with hydrochloric acid to form sodium chloride and water.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

4. Reaction of Metallic Oxides with Acids:

When metallic oxides react with acids, formation of salt and water takes place.

Metal Oxide + Acid → Salt + Water

For Example: Copper oxide reacts with hydrochloric acid, formation of copper chloride and water take place

$CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$

5. Reaction of Non-Metallic Oxides with Basess:

When non-metallic oxides react with bases, formation of salt and water takes place.

Non-Metal Oxide + Base → Salt + Water

For Example: Carbon dioxide reacts with calcium hydroxide, formation of calcium carbonate and water take place

$Ca(OH)_2(s) + CO_2(aq) + \rightarrow CaCO_3(s) + H_2O(l)$

Common Properties of All Acids and Bases [Board 2024]

• Acids give hydrogen gas when they react with metal. This shows that all acids contain hydrogen.

• When acids are dissolved in water they dissociate as H^+ ions. The dissociation of hydrogen ions in aqueous solution is the common property of all acids. As a result, an acid shows acidic behaviour.

$HCl\ (aq) \rightarrow H^+(aq) + Cl^-(aq)$

$HNO_3(aq) \rightarrow H^+(aq) + NO_3^-\ (aq)$

$CH_3COOH\ (aq) \rightarrow H^+\ (aq) + CH_3COO^-\ (aq)$

• As H^+ ion cannot exist alone, it combines with water molecules and forms H_3O^+ (hydronium) ions.

Example:

$HCl + H_2O \rightarrow H_3O^+ + Cl^-$

$H^+ + H_2O \rightarrow H_3O^+$

Thus, acids can also be defined as substances, which when dissolved in water ionise to produce hydrogen ions, $H^+\ (aq)$.

• Similarly, substances which when dissolved in water ionise to produce hydroxide ions, $OH^-\ (aq)$.

Examples: When sodium hydroxide is dissolved in water, it dissociates into hydroxide and sodium ion.

$NaOH(s) \xrightarrow{H_2O} Na^+(aq) + OH^-\ (aq)$

4. Reaction of Acids with Metal Oxides:

Metal oxides react with acids to give salt and water.

Metal oxide + Acid → Salt + Water

Example: Copper oxide reacts with dilute hydrochloric acid to form copper chloride (salt) and water.

$CuO + 2HCl(dil.) \rightarrow CuCl_2 + H_2O$

Copper oxide Copper chloride

Copper oxide is black in colour. When dilute hydrochloric acid is added to it, the colour of the solution becomes blue-green due to the formation of copper chloride.

How Strong Are Acid or Base Solutions?

• Strength of an acid or base depends on the number of H^+ ions or OH^- ions produced by them respectively.

• Based on its ability to dissociate into ions in the solution, acids and bases are classified as strong acid or base and weak acid or base.

(a) Strong Acids: Acids that completely dissociate in water to produce large amount of hydrogen ions are called strong acids. For example, hydrochloric acid (HCl), sulphuric acid (H_2SO_4), nitric acid (HNO_3) are strong acids as they get completely ionised in water to form ions.

$HCl\ (aq) \rightarrow H^+(aq) + Cl^-\ (aq)$

(b) Weak Acids: Acids which get partially ionised in water to produce small amount of hydrogen ions are known as weak acids. For example, acetic acid partially dissociates in water to produce small amount of hydrogen ions.

$CH_3COOH\ (aq) \rightarrow H^+\ (aq) + CH_3COO^-(aq)$

(c) Strong Bases: Bases which completely ionise in water to produce large amount of hydroxide ions are called strong bases. Examples include NaOH, KOH, etc.

(d) Weak Bases: Bases which partially dissociate in water to furnish lesser amount of hydroxide ions are called weak bases. Examples include ammonium hydroxide (NH_4OH) and calcium hydroxide $Ca(OH)_2$.

pH Scale

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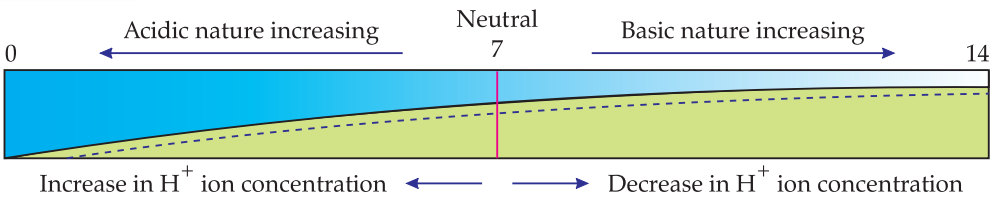
pH

The concentration of H^+ is generally small, therefore concentrations of H^+ is expressed in terms of pH.

pH: A scale for measuring H+ ion concentration in a solution..

Key Diagram



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Importance of pH in Everyday Life: [Board 2024]

- Plants and animals are pH sensitive.
- Our body works within the pH range of 7–7.8.
- When pH of rainwater is less than 5.6, it is called acid rain. Acid rain is rain or any other form of precipitation that is unusually acidic, meaning that it has elevated levels of hydrogen ions (low pH).
- Plants require a specific pH range for their healthy growth.
- pH of stomach is 1.5–3.0 due to the secretion of HCl. In case of indigestion, acidity increases, which can be neutralised by antacids like milk of magnesia.
- Tooth decay starts when the pH of the mouth is lower than 5.5. To prevent tooth decay, toothpastes which are basic in nature are used to neutralise the excess acid.
- Many plants and animals produce certain acids to defend themselves. For example, bee stings leave an acid into the skin, which causes pain and irritation. If a mild base like baking soda is applied to the stung area, it gives relief.

Key Facts

When you mix acid with water, it's extremely important to add the acid to the water rather than the other way around.
Just Remember: If we add water to acid, the high concentration of acid may produce a violent exothermic reaction.

2 Salts, Their Properties and Uses

Concept covered: • Salts • Types of salts • Preparation and uses of Sodium Hydroxide, Bleaching powder, Baking soda, Washing soda and Plaster of Paris.

- **Salts:** A salt is an ionic compound that results from the **neutralisation reaction** of an acid and a base. Salts are composed with the equal numbers of cations and anions, so that, the product is electrically neutral.

Neutralisation reaction: The reaction in which base or basic oxide reacts with acid or acidic oxide is called neutralisation reaction. Example : $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$.

Types of Salts:

(a) Neutral Salts: Salts produced by the reaction of a strong acid and a strong base are neutral in nature with the pH value 7. For example, sodium chloride formed by a reaction between sodium hydroxide (strong base) and hydrochloric acid (strong acid).

$$NaOH\ (aq) + HCl\ (aq) \rightarrow NaCl\ (aq) + H_2O\ (l)$$

(b) Acidic Salts: Salts formed by a reaction between a strong acid and a weak base are acidic in nature with the pH value of less than 7. For example, ammonium hydroxide. It is a salt of hydrochloric acid (strong acid) and ammonium hydroxide (weak base).

$$NH_4OH\ (aq) + HCl\ (aq) \rightarrow NH_4Cl\ (aq) + H_2O\ (l)$$

(c) Basic Salts: Salts formed by reaction of a strong base and a weak acid are basic in nature with the pH value of more than 7. For example, sodium carbonates. It is a salt of carbonic acid (weak acid) and sodium hydroxide (strong base).

Acid and Base
Scan Me!

$$H_2CO_3\ (aq) + 2NaOH\ (aq) \rightarrow Na_2CO_3\ (aq) + 2H_2O\ (l)$$

- **Common Salt (NaCl):**

Preparation: $NaOH + HCl \rightarrow NaCl + H_2O$

Properties:

$$2NaCl\ (aq) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + Cl_2\ (g) + H_2\ (g)$$

Uses of common salt:

(a) Used as daily food.

(b) Used as preservative.

(c) Used in manufacture of metal (Na) and gas (Cl_2) in molten state by electrolysis.

- **Sodium Hydroxide (NaOH)** [Board 2024]

Preparation: $2NaCl\ (aq) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + Cl_2\ (g) + H_2\ (g)$

The process is called chlor-alkali process because of the products formed for chlorine and alkali for sodium hydroxide.

Uses:

(a) Sodium hydroxide is used in making of paper, soap and detergents, for de-greasing metals, etc.

(b) Chlorine gas is used in water treatment, manufacturing of PVC, pesticides, etc.

(c) Liquid hydrogen is used as rocket fuel, in hydrogenation process of oil to produce vegetable *ghee* (margarine) and in making of ammonia for fertilisers.

- **Bleaching Powder ($CaOCl_2$):**

Preparation: It is produced by the action of chlorine on dry slaked lime.

$$Cl_2 + Ca(OH)_2 \rightarrow CaOCl_2 + H_2O$$

Properties:

(a) It has a strong smell of chlorine.

(b) Soluble in water.

(c) It loses chlorine by the action of carbon dioxide.

Water of Crystallisation and Plaster of Paris
Scan Me!

Uses:

(a) Bleaching cotton and linen in textile industry.

(b) Bleaching wood pulp in paper factories.

(c) Oxidising agent in chemical industries.

(d) Disinfecting drinking water.

- **Baking Soda (Sodium Hydrogen Carbonate) ($NaHCO_3$):**

Preparation: $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$ (Baking soda)

Properties:

(a) It is mild non-corrosive base.

(b) When it is heated during cooking, the following reaction takes place.

$$2NaHCO_3(s) \xrightarrow{\Delta} Na_2CO_3(s) + H_2O(g) + CO_2(g)$$

Uses:

(a) For making baking powder (mixture of baking soda and tartaric acid). When baking powder is heated or mixed with water, CO_2 is produced which causes breads and cakes to rise, making them soft and spongy.

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$NaHCO_3 + H^+ \rightarrow CO_2 + H_2O$ + Sodium salt of an acid [Board 2020]

(b) An ingredient in antacid.

(c) Used in soda acids, fire extinguishers.

- **Washing Soda ($Na_2CO_3.10H_2O$):** [Board 2024]

Preparation: Re-crystallisation of sodium carbonate gives washing soda. It is a basic salt.

$$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3.10H_2O$$

Properties:

(a) Transparent crystalline solid.

(b) It has 10 molecules of water of crystallisation.

(c) It dissolves in water and the aqueous solution is alkaline.

(d) It liberates carbon dioxide when treated with hydrochloric acid and sulphuric acid.

Uses:

(a) In glass, soap and paper industry.

(b) Manufacture of borax.

(c) It can be used as cleaning agent.

(d) It can be used to remove permanent hardness of water.

- **Plaster of Paris (Calcium Sulphate Hemihydrate) ($CaSO_4.\frac{1}{2}H_2O$):** [Board 2024]

Preparation: When gypsum $CaSO_4.2H_2O$ is heated at 373K, it loses water molecules and becomes Plaster of Paris (POP).

It is a white powder and on mixing with water it changes to gypsum.

$$CaSO_4.2H_2O \rightarrow CaSO_4.\frac{1}{2}H_2O + 1\frac{1}{2}H_2O$$

Properties: POP reacts with water to form gypsum.

$$CaSO_4.\frac{1}{2}H_2O + 1\frac{1}{2}H_2O \rightarrow \underset{\text{Gypsum}}{CaSO_4.2H_2O}$$

Uses:

(a) Doctors use POP to support fractured bones.

(b) It is also used to make toys and material for decoration.

Water of Crystallisation

- Water molecules present in the crystal structure of salt are called water of crystallisation and such salts are called hydrated salts. Water of crystallisation is the fixed number of water molecules present in one formula unit of a salt.

Examples:

- **Copper Sulphate Pentahydrate ($CuSO_4.5H_2O$):** It has five water molecules in one formula unit of copper sulphate (blue vitriol).
- **Sodium Carbonate decahydrate ($Na_2CO_3.10H_2O$):** It has ten molecules of water as water of crystallisation.
- **Gypsum ($CaSO_4.2H_2O$):** It has two molecules of water as water of crystallisation.

Chapter-3
Metals and Non-Metals

1 Properties of Metals and Non-Metals

Concept covered: • Physical properties of metals and non-metals • Chemical properties of metals and non-metals • Reaction with oxygen • Reaction with water • Reaction with acids • Reaction with other metal salt solutions • Reaction between metals and nonmetals (ionic bond formation).

- **Physical Properties of Metals and Non-metals:**

Property	Metals	Non-Metals
1. Lustre	Metals have shining surface.	They do not have shining surface. • Except iodine.
2. Hardness	They are generally hard. • Except sodium, lithium and potassium which are soft and can be easily cut with knife.	Generally soft. • Except diamond, a form of carbon which is the hardest natural substance.
3. State	Exist as solids. • Except mercury that exists as liquid.	Exist as solids or gases. • Except bromine that exists as liquid.
4. Malleability	Metals can be beaten into thin sheets. • Gold, silver and aluminium are the most malleable metals.	Non-metals are non-malleable. • They are brittle.
5. Ductility	Metals can be drawn into thin wires.	They are non-ductile.
6. Conductor of heat & electricity	Metals are good conductors of heat and electricity.	Non-metals are poor conductors of heat and electricity. • Except graphite.
7. Density and melting point	Generally, metals have high density and high melting point. • Except sodium, potassium, rubidium, cesium and Gallium	Non metals have low density and low melting point.
8. Sonorous	Metals produce a sound on striking a hard surface.	They are not sonorous.
9. Oxides	Metallic oxides are basic in nature.	Non-metallic oxides are acidic in nature.

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• **Chemical Properties of Metals:** [Board, 2019]

(A) Reaction of Metals with Air: Metals combine with oxygen to form metal oxides.

Metals + O_2 → Metal oxide

Examples:

(i) $2Cu + O_2 \rightarrow 2CuO$

Copper (II) oxide (black)

(ii) $4Al + 3O_2 \rightarrow 2Al_2O_3$

Aluminium oxide

(iii) $2Mg + O_2 \rightarrow 2MgO$

Magnesium Oxide

Different metals show different reactivity towards O_2.

➢ Na and K react so vigorously with oxygen that they catch fire if kept in open. So, they are kept immersed in kerosene. [Board, 2024]

➢ Surfaces of Mg, Al, Zn and Pb are covered with a thin layer of oxide which prevents them from further oxidation.

➢ Fe does not burn on heating but iron fillings burn vigorously.

➢ Cu does not burn but is coated with black copper (II) oxide.

➢ Au and Ag do not react with oxygen.

➢ **Amphoteric Oxides:** Metal oxides which react with both acids as well as bases to produce salt and water are called **amphoteric oxides**. [Board, 2024]

Examples: $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

Aluminium chloride

$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$

Sodium aluminate

(B) Reaction of Metals with Water: Metals react with water to produce metal hydroxide and hydrogen gas. This metal oxide further react with water to give metal hydroxide.

Metal + Water → Metal oxide + Hydrogen

$2Na + H_2O \rightarrow Na_2O + H_2 (\uparrow)$

Examples:

Metal oxide + Water → Metal hydroxide

$MgO + H_2O \rightarrow Mg(OH)_2$

Magnesium Hydroxide

• Sodium and potassium react vigorously with cold water. [Board, 2024]

$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g) + Heat$

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g) + Heat$

• Magnesium metal reacts with hot water to produce magnesium hydroxide and hydrogen gas.

$Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2$

• Aluminium and zinc react with steam to produce metal oxide and hydrogen gas.

• Metals like silver, gold, copper and lead does not react with water.

Physical properties of metals and non-metals

Scan Me!

(C) Reactions of Metals with Acid:

• Metal + Dil. Acid ⟶ Salt + Hydrogen gas

e.g., $Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2 (\uparrow)$

• Copper, mercury and silver don't react with dilute acids.

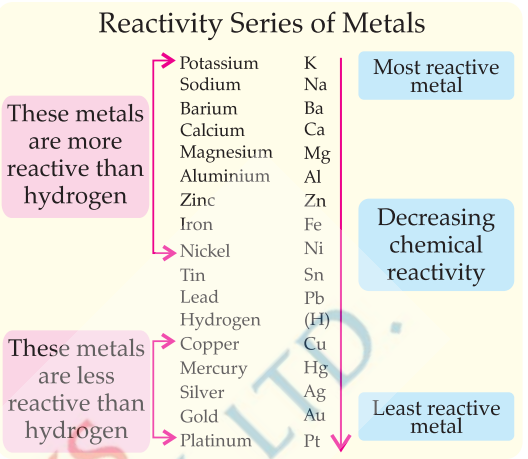
(D) Reaction of Metals with Solutions of Other Metal Salts:

Metal A + Salt solution B → Salt solution A + Metal B

➢ Reactive metals can displace less reactive metals from their compounds in solution form.

$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

• **Reactivity or Activity Series of Metals:** All metals do not react with the same rate. Some react very fast, some react moderately, whereas others react very slowly. The series of metals in decreasing order of reactivity is called reactivity series or activity series of metals. The metals at the top (K at the top most) are most reactive whereas metals at the bottom (Pt at the extreme bottom) are least reactive. [Board, 2024]



MNEMONICS

Concept: Activity Series of Metals

Mnemonics: Popular Scientists Can Make A Zoo In The Low Humid Country More Satisfactorily

Interpretations:

Arrange vertically:

Potassium
Sodium
Calcium
Magnesium
Aluminium
Zinc
Iron
Tin
Lead
Hydrogen
Copper
Mercury
Silver

• **Reaction of Non-Metals:**

➢ **Reaction with oxygen:** Non-metals react with oxygen to form acidic oxides. e.g.,

$C + O_2 \longrightarrow CO_2$

➢ **Reaction with Water:** Non-metals do not react with water but their oxides (like carbon di oxide or sulphur di oxide) do react with water to form acids.

$CO_2 + H_2O \rightarrow H_2CO_3$

➢ **Reaction with dil. acids:** No reaction.

➢ **Reaction with Salt Solutions:** A more reactive non-metals will displace less reactive non-metal from its salt solution.

➢ **Reaction with Chlorine:** Non-metals react with chlorine to form their respective chlorides. e.g.,

$H_2 + Cl_2 \longrightarrow 2HCl$

➢ **Reaction with Hydrogen:** Non-metals react with hydrogen to form their respective hydrides. e.g.,

$H_2 + S \longrightarrow H_2S$

• **Aqua Regia** It is a strong oxidising agent due to the formation of NOCl (Nitrosyl chloride) and chlorine produced by reaction of two acids.

$HNO_3 + 3HCl \rightarrow NOCl + Cl_2 + 2H_2O$

It can dissolve gold and platinum.

Aqua regia: Mixture of conc.HCl and HNO_3 in the ratio 3: 1.

• **Reaction Between Metal and Non-metals:**

➢ Reactivity of an element is the tendency to attain completely filled valence shells.

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➢ Atoms of metals can lose electrons from valence shells to form cations while atoms of non-metals can gain electrons in valence shells to form anions.
➢ Opposite-charged ions attract each other and held by strong electrostatic forces of attraction.
➢ **Let us understand formation of NaCl with the help of Lewis structures:**

Example 1

Step 1: Atomic number of sodium (Na) is 11. Electron in the last shell is 1. (2, 8, 1)
Step 2: Atomic number of chlorine (Cl) is 17. Electrons in the last shell are 7. (2, 8, 7)
Step 3: So, Na gives 1 electron to chlorine atom.
Step 4: This involves the complete transfer of electrons. So, it is an ionic compound.

$$Na \rightarrow Na^+ + e^-$$
2,8,1 2, 8
(Sodium cation)

$$Cl + e^- \rightarrow Cl^-$$
2,8,7 2,8,8
(Chloride anion)

Example 2

Another example of a reaction between a metal and a non-metal is the combination of magnesium (Mg), a metal, with oxygen (O_2), a non-metal, to form magnesium oxide (MgO). The balanced chemical equation for this reaction is:

$$2Mg + O_2 \rightarrow 2MgO$$

In this reaction:
1. Magnesium (Mg), a metal, reacts with oxygen (O_2), a non-metal.
2. Each magnesium atom loses two electrons to achieve a stable electronic configuration, forming Mg^{2+} ions.
3. Oxygen gains two electrons to achieve a stable electronic configuration, forming O^{2-} ions.
4. The oppositely charged ions combine in a 1:1 ratio to form the ionic compound magnesium oxide (MgO).
5. This reaction is an example of the formation of an ionic compound through the combination of a metal and a non-metal, resulting in the transfer of electrons from the metal to the non-metal.

2 Ionic Compounds, Metallurgy and Corrosion

Concept covered: • Ionic compounds • Occurrence of Metals • Extraction of Metals • Corrosion • Alloy.

• Ionic Compounds

The compounds formed by the transfer of electrons from a metal to a non-metal are called ionic compounds or electrovalent compounds. [Board, 2019]

Ionic compounds and their properties
Scan Me!

• Properties of Ionic Compounds

(i) Physical nature: They are solid and hard, generally brittle.
(ii) Melting and Boiling Point: They have high melting and boiling points.
(iii) Solubility: Generally soluble in water and insoluble in organic solvents such as kerosene, petrol, etc.
(iv) Conduction of electricity: Ionic compounds conduct electricity in molten and solution form but not in solid state. [Board, 2024]

• Occurrence of Metals

➢ **Minerals:** The elements or compounds which occur naturally in the earth's crust are called minerals.
➢ **Ores:** Minerals that contain very high percentage of particular metal and the metal can be profitably extracted from it, such minerals are called ores. All ores are minerals but every mineral is not an ore

• Metals on the Basis of Reactivity Can Be Grouped Into Three Categories:

➢ Metals at the bottom of the activity series are least reactive and are often found in free state. e.g., *gold, silver, platinum and copper.*

These metals are very unreactive. The oxides of these metals can be reduced to metals by heating alone. For example, cinnabar (HgS) (an ore of mercury). When it is heated in air, it is first converted into mercuric oxide which is further reduced to mercury on heating.

$$2HgS(s) + 3O_2(g) \xrightarrow{\Delta} 2HgO(s) + 2SO_2(g)$$
$$2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)$$

➢ Metals at the top of the activity series are so reactive that they are not found in nature as free state. e.g., K, Na, Ca, Mg and Al.
➢ Metals in the middle of the activity series are moderately reactive. They are found in the earth's crust as oxides, sulphides and carbonates. e.g., Zn, Fe, Pb, etc. [Board, 2024]

The highly reactive metals are used as reducing agents because they can displace metals of lower reactivity from their compounds.
For example:
$3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + heat$
Thermite reaction:
$Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + heat$
The amount of heat evolved is so large that the metals are produced in the molten state. This reaction is used to join railway tracks or cracked machine parts.

• Extraction of Metals:

It is the process of obtaining pure metal from its ore. Extraction of metals can be classified into three steps: [Board, 2019]
➢ Enrichment of ores or concentration of ores.
➢ Extraction of metal from the concentrated ores.
➢ Refining of metal.

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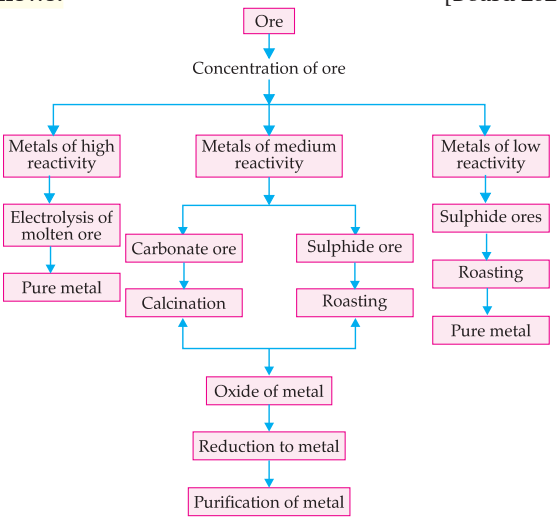
• **Steps Involved in extraction of metals from ores are as follows:** [Board 2024]



• **Metallurgy:** Extracting metals from their ores and then refining them for use is known as metallurgy.

• **Corrosion:** It is the deterioration of a metal as a result of chemical reactions between it and surrounding environment. For example, [Board, 2024]

➢ Silver reacts with sulphur in air to form silver sulphide and articles become black.

➢ Copper reacts with moist carbon dioxide in air and forms green coat of copper carbonate.

➢ Iron acquires a coating of brown flaky substance called rust.

➢ Rust is hydrated iron (III) oxide, i.e., $Fe_2O_3.xH_2O$

• **Prevention of Corrosion:** By painting, oiling, greasing, galvanising and by making alloys.

• **Galvanisation:** It is the process which involves coating of iron with zinc. The oxide thus formed is impervious to air and moisture thus protects further layers from getting corroded.

Galvanisation: Coating of metal with zinc.

• **Alloys:** These are homogeneous mixture of metals with metals and non-metals. For example,

➢ **Stainless steel:** Alloy of iron-carbon, nickel and chromium.

➢ **Brass:** Alloy of copper and zinc.

➢ **Bronze:** Alloy of copper and tin. [Board, 2024]

➢ **Solder:** Alloy of lead and tin.

• **Amalgam:** If one of the metals is mercury, then the alloy is known as amalgam, e.g., sodium amalgam and silver amalgam.

Chapter-4
Metals and Non-Metals

1 Carbon and Its Properties, Homologous Series

Concept covered: • Covalent bonding in carbon compounds • Versatile nature of carbon • Homologous series • Nomenclature of carbon compounds containing functional groups • Difference between saturated and unsaturated hydrocarbons • Chemical properties of carbon compounds.

• **Properties of Carbon**

➢ The element carbon is a non-metal. Its symbol is C.

➢ Carbon is a versatile element. The percentage of carbon in the earth's crust in the form of mineral is 0.02% and in the atmosphere as CO_2 is 0.03%.

➢ All the living things like plants and animals are made up of carbon-based compounds.

➢ Carbon always forms covalent bonds.

➢ The atomic number of carbon is 6.

➢ **Electronic Configuration:**

	K	L
C (6)	2	4

Carbon Compounds Scan Me!

• How carbon attains noble gas configuration?

(i) Carbon is tetravalent in nature. It does not form ionic bond because it has 4 valence electrons, half of an octet. To form ionic bonds, carbon molecules must either gain or lose 4 electrons.

Gain of 4 electrons to form C^{4-} anion is not favourable because it would be difficult for the nucleus with six protons to hold on to ten electrons, that is, four extra electrons.

Loss of 4 electrons to form C^{4+} cation is not favourable as it would require a large amount of energy to remove four electrons, leaving behind a carbon cation with six protons in its nucleus holding on to just two electrons.

(ii) Carbon forms covalent compounds (by sharing electrons).

The atoms of other elements like hydrogen, oxygen, nitrogen and chlorine also forms bonds by sharing electrons.

(iii) The bond formed by sharing of electrons between same or different atoms is covalent bond. In a covalent bond, the shared pair of electrons belongs to the valence shell of both the atoms.

What are Covalent Bonds? Scan Me!

• **Conditions for the Formation of a Covalent Bond:**

(i) The combining atoms should have 4–7 electrons in their valence shell.

(ii) The combining atoms should not lose electrons easily.

(iii) The combining atoms should not gain electrons readily.

(iv) The difference in electronegativity (It is the tendency to attract shared pair electrons) of two bonded atoms should be low.

• **Properties of Covalent Compounds:**

(i) Physical state: The covalent compounds exist as gases liquids, and soft solids.

(ii) Solubility: They are generally insoluble in water and other polar solvents but soluble in organic solvents such as benzene, toluene, etc.

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(iii) Melting and boiling points: They generally have low melting and boiling points.

(iv) Electrical conductivity: They are non-conductors of electricity in solid, molten, or aqueous state.

• **Steps for Writing the Lewis Dot Structures of a Covalent Compound:**

(i) Write the electronic configuration of all the atoms present in the molecule.

(ii) Identify how many electrons are needed by each atom to attain noble gas configuration.

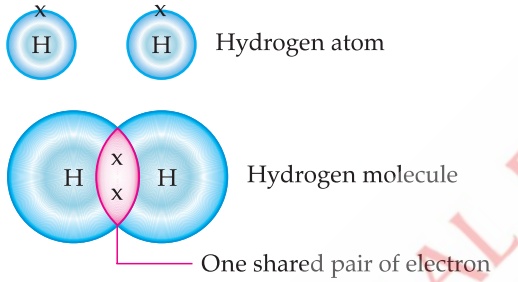
(iii) Share the electrons between atoms in such a way that all the atoms in a molecule have noble gas configuration.

(iv) Keep in mind that the shared electrons are counted in the valence shell of both the atoms sharing it.

• Lewis dot structures reflect the electronic structures of the elements, including how the electrons are paired. In Lewis dot structures each dot represents an electron. A pair of dots between chemical symbols for atoms represents a bond.

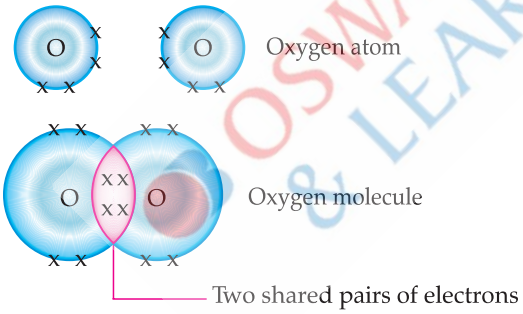
Example 1

(i) H_2



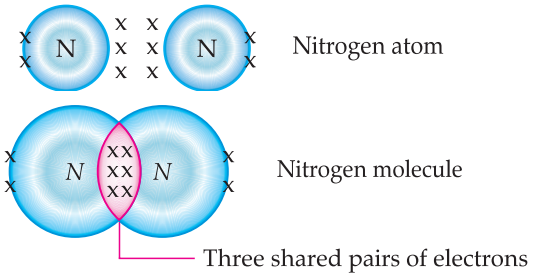
H – H: Single bond between hydrogen atoms

(ii) O_2



O = O: Double bond between oxygen atoms

(iii) N_2



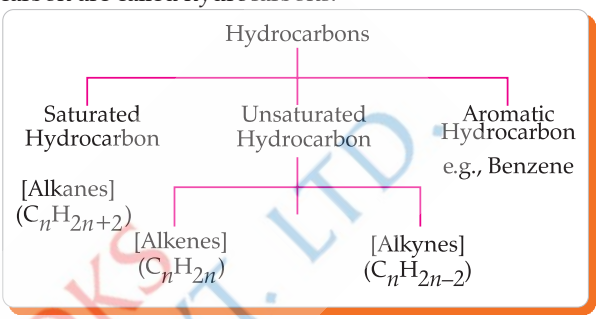
N ≡ N: Triple bond between nitrogen atoms

• **Versatile Nature of Carbon:** Carbon can form large number of carbon compounds. The factors that enable carbon to form a large number of compounds are **catenation** and **tetravalency.** [Board, 2022]

• **Catenation:** It is the unique ability of elements to form long, straight or branched chains and rings of different sizes. Carbon shows maximum catenation in the periodic table.

• **Tetravalency:** It is the state of an atom in which there are four electrons available with the atom for covalent chemical bonding.

• **Hydrocarbon:** Compounds made up of hydrogen and carbon are called hydrocarbons.



MNEMONICS

Concept: Saturated and unsaturated compounds

Mnemonics: Thank You DeSa.

Interpretations:

T: Triple bond, **Y:** Alkyne, **D:** Double bond, **e:** Alkene

S: Single bond, **A:** Alkane

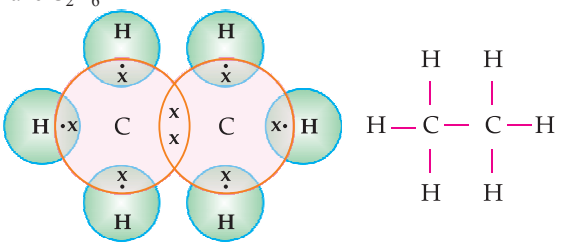
Tips&Trick T&T

single, double, triple bond, as alphabetically.

a- single bond e- double bond y- triple bond

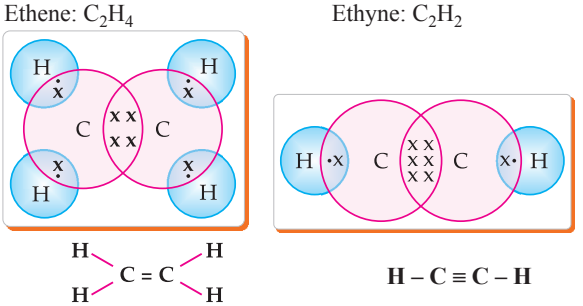
• **Electron Dot Structure of Saturated Hydrocarbons:**

Ethane C_2H_6



• **Electron Dot Structure of Unsaturated Hydrocarbons:**

Ethene: C_2H_4 Ethyne: C_2H_2



H – C ≡ C – H

• **On the Basis of Structures, Hydrocarbons Can Be:**

(i) Straight chain hydrocarbons: Propane, butane, etc.

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(ii) Branched chain hydrocarbon: Iso-butane, iso-pentane, etc.

(iii) Cyclic hydrocarbons: Cyclohexane C_6H_{12}, benzene C_6H_6, etc.

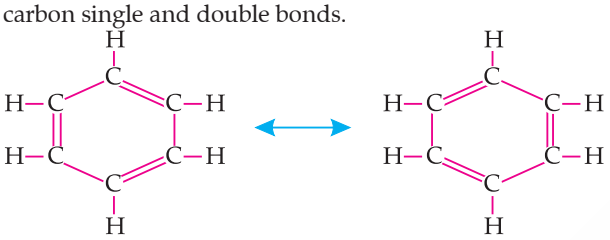
• **Cyclic or Closed Chain Hydrocarbons:** These are the hydrocarbons which have carbon–carbon closed chain.

They are classified as:

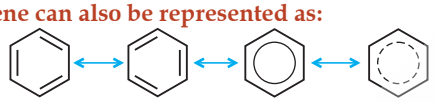
(i) Alicyclic hydrocarbons: These are the hydrocarbons which do not have benzene ring in their structures.

(ii) Aromatic hydrocarbons: The hydrocarbons which have benzene ring in their structures. When hydrogen bonded to carbon of benzene is substituted with halogens, radicals, or other functional groups, the derivatives are called aromatic compounds.

• **Benzene:** It is an aromatic hydrocarbon which has the molecular formula C_6H_6. It has alternating carbon–carbon single and double bonds.



• **Benzene can also be represented as:**



• **IUPAC name of hydrocarbon consists of two parts. It involves:**

(i) Word root: Number of carbons in the longest carbon chain.

Number of carbon atoms	Word root (Greek name)
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

(ii) Suffix: It depends on the type of carbon-carbon bond. For a single bond suffix is – ane; for double bond, suffix is – *ene*; and for triple bond suffix is – *yne*.

• **Types of Formulae for Writing Hydrocarbons:**

(i) Molecular formula: It involves the actual number of each type of atom present in the compound.

(ii) Structural formula: The actual arrangement of atoms is written in structural formula.

(iii) Condensed formula: It is the shortened form of the structural formula.

Heteroatom: An atom other than carbon or hydrogen atom.

• In hydrocarbon chain, one or more hydrogen atom is replaced by other atoms in accordance with their valencies. These are **heteroatoms.**

• These heteroatoms or group of atoms which make carbon compounds reactive and decides its properties are called functional groups.

• **Some Important Functional Groups in Carbon Compounds Are:**

Heteroatoms	Functional group [Board, 2024]	Formula of functional group	Prefix/suffix
Cl/Br	Halo (Chloro/Bromo)	— Cl, — Br, — I	Chloro/Bromo/Iodo
Oxygen	1. Alcohol	— OH	— ol
	2. Aldehyde	— CHO	— al
	3. Ketone	$-C(=O)-$	— one
	4. Carboxylic acid	$-C(=O)-OH$	— oic acid
Double bond	1. Alkene group	> C = C <	— ene
Triple bond	2. Alkyne group	— C ≡ C —	— yne

• Homologous Series: A series of organic compounds in which every succeeding member differs from the previous one by – CH_2 or 14 a.m.u. by mass is called homologous series. The molecular formula of all the members of a homologous series can be derived from a general formula. [Board 2024]

Homologous series

Scan Me!

• **Properties of a Homologous Series:** As the molecular mass increases in a series, physical properties of the compounds show a variation, but chemical properties which are determined by a functional group remain the same within a series.

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MNEMONICS

Concept 1: Homologous series

Mnemonics: **M**onkeys **E**at **P**eeled **B**ananas

Interpretations:
M: Methane (1C), **E: E**thane (2C),
P: Propane (3C), **B: B**utane (4C)

Concept 2: Reactions in saturated and unsaturated compounds

Mnemonics: **SaSUnA**

Interpretations: **Sa: Sa**turated, **S: S**ubstitution,
Un: Unsaturated, **A: A**ddition

- **Homologous Series of Alkanes:** General formula: C_nH_{2n+2}, where n = number of carbon atoms. e.g., CH_4, C_2H_6, C_3H_8.
- **Homologous Series of Alkenes:** General formula: C_nH_{2n}, where n = number of carbon atoms. e.g., C_2H_4, C_3H_6, C_4H_8.
- **Homologous Series of Alkynes:** General formula: C_nH_{2n-2}, where n = number of carbon atoms. e.g., C_2H_2, C_3H_4, C_4H_6.
- **Chemical Properties of Carbon Compounds**

(a) Combustion: Carbon compounds burn in air to give carbon dioxide, water, heat and light.

$$CH_4 + 2O_2 \xrightarrow{\text{Combustion}} CO_2 + 2H_2O + \text{Heat} + \text{Light}$$

➢ Carbon and its compounds are used as fuels because they burn in air releasing a lot of heat energy.

➢ Saturated hydrocarbons generally burn in air with blue and non-sooty flame.

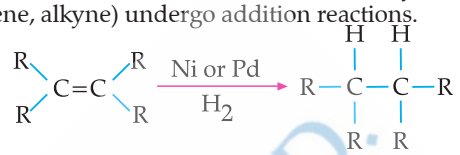
➢ Unsaturated hydrocarbons burn in air with yellow sooty flame because the percentage of carbon is higher than saturated hydrocarbons which do not completely oxidise in air.

(b) Oxidation: Alcohols can be converted into carboxylic acid in the presence of oxidising agent like alkaline $KMnO_4$ (potassium permanganate) or acidic potassium dichromate $K_2Cr_2O_7$.

$$CH_3CH_2OH \xrightarrow[\text{Acidic } K_2Cr_2O_7]{\text{Alkaline } KMnO_4 \text{ Or}} CH_3COOH$$

Ethanol — Ethanoic acid

(c) Addition Reaction: Unsaturated hydrocarbons (alkene, alkyne) undergo addition reactions.



In unsaturated hydrocarbons, hydrogen is added in the presence of the catalyst: palladium or nickel. Vegetable oils are converted into vegetable ghee using this process. It is also called hydrogenation of vegetable oils.

(d) Substitution Reaction: Saturated hydrocarbons undergo substitution reaction in the presence of sunlight. [Board, 2024]

$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

2 Ethanol, Ethanoic Acid, Soaps and Detergents

Concept covered: • Properties and Uses of Ethanol and Ethanoic Acid • Soaps and Detergents.

- **Ethanol:** Ethanol is commonly known as alcohol. It is the second member of the alcohol series. The molecular formula of ethanol is C_2H_5OH.
- **Chemical Properties:** [Board, 2022]

(i) Reaction with sodium: Formation of sodium ethoxide and hydrogen. [Board, 2024]

$$2CH_3CH_2OH + 2Na \rightarrow 2CH_3CH_2ONa + H_2$$

(ii) Reaction with acid: Formation of ester (ethyl ethanoate) – a sweet-smelling ester. This process is called esterification. [Board, 2024]

$$CH_3COOH + C_2H_5OH \xrightarrow{\text{conc. } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

➢ **Uses:** In preparation of soap, cosmetics, alcoholic beverages, medicines, laboratory reagent, etc.

- **Ethanoic acid:** The common name of ethanoic acid is acetic acid and it belongs to the group of acids called carboxylic acid. It is the second member of the series. The molecular formula of the compound is CH_3COOH.

➢ Vinegar — 5-8 % solution of acetic acid in water.

➢ Glacial acetic acid — Pure acetic acid

➢ **Chemical Properties of Ethanoic Acid**

(i) Reaction with sodium carbonate: [Board, 2024]

$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$$

(ii) Reaction with sodium hydrogen carbonate:

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$$

Brisk effervescence marks the presence of carbon dioxide.

(iii) Reaction with NaOH:

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

(iv) Reaction with ethanol (Esterification): [Board, 2024] [Board 2023]

$$CH_3COOH + CH_3CH_2OH \xrightarrow{\text{conc. } H_2SO_4} CH_3COOC_2H_5 + H_2O$$

Note:
Conc. H_2SO_4 is used here as dehydrating agent. This reaction is reversible. The conc. H_2SO_4 absorbs formed water and stops the reversibility of the reaction.

Ester: Ester is a sweet-smelling substance and is produced as a result of the reaction of an acid such as ethanoic acid and an alcohol such as ethanol in the presence of an acid catalyst. It is used in making perfumes and flavouring agents.

- **Soap and Detergents**

➢ Soap is sodium or potassium salt of long-chain carboxylic acid. e.g., $C_{17}H_{35}COONa^+$

➢ On hydrolysis, **ester** gives parent alcohol and sodium salt of carboxylic acid. Alkaline hydrolysis of ester is called saponification. [Board 2017]

➢ Soaps are effective only in soft water.

➢ Detergents are ammonium or sulphonate salts of long chain of carboxylic acid.

➢ Detergents are effective in both hard and soft water.

Soaps and Detergents
Scan Me!

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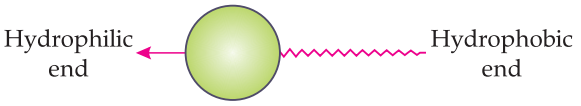
➢ Soap molecule has:

(i) An ionic (hydrophilic) part

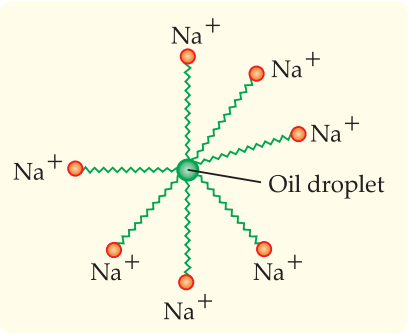
(ii) A long hydrocarbon chain (hydrophobic) part



Structure of the soap molecule

• Cleansing Action of Soap: Most dirt is oily in nature. The hydrophobic end of a soap molecule attaches itself with dirt and the ionic end is surrounded with molecules of water. This result in formation of a radial structure called micelles. [Board 2024]

➢ Soap micelles help to dissolve dirt and grease in water and the cloth gets cleaned. [Board 2025]



➢ The magnesium and calcium salts present in hard water reacts with soap molecules to form insoluble product called scum. This scum create difficulty in cleansing action.

➢ Detergents prevent the formation of the insoluble scum with hard water and clothes get cleaned effectively.

UNIT-II : WORLD OF LIVING

Chapter-5
Life Processes

1 Nutrition

Concept covered: • Modes of nutrition: Autotrophic and heterotrophic • Process of photosynthesis • Nutrition in human beings • Human digestive system.

Introduction

• The basic functions performed by living organisms for their survival and body maintenance are called life processes.

• All living things perform certain life processes like nutrition, excretion, respiration, circulation and reproduction, etc.

• Energy required to carry out the different life processes, is obtained from carbon-based food sources through nutrition.

• **There are two modes of nutrition:**

(i) Autotrophic nutrition: It is a kind of nutrition in which organism prepare its own food by a process called **photosynthesis**. For example, Green plants and some **photosynthetic bacteria**. [Board, 2020]

Photosynthesis: is the process by which plants synthesise food from water and carbon dioxide using chlorophyll and sunlight.

Photosynthetic bacteria: are microorganisms that use light energy and pigmentse like chlorophyll to synthesize food through photosynthesis.

(ii) Heterotrophic nutrition: It is a kind of nutrition in which an organism obtains food from another organism. For example, Animals and fungi. It is of three types: Holozoic (e.g., *Amoeba*, animals), Saprophytic (e.g., fungi) and Parasitic (e.g., *Cuscuta*, ticks and mites).

• Green plants manufacture their own food by the process of photosynthesis. They utilise CO_2 and H_2O (water) in presence of sunlight to produce glucose ($C_6H_{12}O_6$) as food, with the help of chlorophyll. Oxygen (O_2) is released as a by-product.

• **Main Events of Photosynthesis are:**

(i) Absorption of light energy by chlorophyll.

(ii) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.

(iii) Reduction of CO_2 to form carbohydrates. [SQP 2024-25]

• Site of photosynthesis in the leaf is chloroplast. It contains a green colour pigment called chlorophyll.

• Plants carry out exchange of gases with surrounding atmosphere through stomata.

• The opening and closing of stomatal pores are controlled by the turgidity of guard cells.

• When guard cells uptake water from surrounding cells, they swell to become turgid body. This enlarges the pore in between and causes stomatal opening.

• When water is released, guard cells become flaccid. This closes the pore in between, causing stomatal closing.

• *Amoeba* is a unicellular organism, which follows holozoic mode of nutrition. *Amoeba* captures food with the help of temporary finger-like processes called pseudopodia.

Alimentary canal: It is a continuous tube that runs from the mouth to the anus and is responsible for the digestion and absorption of food.

• Human Digestive System: The human digestive system comprises of alimentary canal and associated digestive glands. [Board, 2020]

Salivary gland: It secretes an enzyme called salivary amylase (or ptyalin), which breaks down starch into simple sugars.

• Various regions are specialised in alimentary canal to perform different functions.

• Associated glands include **salivary gland,** gastric gland, liver and pancreas.

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MNEMONICS

Concept: Components of the Human Alimentary Canal

Mnemonics:

MOSS DJ I LA – Remember this as "Kate **MOSS** is a **DJ** In **LA**"

Interpretations:

M = Mouth	**O** = Oesophagus
S = Stomach	**S** = Small Intestine
D = Duodenum	**J** = Jejunum
I = Ileum	**L** = Large Intestine
A = Anus	

Mouth → Intake of whole food.

↓

Teeth → Chewing/grinding of food.

↓

Rolling of food + Tasting of food + Swallowing/Secrete saliva + Mucus

Starch —(Salivary amylase [Saliva])→ Simpler Sugar

Oesophagus → Taking food from mouth to stomach via oesophagus (**peristaltic movements**)

↓

Stomach → Gastric glands —Secrete→ Gastric juice

- Pepsin (Enzyme that breaks down proteins)
- HCl (Makes medium acidic)
- Mucus (Protect inner lining of the stomach)

Small Intestine →

(a) Receives secretion from:

- Liver (Secretes Bile) → Helps in emulsification of fats
- Pancreas → Pancreatic juice
  - Trypsin: Proteins → Peptones
  - Lipase: Fats → Lipids

(b) Intestinal enzyme convert:

- Complex Carbohydrates → Glucose
- Fats → Fatty acid + Glycerol
- Proteins → Amino acids

(c) Villi (finger like projections) → Helps in absorption of digested food into the blood

↓

Large Intestine → Absorb excess of water

→ The rest of the material is removed from the body via the anus.

[Board, 2025]

Digestion in Human beings

Scan Me!

Peristaltic movement: Peristaltic movement is the involuntary, rhythmic contraction and relaxation of muscles in the digestive tract that moves food forward.

Key Facts

- Small intestine is the site of the complete digestion of carbohydrates, proteins and fats.
- Bile is secreted by liver and stored in gall bladder.
- Emulsification of fats is conversion of large fat molecules into very fine fat globules.

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2 Respiration

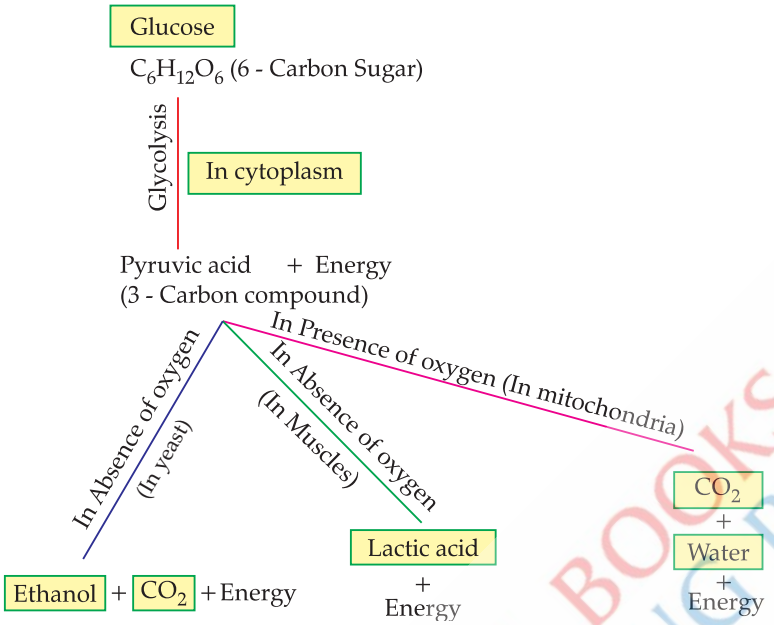
Concept covered: • Breakdown of glucose by various pathways • Types of respiration • Human respiratory system • Process of breathing • Respiration in plants • Respiration in animals.

• Respiration is the process in living organisms, which involves:

(i) Breathing (Gaseous exchange): Intake of oxygen from the atmosphere and release of CO_2.

(ii) Cellular Respiration: Breakdown of glucose in order to release energy inside the cell. **[Board, 2020]**

• **Breakdown of Glucose by Various Pathways:** **[Board, 2025]**



Key Facts

- The breakdown of sugars by yeast to make alcohol in the absence of air (oxygen) is called fermentation.
- The accumulation of lactic acid in muscles(a product of glucose breakdown in absence of oxygen) causes muscle cramps.

• **Types of Respiration:**

Respiration	
Aerobic	**Anaerobic**
Takes place in the presence of oxygen.	Takes place in the absence of oxygen.
Occurs in mitochondria.	Occurs in cytoplasm.
End products are CO_2 and H_2O.	End products are alcohol or lactic acid.
More amount of energy is released.	Less amount of energy is released.
Examples: Most plants and animals.	Examples: Muscle cells, bacteria, yeast and parasitic worms, etc.

• **Human Respiratory system:** Respiratory system in human serves to provide oxygen to all body cells and removes harmful carbon dioxide from the body.

Human Respiratory system: It consists of organs and tissues that facilitate the exchange of oxygen and carbon dioxide between the body and the environment.

• It comprises nostrils, nasal cavity, pharynx, larynx, trachea, bronchi, bronchiole, alveoli, blood capillaries and lungs.

• **Breathing involves two main processes:**

(a) Inspiration (Breathing IN): Inspiration is the active intake of air from atmosphere into lungs.

The path followed by fresh air (oxygen) is:

Outside → External nares → Nasal cavity → Internal nares → Pharynx → Glottis → Larynx → Trachea → Bronchi → Bronchioles → Alveolar duct → Alveoli

(b) Expiration (Breathing OUT): It is the passive expelling of air from the lungs.

The path followed by foul air (carbon dioxide) is:

Alveoli → Alveolar duct → Bronchioles → Bronchi → Trachea → Larynx → Glottis → Pharynx → Internal nares → Nasal cavities → External nares → Outside

• The alveoli of lungs are richly supplied with blood and are the actual sites where exchange of gases (O_2 and CO_2) occurs between blood and atmosphere.

• During breathing, a residual volume of air remains in the lungs to allow continuous gas exchange—oxygen absorption and carbon dioxide release.

• In humans, the respiratory pigment haemoglobin carries oxygen from lungs to different tissues of the body. **[Board, 2024, 225]**

Respiration in Plants

Scan Me!

• **Respiration in Plants:** In plants, gaseous exchange takes place through stomata in leaves, lenticel in stems, general surface of roots.

• **Respiration in Animals:** **[Board, 2018]**

(i) Unicellular animals: Diffusion

(ii) Earthworm: Skin.

(iii) Aquatic animals: Gills which extract dissolved oxygen in water.

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Key Facts

Trachea contains rings of cartilage which ensure that air passage does not collapse.

Respiration in plants occurs all through out the day, but the photosynthesis process occurs in the daytime, in the presence of sunlight only.

(iv) Insects: Tiny holes called spiracles.

(v) Land animals: Lungs

➢ Terrestrial organisms use atmospheric oxygen for respiration.

➢ Aquatic organisms use oxygen dissolved in water.

• The rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms. [Board, 2024]

3 Circulation and Transportation

Concepts covered: • Human circulatory system • Double circulation system in humans • Circulation of blood in animals • Transportation in Plants • Transpiration

Human Circulatory System

Human Circulatory System

Scan Me!

• The circulatory system in human beings consists of: a circulatory medium (blood and lymph), blood vessels (veins, arteries and capillaries) and heart.

• Heart is a muscular organ which is composed of cardiac muscles. It is the main pumping organ which pumps the blood to all parts of the body.

• Human heart is four chambered. It is composed of: right atrium, right ventricle, left ventricle and left atrium. The chambers are separated by a muscular wall that prevents the mixing of the blood rich in oxygen with the blood rich in carbon dioxide.

• Blood pressure is the force that the blood exerts on the blood vessels.

• Humans have double circulation system. Blood travels twice through the heart in one complete cycle of the body.

• Oxygen-rich blood from the lungs enters the left atrium, which contracts to push blood into the left ventricle. The left ventricle then pumps it to the body. Deoxygenated blood from the body enters the right atrium, which sends it to the right ventricle. The right ventricle pumps it to the lungs for oxygenation.

• **Pulmonary Circulation:** Blood moves from the heart to the lungs and back to the heart.

• **Systemic Circulation:** Blood moves from the heart to rest of the body and back to the heart.

• Blood is a fluid connective tissue. It comprises four components- plasma, RBCs, WBCs, and platelets.

• Lymph is a colourless fluid and has less protein than blood plasma. It forms when fluid escapes from the blood capillaries into the intercellular spaces. [Board, 2024, 2025]

• **Blood Vessels:** There are three types of blood vessels: arteries, veins and capillaries.

• **Differences between arteries and veins:**

	Arteries		Veins
1.	Carry oxygenated blood from heart to different body parts except pulmonary artery.	1.	Carry deoxygenated blood from different body parts to the heart except pulmonary vein.
2.	Also, called distributing vessel.	2.	Also, called collecting vessel.
3.	Walls thick, elastic and muscular.	3.	Thin, non-muscular and less elastic.
4.	Deep seated within the body.	4.	Superficial as compared to arteries.
5.	Have no valves.	5.	Have **valves**, which prevent backward flow of blood.

Key Facts

Valves ensures that blood does not flow backward when the atria and ventricles contract.

• Amphibians and many reptiles have a three-chambered heart, allowing some mixing of oxygenated and deoxygenated blood. Fish have a two-chambered heart, where blood is pumped to the gills for oxygenation and then directly circulated to the body. This is single circulation, meaning blood passes through the heart only once per cycle.

• **Transportation in plants:**

There are two main conducting channels in vascular plants. These are xylem and phloem. [Board, 2025]

	Xylem		Phloem
1.	Transports water and minerals from the roots to upper parts of the plant.	1.	Transports product of photosynthesis from leaves to the non-photosynthesising parts of the plants such as root and stem.
2.	No energy is used for transport.	2.	Energy is used from ATP for transport.
3.	On maturity, the xylem becomes dead tissue and gives mechanical support to the plant.	3.	Phloem exists as living soft tissue.

• **Transpiration:** It is the process of loss of water as vapours from aerial parts of the plant. [Board, 2019]

• **Translocation:** Transport of food from leaves (food factory) to different parts of the plant is called translocation. [Board, 2024]

Transportation in Animals and Plants

Scan Me!

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4 Excretion

Concept covered: • Excretion in human • Structure of Nephron • Urine formation • Artificial Kidney • Excretion in plant

Excretion in Human beings: [Board, 2020]

• During **excretion,** the harmful metabolic nitrogenous wastes like urea and uric acid generated are removed from the body.

• **Human Excretory System:** Excretory system of human beings includes a pair of kidney, a urinary bladder, a pair of ureter and a urethra.

• **Nephrons:** Each kidney contains many filtration units called nephrons. Nephrons are the basic filtration units of kidneys. They carry out filtration, selective reabsorption and tubular secretion to form urine in kidney, which is then passed out through the urethra, via the ureters and urinary bladder.

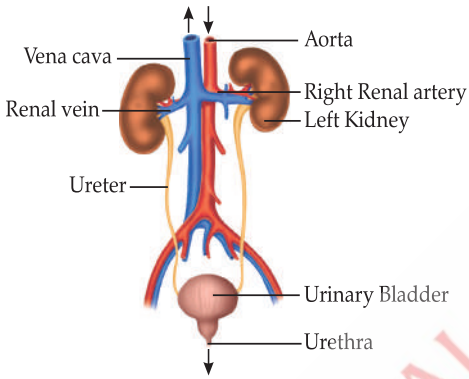

Fig 5.1 : Human Excretory system

• A nephron is made up of a cluster of thin walled capillaries called glomerulus which is associated with a cup like structure called as Bowman's capsule and a long tube. which continues through different segments and ultimately connects to the collecting duct.

Excretory System and Nephron
Scan Me!

• The renal artery brings oxygenated blood to the kidneys along with the nitrogenous wastes like urea and uric acid and many other substances. [Board 2024]

• The blood gets filtered through the glomerulus and this filtrate enters the tubular part of nephron.

• As this filtrate moves down the tubular part, glucose, amino acids, salts and excess of water gets selectively re-absorbed by the blood vessels surrounding the tubules.

• The amount of water re-absorbed depends upon:
➢ How much excess of water is there in the body and,
➢ How much nitrogenous wastes need to be excreted out.

• The fluid now flowing in the tubular part is urine, which gets collected in collecting ducts of nephrons.

• These collecting ducts together leave the kidney at a common point by forming the ureter.

• Each ureter drains the urine in the urinary bladder where it is stored until the pressure of expanded bladder leads to an urge to pass it out through urethra.

• This bladder is a muscular structure which is under nervous control.

• 180 litres of filtrate is formed daily but only 2 litres is excreted out as urine so the rest is re-absorbed in the body.

Key Facts

The **expulsion of urine** from the body is known as micturition.

• **Urine formation in Kidneys:**

Urine formation involves three steps: [Board 2024]

(i) Glomerular filtration: Nitrogenous wastes, glucose, water, amino acids filter from the blood into Bowman's capsule of the nephron.

(ii) Tubular reabsorption: Useful substances from the filtrate are re-absorbed back by capillaries surrounding the nephron.

(iii) Secretion: Urea, extra water and salts are secreted in the tubule which open up into the collecting duct and then into the ureter.

• **Haemodialysis:** In case of kidney failure, haemodialysis is the process of purifying blood by an artificial kidney.

• **Excretion in plants:** In plants, excretion of oxygen, CO_2 and water takes place through stomata by the process of transpiration.

• Plants also manage waste by storing it in vacuoles, leaves that later fall off, or in resins and gums within old xylem. Some wastes are also excreted into the surrounding soil. [Board 2025]

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Chapter-6
Control and Co-ordination in Animals and Plants

1 Control and Co-ordination in Animals

Concepts covered: • Nervous System • Voluntary and involuntary action • Reflex action • Animal hormone

• Control and coordination is brought about in all animals with the help of two main systems: Nervous system and Endocrine system.

• **Nervous System:** It is the system of conducting tissues that receives the stimulus and transmits it to other parts of the body forming a network of nerves. It is involved in receiving information (sensation) and generating responses to that information (motor response).

The units which make up the nervous system are called nerve cells or neurons.

• **A Typical Neuron Consists of Following Parts:**

(i) Cyton or Cell Body: It is star shaped which contains nucleus with abundant cytoplasm called neuroplasm. The information acquired by it travels as an electrical impulse.

(ii) Dendrite: The hair-like structure protruding out from margins of cell body is called dendrite. It receives the nerve impulses.

(iii) Axon: It is the longest fibre on the cell body. It ends in several hairs like structures called axon terminals, which transmits electrical impulse from cell body to dendrite of next neuron.

(iv) Myelin Sheath: It is a protective insulating layer made of lipids and proteins, wrapped around the axon.

(v) Synapse: It is the point of contact between the nerve ending of one neuron and dendrite of an other neuron. It is the site where an electrical signal is converted into chemical signal for onward transmission to next neuron. [Board, 2023]

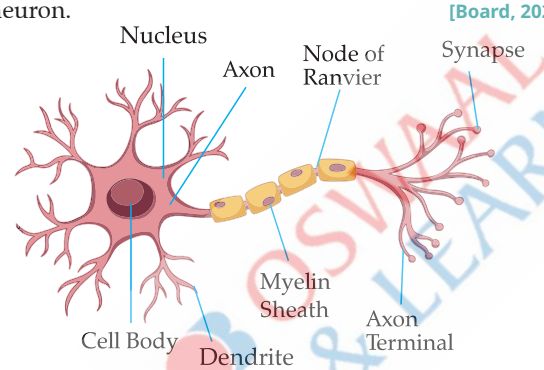

Fig.: 6.1 Structure of a Neuron

• A chemical synapse formed by the contact between a motor neuron and a muscle fibre is called a **neuromuscular junction**. [Board, 2024]

• **Functioning of Neuron (Pathway of signal transmission):** Dendrites → Cell body → Axon → Nerve endings at the tip of axon → Synapse → Dendrite of next neuron

• The receptors pass the information to the brain through a type of nerve cells called **sensory neurons**.

• Motor neurons transmits the information from the brain to the effector organs, mainly muscles and glands.

• **Nerve Impulse:** It is the transmission of information in the form of electrical signals along the neuron and chemical signals across synapses. These impulses are received by dendrites and carried toward the cell body.

• **Voluntary Action:** These are the actions which need thinking and are performed knowingly, i.e., these are controlled by conscious thought. For example, Speaking to a friend, writing a letter, etc.

• **Involuntary Action:** These are not under the control of the will of an individual and are automatic responses to a stimulus which is not under the voluntary control of the brain. It occurs without the conscious choice of an organism. For example, breathing, digestion, blinking of eyes etc.

• **Reflex Action:** It is quick, sudden and immediate response of the body to a stimulus. For example, knee jerk, withdrawal of hand on touching hot object. [Board 2024, 2025]

• **Reflex Arc:** The pathway through which nerve impulse travels to produce a reflex action is called reflex arc. It involves sensory input, signal processing and motor output, bypassing the brain for a faster response. [Board, 2023, 2024]

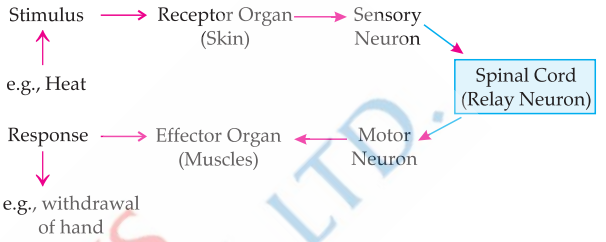

Fig.: 6.2 Reflex pathway

• **Example:** When exposed to heat (stimulus), sensory receptors in the skin detect the high temperature. A nerve impulse travels through sensory neurons to the spinal cord, triggering an immediate response. Motor neurons send signals to the muscles, causing the withdrawal of the hand, protecting the body from harm.

MNEMONICS

Concept: Reflex Arc
Mnemonics: RACEE
Interpretations:
R: Receptor
A: Afferent or Sensory Nerve
C: Centre (Spinal cord)
E: Efferent or Motor Nerve
E: Effector

• **Stimulus and Response:** Stimulus is an observable or detectable change in the external or internal environment to which an organism reacts while response is the final reaction after reflex action.

• **Responses Are of Three Main Types:**

(a) Voluntary: Controlled by forebrain. For example, talking, writing, walking, etc.

(b) Involuntary: Controlled by midbrain and hindbrain. For example, heartbeat, vomiting, respiration, etc.

(c) Reflex Action: Controlled by spinal cord. For example, Withdrawal of hand on touching a hot object.

• **Need of Reflex Actions:** In some situations such as touching a hot object, pinching, etc., we need to act quickly, otherwise our bodies would be harmed. Hence, this response is generated from spinal cord instead of brain.

• **Human Nervous System:** The nervous system of vertebrates (including humans) is divided into the central nervous system (CNS) and the peripheral nervous system (PNS). [Board, 2016, 2025]

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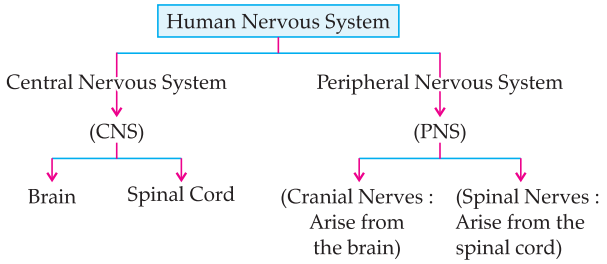

• **Human brain** is the main coordinating centre of the body. It has three major parts: Forebrain, midbrain and hind brain. [Board, 2023]

• The brain is protected by the skull called the cranium and is surrounded by three membranes called the meninges. [Board 2024]

Human Brain
Scan Me!

(a) Forebrain: It is the most complex or specialised part of the brain. It consists of **cerebrum**. The main functions of forebrain are as follows:

(i) Main thinking part of the brain.

(ii) Controls the voluntary actions.

(iii) Stores information (memory).

(iv) Receives sensory impulses from various parts of the body and integrate it.

(v) It is the centre associated with hunger. [Board, 2025]

(b) Midbrain: Controls involuntary actions.

(c) Hindbrain: It has three parts:

(i) Cerebellum: Controls posture and balance, precision of voluntary actions, e.g., picking pen. [Board, 2025]

(ii) Medulla: Controls involuntary actions, e.g., blood pressure, salivation, vomiting. [Board, 2025]

(iii) Pons: Controls voluntary actions and helps in regulation of respiration.

• **Peripheral Nervous system:** There are 31 pairs of spinal nerves and 12 pairs of cranial nerves in humans.

• Spinal cord is a cylindrical structure and a part of the central nervous system. It is made up of nervous tissue that extends from **medulla oblongata** in the brain stem to the lumbar region of vertebral column. It functions primarily in the transmission of nerve signals.

The medulla oblongata, pons varolii and midbrain are collectively known as the brainstem.

• **Endocrine System:** Endocrine system comprises endocrine glands and its secretions called hormones.

• Hormones are **chemical messengers** secreted in very small amounts by specialised tissue called ductless glands. They act on target tissues/organs usually away from their source.

• Hormones assist the nervous system in control and coordination.

• **Endocrine Glands with the Hormones Names and Their function to the target site in Humans:**

No.	Gland	Hormones	Functions	Target Site
1.	Hypothalamus	(i) Releasing hormones (RH) (ii) Growth hormone inhibiting hormones	Regulates secretion of pituitary hormones.	Pituitary gland
2.	Pituitary Gland	(i) Growth hormone (GH)	Stimulates growth; deficiency causes dwarfism, excess causes gigantism.	Various body cells which undergo growth
3.	Thyroid Gland	(i) Thyroxine (ii) Triiodothyronine (iii) Thyrocalcitonin (TCT)	Regulate basal metabolic rate, RBC formation. Regulate calcium level.	Body cells, and heart
4.	Adrenal Gland	(i) Adrenaline (ii) Noradrenaline (iii) Corticoids	Increase alertness, pupillary dilation, pilo-erection, sweating and heart rate.	Body tissues and heart
5.	Pancreas	(i) Insulin	Lowers blood glucose by promoting glycogenesis and cellular glucose uptake.	Liver, muscles and tissues
		(ii) Glucagon	Raises blood glucose by stimulating gluconeogenesis and glycogenolysis.	
6.	Testis	(i) Testosterone (ii) Androgens	Plays a role in development of male reproductive tissues and promotes secondary sexual characters in males.	Male body tissues
7.	Ovary	(i) Estrogen (ii) Progesterone	Helps in the development of female secondary sexual characters. Support pregnancy.	Female body tissues

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• **Hormonal Disorders:** All hormones are secreted in precise quantities. Its hypo-(less) or hyper-(more) secretion may lead to different disorders. For example:

(a) Dwarfism: Hyposecretion (Deficiency) of growth hormone causes a disease called dwarfism.

(b) Gigantism: Hypersecretion of growth hormone causes a disease called gigantism or excessive growth.

(c) Goitre: Deficiency of iodine causes a disease called goitre which results from decreased secretion of the thyroid hormone (thyroxine).

• **Importance of Iodine:** Thyroid gland needs iodine to make thyroxine, which helps in regulating the metabolism of carbohydrates, fats and proteins. [Board 2024]

(d) Diabetes: Deficiency of insulin hormone causes diabetes. Diabetes can be treated by injecting insulin hormone in the patient's body.

• **Feedback Mechanism:** The excess or deficiency of hormones have a harmful effect on our body. Feedback mechanism makes sure that hormones are secreted in precise quantities and at the right time.

For example, if the sugar levels in blood rise, they are detected by the cells of the pancreas which respond by producing more insulin. As the blood sugar level falls, insulin secretion is reduced.

2 Control and Co-ordination in Plants

Concept covered: • Tropic movements in plants • Plants hormones-Auxins, Gibberellins, Cytokinins, Abscisic acid and Ethylene.

Introduction

• All the living organisms respond and react to the changes that happens in the environment around them.

• The changes in the environment to which the organisms respond and react are called stimuli such as light, heat, cold, smell, touch, etc.

• Both plants and animals respond to stimuli but in a different manner.

• **Plant Movements:** Unlike animals plants do not locomote but they show the movements of the individual plant parts like shoot, root, etc., due to some external stimuli like light, force of gravity, chemical substance, water, etc.

• **Tropic Movement:** It is the movement because of the directional growth of a plant organ in response to an external stimulus. Growth towards the stimulus is positive tropism and growth away from the stimulus is negative tropism. [Board 2024]

• **Plants Show Two Different Types of Movement:** [Board, 2024]

(A) Growth-dependent Movements (Tropic movements): These are directional movements of plants caused due to the presence of a specific stimulus. It is of four types: [Board, 2023]

Coordination in Plants
Scan Me!

(i) Phototropism: Growth movements of plants towards light. For example, Shoots bend or move toward light (positively phototropic) and roots move away from light (negatively phototropic).

(ii) Geotropism: Movement towards gravity. For example, roots of a plant are positively geotropic while shoots of a plant are negatively geotropic.

(iii) Chemotropism: Movement towards chemicals. For example, Growth of pollen tube towards ovule.

(iv) Hydrotropism: Movement towards water. For example, Movement of plant roots towards water. It is positively hydrotropism. [Board 2024]

(B) Growth-Independent Movements: These are directionless movements of plants caused due to variations in the outside environment. For example, Folding or drooping of leaves of 'Touch me not plant' on touching it. This is known as seismonasty.

• **Plant Hormones:** Plant hormones or phytohormones are the chemical compounds produced naturally in plants which control the growth and other physiological functions' at a site, far away from the place of secretion. They are required in very small amount and help to coordinate growth, development and responses to the environment.

Hormones: Chemical substances that regulate activities and growth in organisms, acting in small concentrations at distant sites from their production

• **Main Plant Hormones are:** [Board, 2019]

(a)	**Auxins**	**Synthesised at shoot tip. Promotes cell elongation. Involved in tropic movements (e.g., phototropism and gravitropism)**
(b)	**Gibberellin**	**Stimulate stem elongation Play a role in breaking seed dormancy and initiating germination.**
(c)	**Cytokinins**	**Promotes cell division. Found in higher concentrations in regions of active growth, such as fruits and seeds.**
(d)	**Abscisic Acid**	**Inhibits growth. Induces leaf wilting and stomatal closure under stress. Also called as stress hormone.**
(e)	**Ethylene ($H_2C=CH_2$)**	**A gaseous hormone which helps in ripening of fruits.** Promotes senescence (ageing) and abscission of leaves.

MNEMONICS

Concept: Plant Hormone

Mnemonics: A CAGE

Interpretations:

A–Auxins, **C–C**ytokinins, **A–A**BA (Abscisic Acid)

G–Gibberellins, **E–E**thylene

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Chapter-7
Reproduction

1 Asexual Reproduction and Vegetative Propagation

Concept covered: • Reproduction and its type • Various types of asexual reproduction

Introduction

• Reproduction is the process by which living organisms produce new individuals similar to themselves. It ensures continuity of life on the earth.

• Nucleus of the cell contains DNA (Deoxyribonucleic acid), which is the hereditary material.

• DNA replicates and forms new cells. Sometimes, slight differences during the copying of the DNA, leading to variations. As a result, the new cells are similar but may not be identical to original cell. [Board, 2017]

• Variations arise mainly through sexual reproduction. They are useful for the survival of the individual and species over time. It is the base of evolution.

• **Types of Reproduction**

I. Asexual Reproduction

• A single individual give rise to new individual. [Board, 2018]

• **Gametes** are not formed.

• New individuals are identical to their parents.

Gametes are haploid reproductive cells (sperm in males, eggs in females) that combine during fertilisation to form a zygote.

• Asexual reproduction is seen in most of the lower organisms and plants as well.

II. Sexual Reproduction

• Two individuals i.e., one male and one female are needed to give rise to new individual. [Board, 2018]

• Gametes are formed.

• New individual, is genetically similar but not identical to parents.

• It is useful to generate more variations in species. [Board, 2018]

• Adopted by higher organisms.

• **Asexual Reproduction** takes place through fission, fragmentation, regeneration, budding, vegetative propagation and spore formation. These modes of reproduction depends on the body design of the organisms.

(a) Fission: It is of two types - binary fission and multiple fission.

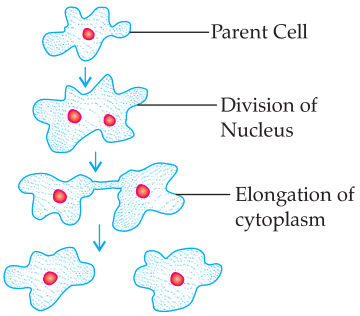

Fig.: 7.1 Binary fission in Amoeba

(i) Binary Fission: It is the division of one cell into two similar or identical cells. The nucleus first divides amitotically into two, followed by the division of the cytoplasm. The cell finally splits into two daughter cells, e.g., *Amoeba* [Board 2022, 2020]

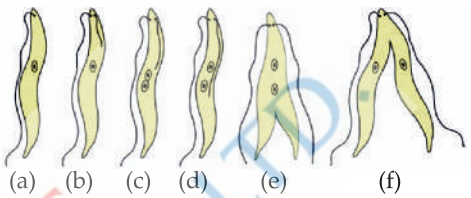

Fig.: 7.2 Binary fission in Leishmania

Some unicellular organisms with more structured bodies like in *Leishmania* (which cause kala-azar), have a whip-like structure at one end of the cell. In such organisms, reproduction through binary fission occurs in a definite orientation with respect to their structure. [Board 2022]

(ii) Multiple Fission: In multiple fission, many individuals are formed from a single individual, e.g., *Plasmodium*. The nucleus divides repeatedly, producing many nuclei and many daughter cells are formed. [Board 2025, 2024, 2020, 2017]

(b) Fragmentation: It takes place in multicellular organisms with simple body organisation such as in *Spirogyra*. In this, the body breaks up into two or more small pieces of fragments upon maturation. These fragments grow into new individuals. [Board, 2024, 2020, 2017]

(c) Regeneration: It is the ability of a fully differentiated organism to give rise to new individuals from its body section. Small cut or broken sections of the organism can grow or regenerate into complete individuals. For example: *Planaria* and *Hydra*. [Board, 2019] [Board 2024, 2017, 2016]

Spirogyra is a filamentous green alga with spiral chloroplasts, commonly found in freshwater environments.

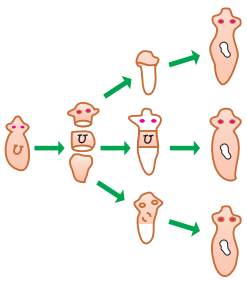

Fig.: 7.3 Regeneration in Planaria

(d) Budding: In budding, a small part of the parent's body develops into a bud, which then separates to form a new organism. *Hydra* reproduces by budding using the regenerative cells. [Board, 2023, 24]

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(e) Vegetative Propagation: In many plants, new plantlets develops from vegetative parts of a plant body such as stem, roots, leaves, etc. [Board, 2022]

➢ **Methods of Vegetative Propagation:**

(i) Natural methods:

- By roots: e.g., *Dahlia*, sweet potato.
- By stems: e.g., Potato, ginger.
- By leaves: e.g., *Bryophyllum* (leaf margins bear buds which develop into plants). [Board 2024, 2017, 2016]

(ii) Artificial Methods:

- Grafting: e.g., Mango
- Cutting: e.g., Rose
- Layering: e.g., Jasmine

MNEMONICS

Concept: Vegetative Reproduction
Mnemonics: Positive **E**xample **B**ased **L**earning
Interpretations:
P - Potato,
E - Eyes,
B - Bryophyllum,
L - Leaf buds

- **Tissue Culture:** Tissue culture is the production or propagation of new plants from isolated plant cells or small pieces of plant tissue in a nutrient medium. This technique is also known as micro-propagation, and *in vitro* culture because it takes place outside the body of the parent plant in a test tube in an artificial environment, e.g., Ornamental plants, orchids.

Advantages of Vegetative Propagation: Early flowering and fruiting, propagation of seedless plants like banana and rose, genetic similarity. [Board, 2025]

(f) Spore Formation: Spores are small bulb-like structures which are covered by thick walls. Under favourable conditions, they germinate and produce new organisms. For example, *Rhizopus*.

[Board, 2024, 2023, 2017, 2016]

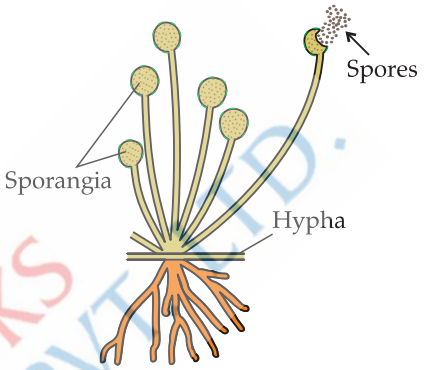

Fig.: 7.5 Spore formation in Rhizopus

2 Sexual Reproduction in Flowering Plants

Concept covered: • Parts of Flower • Pollination and Fertilisation

- **Parts of Flower**

➢ Flowers are main reproductive part of a plant. The main parts of a flower are: sepals, petals, stamens and carpels.

➢ Stamens and carpels are the reproductive parts of a flower which contains the germ cells. The male organ of a flower called '**stamen**' produces the male gamete which are present in the pollen grain. The female part of a flower called 'red' produces the female gamete, which are present in ovules of the plant. [Board, 2019, 2022]

Stamen: Pollen producing reproductive organ of a flower. red: Female reproductive organ of a flower

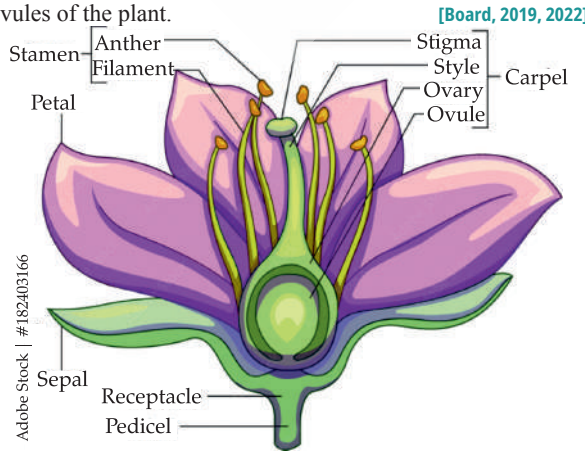

Fig.: 7.6 Parts of flower

MNEMONICS

Concept: Reproductive parts of a flower
Mnemonics:
a CAR belongs to **W**omen but **ST**eering belongs to **Men**
Interpretations:
CARpel : Female organ of the plant.
STaMEN : Male part of the plant

➢ Flowers may be unisexual contains either stamens or pistil/ carpel (e.g., papaya and watermelon) or bisexual contains both stamens and pistil/ carpel (e.g., *Hibiscus* and mustard). [Board, 2017, 2025]

- **Pollination:** It is the transfer of pollen grains from the anther of a stamen to the stigma of a carpel. Pollination is of two types: Self-pollination and cross-pollination. [Board, 2016, 2019]

Pollination
Scan Me!

➢ Self-pollination is the transfer of pollen in the same flower. In cross-pollination, pollen is transferred from one flower to another. [Board, 2017]

➢ Cross-pollination introduces variations in plants because of the mixing of different genes. These variations further increase the adaptability of plants towards the environment or surroundings. [Board, 2017]

➢ The transfer of pollens take place by agents like wind, water or animals.

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➢ After pollination, a pollen tube grows out of pollen grains, through which male germ cell reaches the ovary and fuses with the female germ cell.

➢ **Fertilisation** is the process of fusion of male and female gamete to produce zygote. It occurs inside the ovary.

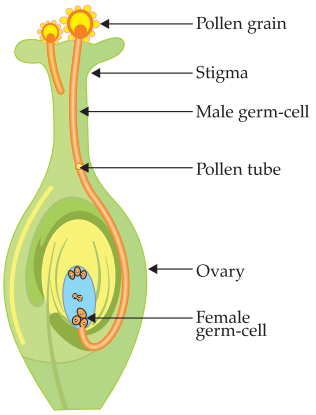


Fig.; 7.7 Germination of pollen on stigma

[Board, 2019, 2024, 2025]

• **Post-fertilisation changes:** [Board, 2017, 2025]

➢ The outer layers of the ovule become impervious and hard and function as a seed coat.

➢ An ovule with an embryo inside is called a seed.

➢ The ovary enlarges and ripens to become a fruit. [Board, 2019]

➢ Other floral parts such as sepals, petals, stamens, styles and stigma may fall off. However, in some cases, they persist in the fruit.

➢ Seed is the final (last) product of sexual reproduction in angiosperms. It is the fertilised ovule that is developed inside a fruit.

➢ Seed protects the future plant, i.e., embryo.

➢ A seed consists of seed coat(s), cotyledon(s) and an embryonal axis.

➢ **Embryonal axis has two parts:** Plumule and radical. Plumule develops into shoot and radicle develops into root. [Board, 2016]

➢ The process of development of seedling from the embryo under suitable conditions such as air, moisture, etc., is known as germination.

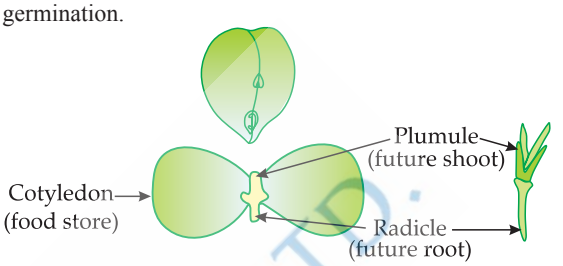


Fig 7.8 Seed germination

3 Reproduction in Human Beings

Concept covered: • Puberty • Male and female reproductive system • Process of fertilisation • Menstrual cycle • Sexually Transmitted Diseases • Methods of contraception.

• Humans have sexual mode of reproduction.

• It needs sexual maturation, which is the period of life when production of germ cells, i.e., ova (female) and sperm (male) start in the body. This period of sexual maturity is called puberty. [Board, 2022]

➢ **Changes at Puberty are:** [Board, 2022]

(a) Common in Boys and Girls:

➢ Thick hair growth in armpits and genital area.

➢ Skin becomes oily, may result in pimples.

(b) In Girls:

➢ Breast size begins to increase.

➢ Girls begin to menstruate.

(c) In Boys:

➢ Thick hair grows on face in the form of beard and moustache.

➢ Voice begins to crack.

• These changes signals that sexual maturity is taking place.

• **Male Reproductive System**

(a) Testes: A pair of testes are located inside **scrotum** which is present outside the abdominal cavity. Scrotum has a relatively 2°C lower temperature needed for the production of sperm.

Scrotum: The external sac that holds and protects the testes, regulating their temperature for sperm production.

[Board, 2025, 2024, 2023, 2017]

• **Functions of Testes:**

➢ Produce male germ cells, i.e., sperms

➢ Testes release male sex hormone (testosterone). Its function is to:

(i) Regulate production of sperms.

(ii) Bring changes at puberty. [Board, 2023]

Male Reproductive System

Scan Me!

MNEMONICS

Concept: Parts of Male Reproductive System

Mnemonics: SEVEn UP

Interpretations:

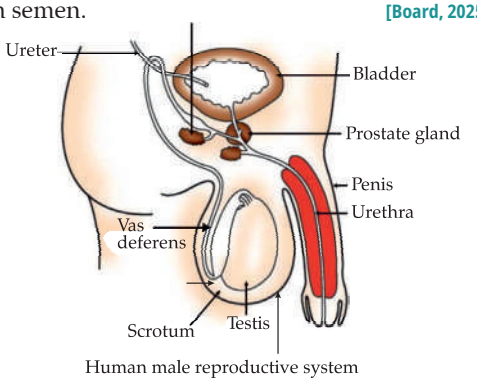
S: Seminiferous tubules, **E:** Epididymis, **V:** Vas deferens, **E:** Ejaculatory duct, **U:** Urethra, **P:** Penis

(b) Vas deferens: It passes sperms from testes towards the urethra. [Board, 2024, 2023]

(c) Urethra: It is a common passage for both sperm and urine. Its outer covering is called penis. It is like a **fibromuscular** long tube which travels through penis.

Fibromuscular: Referring to tissue that is composed of both fibrous and muscular elements, typically found in structures like the walls of certain organs or blood vessels.

(d) Associated Glands: Seminal vesicles and prostate gland are associated glands, which add their secretion to the sperms. This fluid provide nourishment to sperms and make their transport easy. Sperm along with secretion of glands form semen. [Board, 2025, 2023]



Human male reproductive system

Fig.: 7.9

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MNEMONICS

Concept: Accessory glands in Males
Mnemonics: Saint Peters
Interpretations:
S: Seminal vesicle, **P:** Prostate gland

- **Female Reproductive System**

The Female Reproductive System
Scan Me!

(a) Ovary: A pair of ovary is located in both sides of abdomen. [Board, 2017]

➢ Female germ cells, i.e., eggs are produced here. [Board, 2016]

➢ At the time of birth of a girl, thousands of immature eggs are present in the ovary.

➢ At the onset of puberty, some of these eggs start maturing.

➢ Ovary produces and secretes the female sex hormone estrogen and progesterone. [Board, 2016]

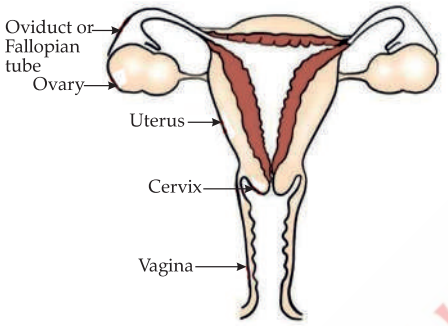

Fig.: 7.10 Human female reproductive system

(b) Oviduct or Fallopian Tube

➢ Receives the egg produced by the ovary and transfer it to the uterus.

➢ Fertilisation, i.e., fusion of gametes takes place here.

➢ The cell which is formed by the fusion of a male gamete and female gamete is called zygote, i.e., it is a 'fertilised ovum' or 'fertilised egg.' The zygote undergo multiple divisions to form an embryo.

(c) Uterus: It is a bag-like structure where development of the foetus takes place.

➢ Uterus opens into vagina through cervix.

➢ Embryo is the stage of development between the zygote or fertilised egg and the newly formed offspring. The embedding of the embryo in the thick inner lining of the uterus is called **implantation**. [Board, 2022, 2024]

• The time period from the development of foetus inside the uterus till birth is called gestation period. The act of giving birth to the fully developed foetus at the end of gestation period is termed as parturition.

• Placenta is a specialised tissue embedded in the uterine wall. It contains villi on the embryo's side and blood spaces on the mother's side. [Board, 2022, 2023]

• Placenta provides oxygen and nutrients to the developing embryo. It also helps in removal of wastes generated by developing fetus. [Board, 2024, 2023]

• Each month, one egg is released, and the uterus prepares to receive a fertilised egg by developing a thick, spongy lining.

• The breakdown and removal of the inner, thick and soft lining of the uterus along with its blood vessels in the form of vaginal bleeding is called menstrual flow or menstruation This occurs when the egg released during the menstrual cycle is not fertilised. [Board, 2025, 2024]

- **Reproductive Health:**

Reproductive health is all those aspects of general health which help a person to lead a normal, safe and satisfying reproductive life.

- **Sexually Transmitted Diseases (STDs)** are the diseases which spread by sexual contact from an infected person to a healthy person. Some common STDs are gonorrhoea, syphilis, warts, HIV-AIDS.

- **Birth Control Methods:** There are different methods which are developed to prevent and control pregnancy. These are also known as methods of contraception. Contraception is the avoidance of pregnancy, which can be achieved by preventing the fertilisation of ova. [Board, 2017, 2024]

- **Methods of Contraception**

(a) Physical Barrier Methods [Board, 2022, 2023]

➢ To prevent union of egg and sperm.

➢ Example include condoms, cervical caps and diaphragm.

(b) Chemical Methods

➢ Example include oral pills.

➢ They alter hormonal balance of body so that eggs are not released.

➢ May have side effects. [Board, 2022, 2023]

(c) Intrauterine Contraceptive Device (IUCD)

➢ They prevent pregnancy by interfering with sperm movement or egg implantation.

➢ Contraceptive devices such as the loop or the copper-T is placed in uterus to prevent pregnancy. [Board, 2022, 2023]

(d) Surgical Methods

➢ In males, the vas deferens is blocked to prevent sperm transfer and the process is called vasectomy.

➢ In females, the fallopian tube is blocked to prevent egg transfer and the process is called tubectomy. [Board, 2025]

MNEMONICS

Concept 1: Birth Control Methods
Mnemonics: SON Is Born
Interpretations: S: **S**urgical; **O:** **O**ral contraceptive;
N: **N**atural; **I:** **I**UD; **B:** **B**arrier

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Chapter-8
Heredity and Evolution

• **Variations** typically arise during sexual reproduction, and are more significant than those in asexual reproduction.

• Minor variations in sexual reproduction are caused by slight inaccuracies in DNA copying. Additionally, crossing over during meiosis contributes to variations.

Variation refers to the differences in characteristics or traits among individuals of the same species.

• Beneficial variations help the species to survive better in the environment.

• Nature selects these beneficial variations thereby leading to **evolution**.

Evolution is the gradual process through which species undergo changes over generations, resulting in the development of new species from common ancestors.

• While sexual reproduction results in offspring with a body design similar to their parents, the offspring are not identical and exhibit considerable variation.

• **Importance of Variation:**

Depending upon the nature of variations, different individuals would have different kinds of advantages.

For example, bacteria that can withstand heat will survive better in a heat wave (Archaebacteria).

Main advantage of variation to species is that it increases the chances of its survival in a changing environment.

Mendel and His Work on Inheritance

• **Gregor Johann Mendel (1822-1884)** started his experiments on plant breeding and hybridisation. He proposed the laws of inheritance in living organisms.

• Mendel is known as the Father of Genetics.

• Plant selected by Mendel was *Pisum sativum* (garden pea). Mendel selected a number of contrasting visible characters of garden peas for his experiment.

• **Seven pairs of** contrasting characters in garden pea plant selected by Mendel were:

Character	Dominant Trait	Recessive Trait
Flower colour	Violet	White
Flower position	Axial	Terminal
Seed colour	Yellow	Green
Seed shape	Round	Wrinkled
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Height of plant	Tall	Dwarf/Short

• Mendel conducted a series of experiments in which he crossed pea plants with different contrasting characteristics to study one character (at a time).

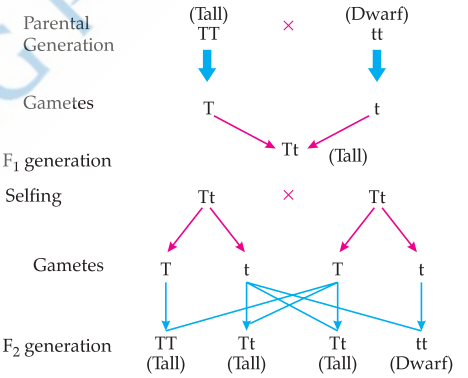
• Monohybrid cross is a cross between two pea plants with one pair of contrasting characters. For example, cross between a tall and a dwarf plant.

• In case of monohybrid cross with pure line breeding varieties of plants, the phenotypic ratio obtained in F_2 generation was 3:1. [Board, 2018, 2024]

Example 1

Cross between tall and dwarf pea plants:

Board 2025, 2024,2019



(a) Phenotypic ratio:- Tall:Dwarf = 3:1

(b) Genotypic ratio:- Pure Tall:Hybrid Tall:Pure Dwarf = 1:2:1

• **Dihybrid cross** is a cross between two plants considering two pairs of contrasting characters. For example, cross between green round seed with a yellow wrinkled seed. [Board 2017]

• In case of dihybrid cross, i.e., involving two pairs of contrasting characters, the phenotypic ratio obtained in F_2 generation was 9:3:3:1.

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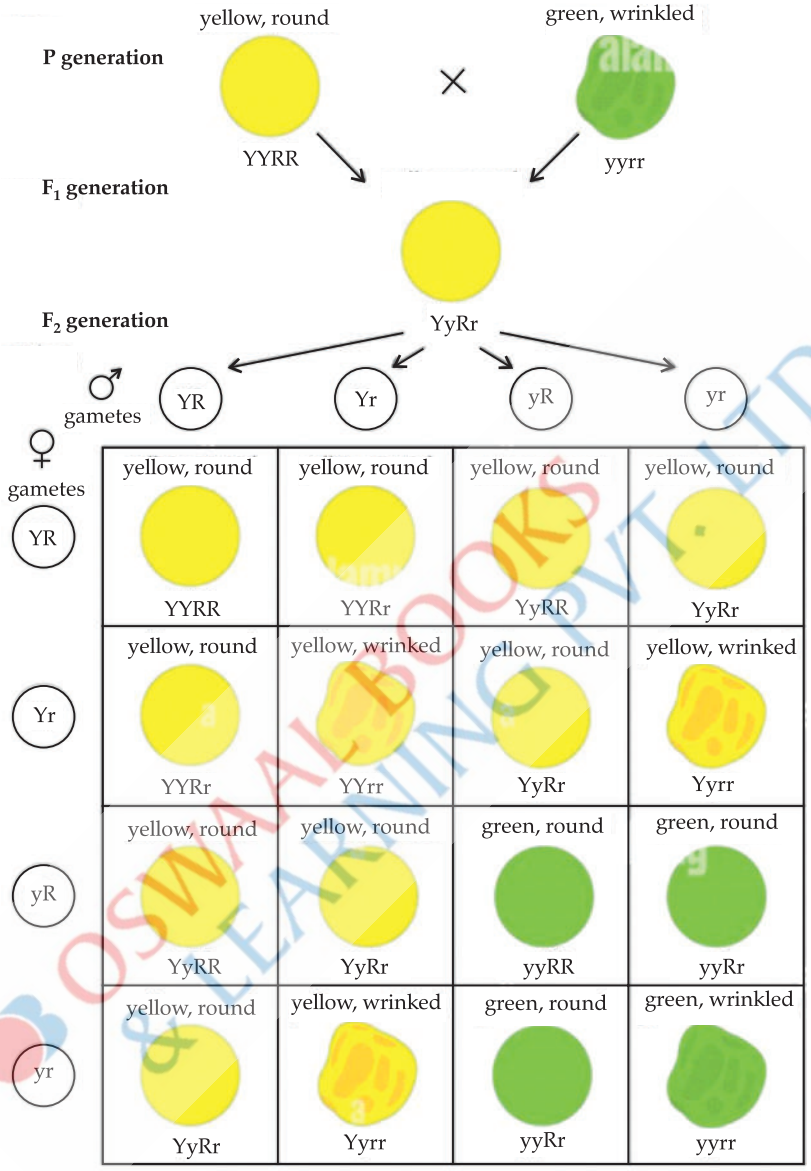
Example 2

Cross between pea plants bearing round yellow seeds with plants bearing wrinkled and green seeds: **[Board 2025, 2024, 2017]**



• Based on the above crosses, Mendel proposed three laws:

• **Law of Dominance:** It states that "when parents having pure contrasting characters are crossed then only one character expresses itself in F_1 generation. This character is the **dominant character** and the character which cannot express itself is called **recessive character**".

[Board, 2022, 2024,2020,2019]

• The homozygous dominant trait is denoted by two capital letters whereas the homozygous recessive trait is denoted by two small letters.

• Alleles are alternate forms of genes. For example, the gene for eye colour has several alleles. Two major alleles are: brown and blue.

Law of Genetics
Scan Me!

• A dominant allele expresses itself in the presence or absence of a recessive trait whereas a recessive allele is able to express itself only in the absence of a dominant trait.

• **Law of Segregation:** It states that every individual possesses a pair of alleles for a particular trait. During gamete formation, a gamete receives only one allele for each trait. A trait may appear dominant or recessive in a particular generation , depending on the combination of alleles inherited.

• **Law of Independent Assortment:** It states that alleles of different characters separate from each other during gamete formation. **[Board, 2017, 2022]**

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• Genes carry information for producing proteins, which in turn control the various body characteristics.

• For a particular trait, the offspring receives one allele from the father and one allele from the mother.

• The combination of the male and female germ cells gives a **diploid zygote**. Thus, the normal diploid number of chromosomes in the offspring is restored. **[Board, 2024]**

Diploid Zygote: A fertilised egg cell formed by the union of two haploid gametes (sperm and egg), containing two sets of chromosomes, one from each parent.

• Different mechanisms are used for sex determination in different species.

Factors Responsible for Sex Determination

- **Environmental**
  - In some animals, the temperature at which the fertilised eggs are kept, decides the gender e.g., in turtle.
- **Genetic**
  - In some animals like human, gender of individual is determined by a pair of chromosomes called sex chromosomes.
  - XX – Female
  - XY – Male

• The process by which sex of a new born individual is determined is called sex determination.

• **Sex Chromosomes:** In human beings, there are 23 pairs of chromosomes. Out of these, 22 chromosome pairs are called autosomes and the last pair of chromosome which helps in deciding sex of the individual is called **sex chromosome**. **[Board 2024]**

• Autosomes are pairs of chromosomes that are identical in appearance and are not involved in sex determination.

• Sex chromosomes are pairs of chromosomes involved in sex determination and are not identical in appearance (e.g., X and Y chromosome in humans).

Sex Determination

Scan Me!

• **Sex determination in human beings:** A male has one X and Y sex chromosomes (XY) while a female has two X sex chromosomes (XX).

• An egg fertilised by X chromosome carrying sperm result into girl child.

• An egg fertilised by Y chromosome carrying sperm results into a boy child.

Thus, sex of the child is determined by the type of sperm that fuses with egg. **[Board, 2025, 2024]**

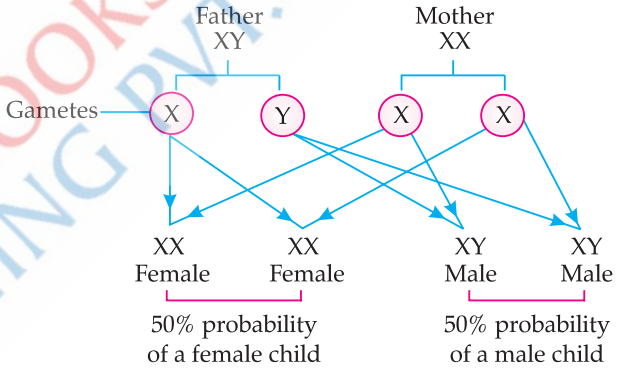

Fig 8.1 : Sex determination in human beings

UNIT-III : NATURAL PHENOMENA

Chapter-9
Light - Reflection and Refraction

1 Reflection of Light, Images Formed by Spherical Mirror

Concept covered: • Reflection of light • Laws of reflection of light • Images formed by plane mirror and spherical mirrors.

• When light falls on object, it may be absorbed, transmitted or get reflected back to the same medium.

Reflection of light: It is the phenomenon of bouncing back of the light rays in the same medium.

Reflection of Light

Scan Me!

Laws of reflection:

(i) The incident ray, the reflected ray and the normal, all lie in the same plane at the point of incidence.

(ii) The angle of incidence is equal to the angle of reflection.

Real image is obtained when the rays of light after reflection or refraction actually meet at some point. It can be obtained on the screen and can be seen with the eye.

Virtual image forms when rays of light do not actually meet, but appear to meet when produced backwards. It cannot be obtained on the screen.

Mirror: A device with only one reflecting surface.
Lens: It is a device which refracit light.

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Image formed by plane mirror: [Board, 2015]

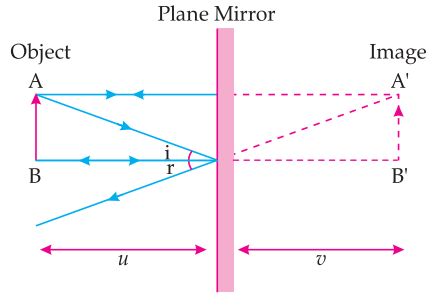

Fig 9.1

Characteristics of image formed by plane mirror:

(i) Virtual and erect.

(ii) Size of image is equal to the size of the object.

(iii) Image is formed as far as behind the mirror as the object is in front of it.

(iv) Laterally inverted.

Lateral inversion: The phenomenon due to which the right side of the object appears as left and the left side of the object appears as right, i.e., the image is inverted sideways.

• A spherical mirror whose reflecting surface is curved inwards is called a concave mirror.

• A spherical mirror whose reflecting surface is curved outwards is called a convex mirror.

• Concave mirror can form both real and virtual images. Convex mirror always form virtual images.

Differentiating between a plane mirror, a concave mirror and a convex mirror, based on image formation:

(i) If the formed image is erect, of same size and equidistant as of object always, then it is a plane mirror.

(ii) If the image is erect, virtual but smaller in size, then it is a convex mirror.

(iii) If the formed image is erect, virtual and magnified when the mirror is close to the object, then it is a concave mirror. [Board, 2024]

• The centre of the reflecting surface of a spherical mirror is called the pole of the mirror, and it is usually represented by P.

• The horizontal line passing through the centre of curvature and pole of the spherical mirror is known as principal axis.

• The centre of curvature of a spherical mirror is the centre of the hollow sphere of glass, of which the spherical mirror is a part and is usually represented by C.

• The radius of curvature of a spherical mirror is the radius of the hollow sphere of glass, of which the spherical mirror is a part and is usually represented by *R*.

• The diameter of the reflecting surface is called its aperture.

• Radius of curvature (R) = 2 × focal length (f).

• **Principal focus:** It is the point on principal axis through which the rays of light, which are parallel to it, pass after reflection or they appear to be arising from this point.

• **Rules for making ray diagrams by concave mirror:** [Board, 2024, 2019]

(i) A ray parallel to the principal axis passes through the principal focus, after reflection.

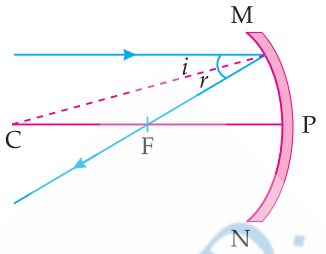

Fig 9.2

(ii) A ray passing through the principal focus of concave mirror becomes parallel to principal axis after reflection.

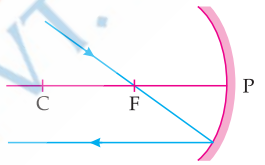

Fig 9.3

(iii) A ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path as it is a normally incident ray at the mirror surface.

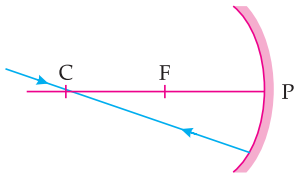

Fig 9.4

(iv) A ray incident obliquely to the principal axis of a concave mirror is reflected obliquely making equal angle with the principal axis.

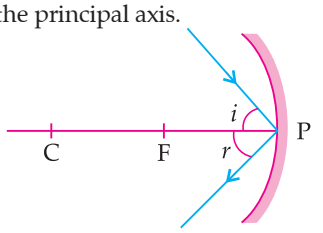

Fig 9.5

Image formation by a concave mirror for different positions of the object:

Position of Object	Position of Image	Size of Image	Nature of Image
At infinity	At the focus F	Highly diminished/ point-sized	Real and inverted
Beyond C	Between F and C	Smaller in size	Real and inverted
At C	At C	Same size	Real and inverted

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Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Highly enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect

Image formation by a convex mirror for different positions of the object: [Board, 2017]

Position of Object	Position of Image	Size of Image	Nature of Image
At infinity	At the focus F, behind the mirror	Highly diminished/point-sized	Virtual and erect
Between infinity and the pole P of the mirror	Between P and F, behind the mirror	Diminished	Virtual and erect

Mirror Formula: [Board, 2024]

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Where, v = Image distance
u = Object distance
f = Focal length

Magnification of Spherical Mirrors:

It is the ratio of the height of the image to the height of the object.

Mirror formula and magnification
Scan Me!

$$m = \frac{\text{Height of image}}{\text{Height of object}}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

If 'm' is negative, image is real.
If 'm' is positive, image is virtual.
If $h_i = h_o$ then $|m| = 1$, i.e., size of image is equal to the size of the object.
If $h_i > h_o$ then $|m| > 1$ i.e., image is enlarged.
If $h_i < h_o$ then $|m| < 1$ i.e., image is diminished.

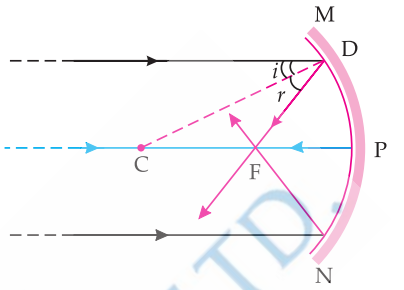
- Magnification of plane mirror is always + 1.
  '+' sign indicates virtual image.
  1' indicates that image is equal to object's size.
- If 'm' is '+ve' and less than 1, it is a convex mirror.
- If 'm' is '+ve' and more than 1, it is a concave mirror.
- If 'm' is '–ve', it is a concave mirror.

Example 1

A student holding a mirror in his hand, directed the reflecting surface of the mirror towards the sun. He then directed the reflected light on to a sheet of paper held close to the mirror.
(a) What should he do to burn the paper?
(b) Which type of mirror does he have?
(c) Will he be able to determine the approximate value of focal length of this mirror from this activity? Give reason and draw ray diagram to justify your answer in this case. AI Ap [Board, 2019]

Concept Applied Approximate determination of focal length of concave mirror from image formation of a distant object.

ANS **(a)** Move the mirror/paper to focus the rays at one point. [½]
(b) Concave mirror. [½]
(c) Yes, the distance between the mirror and the point at which the sharp focussed image of the sun is formed gives approximate focal length. [½ + ½]



[½ + ½] [Marking Scheme, 2019]

Example 2

The image of a candle flame placed at a distance of 30 cm from a mirror is formed on a screen placed in front of the mirror at a distance of 60 cm from its pole. What is the nature of the mirror? Find its focal length. If the height of the flame is 2.4 cm, find the height of its image. State whether the image formed is erect or inverted. Ap [Board, 2017]

Concept Applied Image formation by concave mirror.

ANS The nature of the mirror is concave since the image formed is real.

Given, $u = -30$ cm, $v = -60$ cm, $h_o = 2.4$ cm,
Using mirror formula,

$$\frac{1}{f} = \frac{1}{v}+\frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-60}+\frac{1}{-30}$$

$$= -\frac{3}{60} = -\frac{1}{20}$$

Therefore, $f = -20$ cm

Magnification, $m = -\frac{v}{u} = \frac{\text{Height of image}}{\text{Height of object}}$

$$-\frac{v}{u} = -\frac{-60}{-30} = \frac{h_i}{2.4}$$

$$h_i = \frac{60}{-30} \times 2.4 = -4.8 \text{ cm}$$

The required height of the image is – 4.8 cm.
The image formed by the mirror is inverted. [1 + 1 + 1]

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Commonly Made Error

Calculation error is commonly seen. In many answers, no unit is written for the final answer.

Answering Tip

While solving numerical, it is advisable that the formula need to be written in the beginning. Essential steps needs to be shown, and the final answer needs to be expressed along with a proper unit.

2 Refraction, Lenses and Power of Lens

Concept covered: • Refraction • Laws of refraction • Refractive index • Lens formula • Magnification • Power of lens

• The phenomenon of change in the path of light from one medium to another is called refraction of light.

• The angle formed between the **incident ray** and the normal is called angle of incidence, and the angle formed between the refracted ray and the normal is called angle of refraction.

An incident ray is the ray of light that strikes a surface or boundary of two media before reflection or refraction occurs.

• The cause of refraction is the change in the speed of light as it moves from one medium to another medium.

Refraction of Light

Scan Me!

• Greater the difference in speed of light between the two media across the interface, the greater is the deviation and vice-versa.

• When a ray of light passes from a rarer medium to a denser medium, it bends towards the normal. Hence, the angle of incidence is greater than the angle of refraction.

• When a ray of light passes from a denser medium to a rarer medium, it bends away from the normal. Hence, the angle of incidence is less than the angle of refraction.

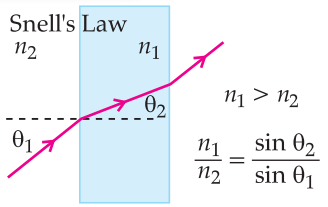
• **Laws of refraction:**

➢ **First law:** The incident ray, the refracted ray and the normal at the point of incidence all lie **in the same plane.**

➢ **Second law:** The ratio of the sine of angle of incidence to the sine of angle of refraction is constant, for the light of a given colour and the given pair of media. This constant is known as the refractive index of second medium with respect to the first. This law is also known as Snell's Law of refraction.

$$n = \frac{\sin i}{\sin r}$$

Key Diagram



• **Refractive index (n):** The ratio of speed of light in a given pair of media, [Board, 2024] [Board, 2023]

$$n = \frac{\text{Velocity of light in medium 1}}{\text{Velocity of light in medium 2}}$$

• **Absolute Refractive Index:** Refractive index of a medium with respect to vacuum or air.

$$n = \frac{c}{v} \text{ where, } c = 3 \times 10^8 \text{ ms}^{-1}$$

• Refractive index of one medium is reciprocal of other's refractive index in a given pair.

$$n_{12} = \frac{1}{n_{21}}$$

• If refractive index of medium 1 w.r.t. air is given as ${}_1n_{air}$ and if refractive index of medium 2 w.r.t. air is given as ${}_2n_{air}$, then refractive index of medium 1 w.r.t. medium 2 = $\frac{{}_1n_{air}}{{}_2n_{air}} = {}_1n_2$

• Refractive index of diamond is the highest till date. It is 2.42. It means speed of light is $\frac{1}{2.42}$ times less in a diamond than in vacuum.

Lens Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ [Board, 2024]

Magnification: $m = \frac{h_i}{h_o} = \frac{v}{u}$ [Board, 2024]

Lens Formula and Magnification

Scan Me!

• **Power of a lens:** It is defined as the reciprocal of focal length in metre. [Board, 2024] [Board, 2019]

• The degree of convergence or divergence of light rays is expressed in terms of power.

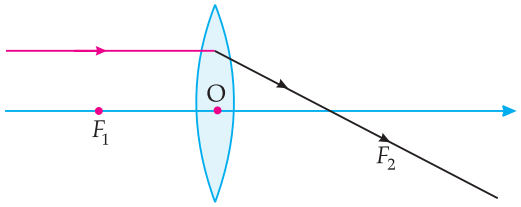
$$\text{Power} = \frac{1}{\text{Focal length (in metre)}} \text{ or, } P = \frac{1}{f}$$

• SI unit of Power is dioptre = (D), 1 D = 1 m^{-1}

• 1 dioptre is the power of lens whose focal length is one metre.

• **Rules for making ray diagrams by convex and concave lens**

(i) A ray of light from a object, parallel to the principal axis, after refraction, (i) passes through the focus for a convex lens (ii) refracted ray meets the focus when extended backward for a concave lens.



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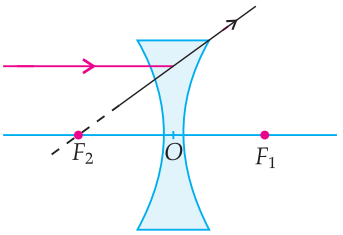
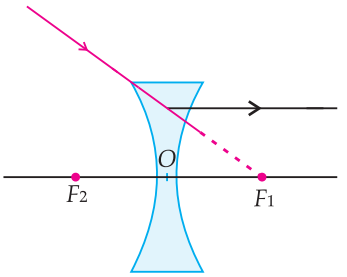
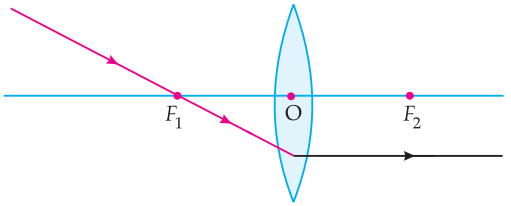
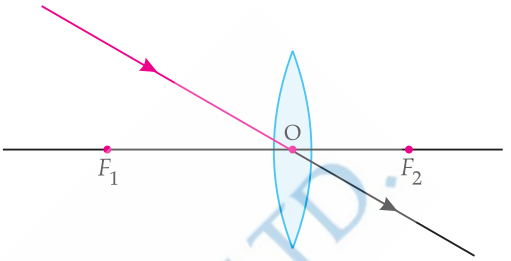


(ii) A ray of light (i) passing through a principal focus of a convex lens or (ii) directed towards the principal focus of a concave lens becomes parallel to the principal axis after refraction.



(iii) A ray of light passing through the optical centre travels undeviated even after refraction for both types of lenses.





• **Nature, position and relative size of the image formed by a convex lens for various positions of the object:**

[Board, 2024]

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At focus F_1	At infinity	Infinitely large or highly enlarged	Real and inverted
Between focus F_1 and optical centre O	On the same side of the lens as the object	Enlarged	Virtual and erect

• **Nature, position and relative size of the image formed by a concave lens for various positions of the object:**

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect

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Chapter-10
Light - Reflection and Refraction

1 Human Eye, Defects of Vision and Its Corrections

Concept covered: • Human eye • Defects of vision and its Corrections.

Human Eye

• Eye is a natural **optical device** using which humans can see objects around them. It forms an inverted, real image on a light sensitive surface called retina.

• Rods and cones are the cells in retina, which are light sensitive. Rods respond to the intensity of light. Cones respond to the illumination of colours. There are around 125 million rod and cone cells. Cells generate signals which are transmitted to the brain through the optic nerve.

An optical device is an instrument that manipulates light to enhance, redirect or analyse visual or optical signals.

Parts of the Human Eye

• **Cornea:** It is the outermost, transparent part where the

maximum refraction takes place.

• **Lens:** The lens, a fibrous, jelly-like structure, focuses light onto the retina, to produce a real, inverted image.

• **Iris:** It is a dark muscular **diaphragm** that controls the size of the pupil.

A diaphragm is a membrane that regulates the amount of light passing through an optical system.

• **Pupil:** It is the window of the eye. It is the central aperture in the iris. It regulates and controls the amount of light entering the eye. [Board 2024]

• **Aqueous humor:** Aqueous humor is a water-like fluid that lies in front of the lens.

• **Vitreous humor:** Vitreous humor is a gel-like substance that lies behind the lens and in front of the retina.

• **Retina:** It is a delicate membrane having enormous number of light sensitive cells. [Board 2024]

• **Ciliary muscles:** These muscles change the shape and size of the eye lens for focussing. [Board 2024]

• **Far point:** The maximum distance at which the object can be seen clearly is far point of the eye. For a normal adult eye, its value is infinity.

• **Optic nerve:** The optic nerve sends the electrical impulses from the retina, at the back of the eyes to the brain.

• **Blind spot:** The point where exit the eye and has no light-sensitive cells. This point is known as "blind spot."

• **Yellow spot:** The point at the middle of the retina has the maximum light-sensitive cells. This point is known as "yellow point".

Structure and Functioning of Human Eye
Scan Me!

• **Near point or least distance of distinct vision:**

• It is the minimum distance at which objects can be seen most distinctively without strain.

• For a normal adult eye, its value is 25 cm.

• Range of human vision is 25 cm to infinity.

• **Accommodation:** The ability of the eye lens to adjust its focal length is called accommodation. Focal length can be changed with the help of ciliary muscles.

[Board 2024] [Board 2019]

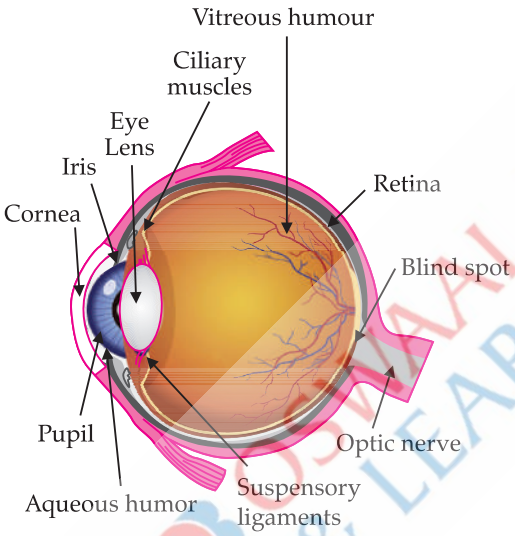


Fig 10.1 Structure of human eye

Myopia (near sightedness) [Board, 2019, 23, 24]

• Distant objects are not clearly visible.

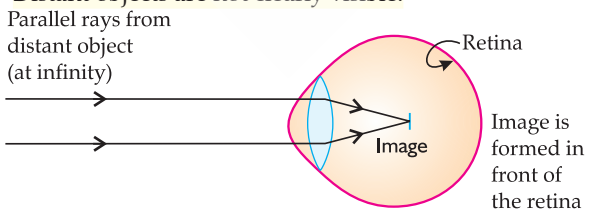
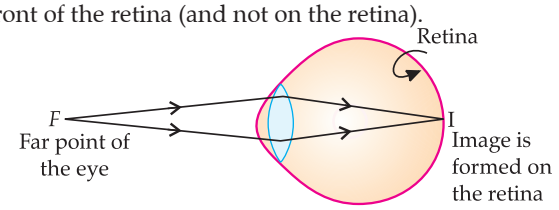
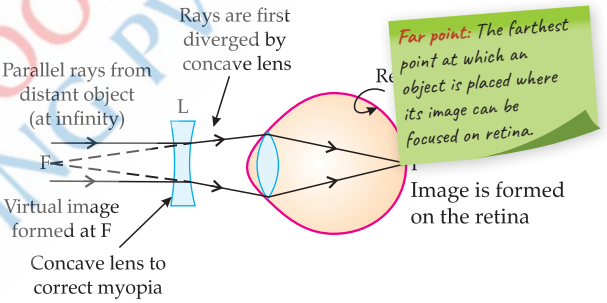


Fig 10.2 (a): Myopic eye

• In a myopic eye, image of distant object is formed in front of the retina (and not on the retina).



• The **Far point** (F) of a myopic eye is less than infinity.



Far point: The farthest point at which an object is placed where its image can be focused on retina.

Fig 10.2 (c): Correction for myopia

• **Correction:** A concave lens placed in front of the eye. It forms a virtual image of distant object at far point (F) of myopic eye. [Board 2024]

• **Cause:** Nearsightedness happens when eyeball gets elongated, or when there are problems with the shape of cornea or lens .

Hypermetropia (Farsightedness):

• The affected person can see far objects clearly but cannot see nearby objects clearly.

• The near point of the eye moves away from 25 cm. [Board 2024]

• Image is formed behind the retina.

• **Correction:** Use of convex lens of suitable power can correct this defect.

• **Cause:** Hypermetropia appears when eyeball becomes too short or when there are problems with the shape of cornea or lens. [Board 2024]

Eye Defects
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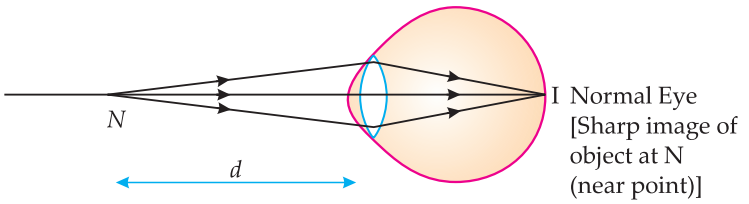

Fig 10.3 (a): Near point of a hypermetropic eye [Board 2024]

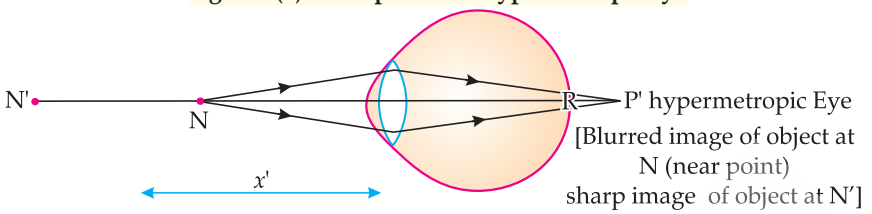

Fig 10.3 (b): Hypermetropic eye

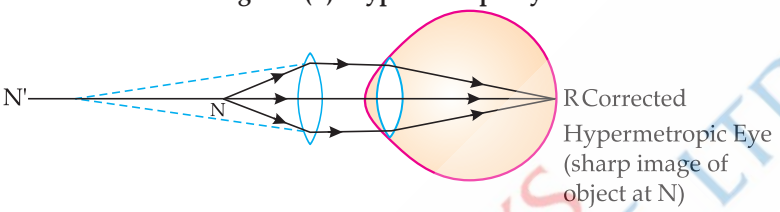

Fig 10.3 (c): Correction for hypermetropic eye

Tips & Trick T&T

While calculating the power of lens required to correct any eye vision:

(i) Use lens formula

(ii) u = 25cm for hypermetropia and distance of infinity for myopia

(iii) v = Distance of defective near point for hypermetropia and distance of defective far point for myopia

Presbyopia (Old age Hypermetropia):

[Board 2024, 2020, 2017]

- It is the defect of vision due to which an old person cannot see nearby objects clearly and even distant objects due to less power of accommodation of the eye.
- The near-point of an old person having presbyopia gradually recedes and becomes much more than 25 cm.
-

Example 1

A student needs spectacles of power –0.5 D for the correction of his vision.

(i) Name the defect in vision the student is suffering from.

(ii) Find the nature and focal length of the corrective lens.

(iii) List two causes of this defect. A [Board, 2019]

ANS **(i)** Myopia [½]

(ii) Concave / diverging lens and [½]

Focal length, $f = \frac{1}{P} - = \frac{1}{(-0.5)} = -2$ m [½]

(iii) (a) Excessive curvature of the eye lens. [½]

(b) Elongation of eye ball. [½] [Marking Scheme, 2019]

Example 2

Due to gradual weakening of ciliary muscles and diminishing flexibility of the eye lens, a certain defect of vision arises. Write the name of this defect. Name the type of lens required by such persons to improve the vision. Explain the structure and function of such lens. AU [Board 2017]

ANS Presbyopia

Bifocal lens

Upper portion/part — Concave/ Diverging lens

— To view far off objects

Lower part — Convex/converging lens

— To facilitate/ view nearby objects [3] [Marking Scheme, 2017]

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2 Refraction of Light Through Prism, Dispersion of Light and Scattering of Light

Concept covered: • Dispersion of light • Atmospheric refraction • Refraction of light using prism • Scattering of light.

• **Refraction:** A triangular glass prism has two triangular bases and three rectangular lateral surfaces. These surfaces are inclined to each other. The angle between its two lateral faces is called the angle of the prism.

• When a polychromatic light enters a prism, different constituent colours, having different speeds, suffer deviations of different amounts and thus get separated from each other.

Dispersion of Light
Scan Me!

• The phenomenon of splitting of white light into its constituent colours on passing through a glass prism is called dispersion of light. [Board, 2023]

• **Spectrum:** Band of coloured components of a light beam is known as spectrum.

• Different colours undergo different deviations on passing through a **prism.**

Prism: It is a five faced solid with a triangular cross-section.

• If a second identical prism is placed in an inverted position with respect to the first prism, all the colours recombine to form polychromatic light.

Example:

Rainbow formation
Scan Me!

• A rainbow is a natural spectrum that appears in the sky after a rain shower. It is caused by dispersion of sunlight by tiny water droplets present in the atmosphere. A rainbow always forms in the direction opposite to that of the Sun. [Board 2024]

• **Atmospheric refraction** is the phenomenon of bending of light on passing through the Earth's atmosphere. [Board 2024]

• As we move above the surface of the Earth, density of air goes on decreasing.

Scattering of Light
Scan Me!

• Light travelling from rarer to denser layers always bends towards the normal.

• Stars twinkle on account of atmospheric refraction. [Board 2024]

• The sun appears to rise two minutes earlier and set two minutes later due to atmospheric refraction.

• The phenomenon in which a part of the light incident on a particle is absorbed and then redirected in different directions is called scattering of light. [Board, 2023]

• Very small particles scatter light of shorter wavelengths better than longer wavelengths.

• The scattering of longer wavelengths of light increases as the size of the particle increases.

• Larger particles scatter light of all wavelengths equally well.

MNEMONICS

Concept: Colours of spectrum

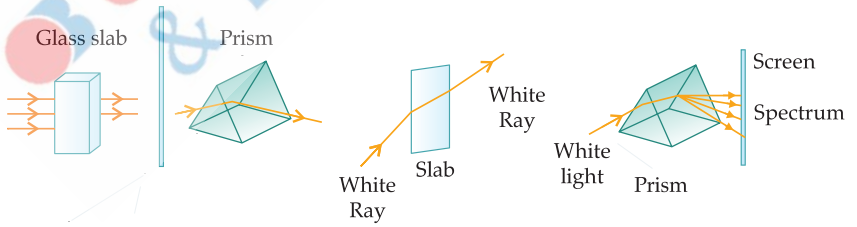
Mnemonics: VIBGYOR

Interpretations:

V: Violet **I:** Indigo **B:** Blue **G:** Green **Y:** Yellow **O:** Orange **R:** Red

Key Diagram

Difference in refraction of light by a glass slab and a glass prism:



Example 2

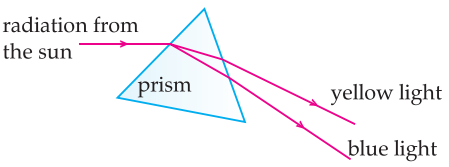
State the phenomena observed in the given diagram. Explain with reference to the diagram, which of the two lights mentioned below will have a higher wavelength? A [SQP, 2022-23]

Concept Applied Dispersion of Light

ANS Dispersion: The splitting of white light into seven colours on passing through a prism. [1]

During dispersion, as the wavelength increases, bending of emergent ray decreases. Yellow has bent less than blue. So, yellow has greater wavelength than blue. [½ + ½]

[3] [Marking Scheme SQP, 2022-23]



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UNIT-IV : EFFECTS OF CURRENT

Chapter-11
Electricity

1 Electric Current, Ohm's Law

Concept covered: • Electric current • Potential difference and electric current • Ohm's law • Resistance • Resistivity • Factors on which the resistance of a conductor depends.

• **Charge** is a fundamental quantity of matter. It may be positive and negative.
• S.I. unit of charge is Coulomb (C).
• **Static and current electricity:** Static electricity deals with the **electric charges** at rest, while the current electricity deals with the electric charges in motion.
• **Electric current:** The electric current is defined as the rate of flow of electric charge through any cross-section of a conductor.

$$\text{Electric current} = \frac{\text{Charge}}{\text{Time}} \quad \text{or} \quad I = \frac{Q}{t}$$

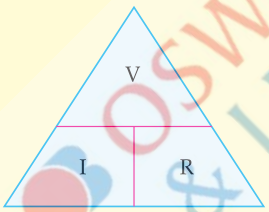
Electric Current: The rate of flow of electric charges, traditionally considered as the flow of positive charges, with current direction opposite to the flow of electrons.

Introduction to Electricity
Scan Me!

• Electric current is a scalar quantity.

MNEMONICS

Concept 1: Current Formula
Mnemonics:



Interpretations:
I = Q/t, Q = I x t, t = Q/I

• **Ampere:** It is the SI unit of current. If one Coulomb of charge flows through any cross-section of a conductor in one second, then current through it is said to be one ampere.
• **Electric circuit:** The closed path along which an electric current flows is called an 'electric circuit'.

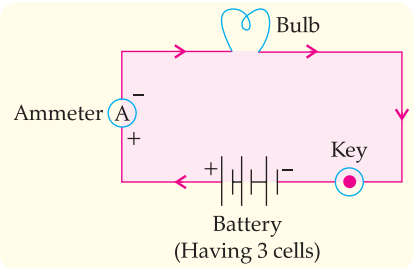
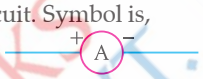


Fig 11.1: A typical electric circuit

• **Conventional direction of current:** Conventionally, the direction of the motion of positive charges through the conductor is taken as the direction of current. The direction of conventional current is opposite to that of the negatively charged electrons.
• **Galvanometer:** It is a device to detect current in an electric circuit.
• **Ammeter:** It is a device to measure current in a circuit. It is a low resistance galvanometer and is always connected in series in a circuit. Symbol is,



Ammeter: A galvanometer with low internal resistance, achieved by placing a low resistor (shunt) in parallel.

• **Electric potential** is the amount of electric potential energy at a point.
• **Potential difference** is the difference in the amount of electric potential energy between two points in an electric circuit.
• It is equal to the work done to move a unit charge from one point to another.

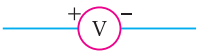
$$\text{Voltage (V)} = \frac{\text{Work done (W)}}{\text{Charge (Q)}}$$

• S.I. unit of potential difference is Volt (V).
• **1 Volt:** When 1 Joule of work is done in carrying one Coulomb charge, then potential difference is called 1 Volt. Mathematically,

1 volt = 1 joule/1 coulomb

$$\mathbf{1\ V = 1\ JC^{-1}}$$

• **Voltmeter:** It is a device to measure the potential difference. It is a high resistance galvanometer and is always connected in parallel to the component across which the potential difference is to be measured. Symbol is, **[Board, 2023]**



• **Ohm's Law:** It states that "the current through a conductor between two points is directly proportional to the potential difference across the conductor provided external conditions remain constant". **[Board, 2024, 2022]**

Ohm's Law
Scan Me!

Ohm's Law: The potential difference across a metallic conductor is directly proportional to the current, applicable under constant physical conditions like temperature.

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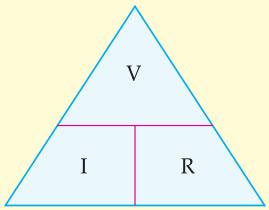
MNEMONICS

Concept 2: Formula of Ohm's Law

Mnemonics:



Interpretations:

To find **V** : Multiply I and R
To find **I** : Divide V and R
To find **R** : Divide V and I

(i) Mathematical expression for Ohm's law:

$$I \propto V$$

$V = IR$ **(where, R = Resistance)**

(ii) V-I graph for Ohm's law: The graph between V and I is always straight line, passing through origin, with slope equals to R.

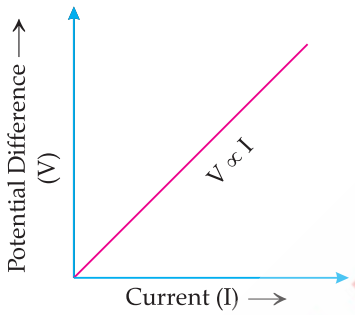


Fig 11.2 : *V-I* Graph [Board, 2023, 24]

• **Resistance (R):** It is the property of a conductor to resist the flow of charges through it. [Board, 2023]

(i) S.I. unit of resistance is ohm (Ω).

(ii) $1 \text{ ohm} = \dfrac{1 \text{ volt}}{1 \text{ ampere}}$

• If 1A current flows through a conductor, when 1V potential difference is applied across it, then the resistance of the conductor is said to be 1 ohm.

• **Factors on which the resistance of a conductor depends:** Resistance of a uniform metallic conductor is,

(i) Directly proportional to the length of the conductor.

(ii) Inversely proportional to the area of cross-section. Combining (i) and (ii), $R = \rho L/A$ [Board, 2024]

(iii) Directly proportional to the temperature. [Board, 2022]

(iv) Depends on nature of the material.

Resistivity of Metal, Alloy, and Insulator: Alloys have higher resistivity than their constituent metals, and insulators have the highest resistivity, followed by alloys and metals.

• **Resistivity (ρ):** The resistance offered by a conductor of unit length and unit cross-sectional area is called resistivity. [Board, 2024, 2022]

Resistance of a conductor

Scan Me!

- Its S.I. unit is ohm-metre (Ω m).
- Resistivity does not change with change in length or area of cross-section, but it changes with change in temperature.
- Range of resistivity of metals and alloys is 10^{-8} to 10^{-6} Ω m.
- Range of resistivity of insulators is 10^{12} to 10^{17} Ω m.
- Resistivity of alloy is generally higher than that of its constituent metals.
- Alloys do not oxidise (burn) readily at high temperature, so they are commonly used in electrical heating devices.
- Copper and aluminium are used for electrical transmission lines as they have low resistivity.

2 Resistance in Series and Parallel Combination, Electric Power and Heating Effects of Electric Current

Concepts covered: • Series combination of resistors • Parallel combination of resistors • Applications of combination of resistors in daily life • Heating effect of electric current and its applications in daily life • Electric power • Interrelation between P, V, I and R.

• **Resistors in series:** When two or more resistors are connected end to end so that same current flows through each one of them, they are said to be connected in series. Here, the total resistance is equal to the sum of the individual resistances. [Board 2024, 2022]

$$R_s = R_1 + R_2 + R_3 + ...$$

• **Resistors in parallel:** When two or more resistors are connected across two points so that each one of them provides a separate path for current, they are said to be connected in parallel. Here, the reciprocal of their combined resistance is equal to the sum of the reciprocals of the individual resistances.

Resistor: Parallel and Series

Scan Me!

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ... \quad \text{[Board 2024, 2023]}$$

• **Heating effect of current:** When an electric current is passed through a conductor, heat is produced in it. This is known as heating effect of current. [Board 2023, 2022]

• **Joule's law of heating:** It states that the heat produced in a conductor is directly proportional to (i) the square of the current I flowing through it, (ii) its resistance R and (iii) the time t, for which current is passed. Mathematically, it can be expressed as: [Board 2024, 2023]

$$H = I^2 Rt \text{ joule} = \frac{I^2 Rt}{4.18} \text{cal} \quad \text{Or} \quad H = VIt \text{ joule} = \frac{VIt}{4.18} \text{cal}$$

• **Practical application of the heating effect of electric current:** It is utilised in the electrical heating appliances such as electric iron, room heaters, water heaters, etc.

• The electric bulb also works on the principle of heating effect of the electric current. When electric current passes through a very thin, high resistance tungsten filament of an electric bulb, the filament becomes white-hot and emits light. [Board, 2019]

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• **Electric energy:** It is the total work done in maintaining an electric current in an electric circuit for a given time. [Board 2023, 2022]

Electric energy,
$W = VIt = I^2Rt$ joule

Electric energy: The energy consumed or produced by an electric circuit, calculated as the product of power and time.

• **Electric Fuse:** It is a safety device that protects electrical appliances in case of short circuit or overloading. [Board, 2024]

(i) Fuse is made up of alloy of copper and tin.

(ii) Fuse is always connected in series with live wire.

(iii) Fuse has a low melting point.

(iv) Current capacity of fuse is slightly higher than that of the appliance.

• **Electric Power:** The rate at which electric energy is consumed or dissipated in an electric circuit/device. [Board 2024, 2023, 2022]

$$P = VI$$

$$P = I^2R = \frac{V^2}{R}$$

MNEMONICS

Concept 3: Electric Power

Mnemonics: 2 IRon men have Super Power.

Interpretations:

2 represents square of I. IRon - I and R

• S.I. unit of power = watt (W)

1 watt = 1 volt × 1 ampere

• **Commercial unit of electric energy = killowatt hour (kWh)**

$1\ kWh = 3.6 \times 10^6\ J$

1 kWh = 1 unit of electric energy

• **Kilowatt hour:** It is the commercial unit of electrical energy. One kilowatt hour is the electrical energy consumed by an appliance of 1000 watts when used for one hour.

• **Watt:** It is the S.I. unit of power. The power of an appliance is 1 watt if 1 ampere of current flows through it on applying a potential difference of 1 volt across its ends.

$$1\ watt = \frac{1\ joule}{1\ second}$$

$$= 1\ volt \times 1\ ampere$$

or $1\ W = 1\ Js^{-1}$

1 kilowatt = 1000 W

• **Power rating:** The power rating of an appliance is the electric energy consumed per second by the appliance when connected across the marked voltage of the mains.

Chapter-12
Magnetic Effects of Electric Current

1 Magnetic Effects of Electric Current

Concept covered: • Magnetic field • Field lines • Field due to a current carrying conductor • Field due to current carrying coil or solenoid • Force on current carrying conductor • Fleming's Left Hand Rule

H.C. Oersted, a Danish physicist was the first to notice the magnetic effects of electric current. According to him, a needle kept near the wire carrying current, deflects due to the magnetic field produced. Any change in the direction of current shows variation in the deflection.

• **Magnetic field:** It is the area around a magnet in which its magnetic force can be experienced.

(i) Its S.I. unit is tesla (T). [Board, 2022]

(ii) Magnetic field has both magnitude and direction. It is a vector quantity.

Magnetic Field: The region around a magnet, where magnetic forces are experienced, measured in Tesla, with stronger fields indicated by closer field lines.

• **Magnetic field lines:** The imaginary lines of magnetic field around a magnet are called magnetic field lines.

• Parallel straight magnetic field lines indicate uniform magnetic field. [Board, 2024]

• **Characteristics of field lines**

(i) Field lines arise from the North pole and end into the South pole of the magnet.

(ii) Field lines are closed curves.

(iii) Field lines never intersect each other as for two lines to intersect, there must be two directions of magnetic field at a point, which is not possible. [Board, 2023]

(iv) Direction of field lines inside a magnet/solenoid is from the South to the North pole.

(v) The relative strength of magnetic field is shown by degree of closeness of field lines. Closer the lines, more will be the strength and farther the lines, less will be the magnetic field strength.

• **Right hand thumb rule:** Imagine you are holding a current carrying straight conductor in your right hand, such that the thumb is pointing towards the direction of current. Then, the fingers wrapped around the conductor gives the direction of magnetic field. [Board, 2024]

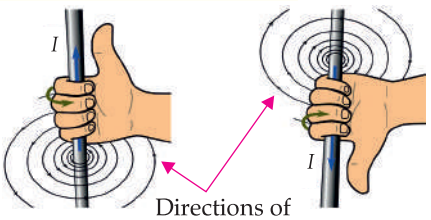

Fig. 12.1

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• **Magnetic field due to current through a straight conductor**

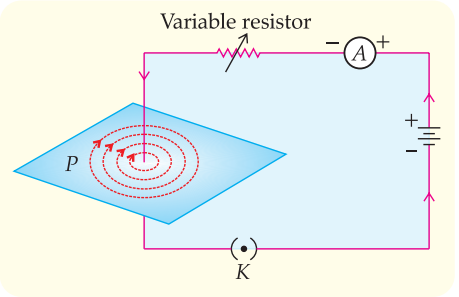


Fig. 12.2: A pattern of concentric circles indicating the field lines of a magnetic field around a straight conducting wire. The arrows in the circles show the direction of the field lines.

➢ It can be represented by **concentric circles** around the conductor at every point on conductor.

Concentric Circles: Circles that share the same centre but have different radii.

➢ Direction can be given by the right hand thumb rule or compass.

➢ Circles are closer near the conductor.

Factors affecting the magnetic field due to a current-carrying straight conductor

➢ Magnetic field ∝ Strength of current [Board, 2024]

➢ Magnetic field $\propto \frac{1}{\text{Distance from the conductor}}$

• Magnetic field due to a current through a circular loop [Board, 2020]

➢ It can be represented by concentric circles at every point.

➢ Circles become larger and larger as we move away.

➢ Every point on wire carrying current would give rise to magnetic field appearing as straight line at centre of the loop.

➢ The direction of magnetic field inside the loop is same.

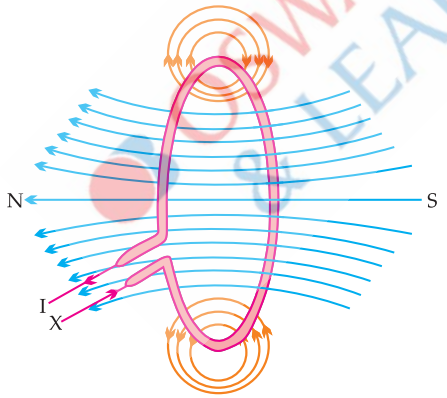


Fig. 12.3 : Field lines through and around a current carrying circular loop

• **Factors affecting magnetic field of a circular current carrying coil/solenoid:**

(i) Magnetic field ∝ Current passing through the coil/solenoid

(ii) Magnetic field $\propto \frac{1}{\text{Distance from coil/solenoid}}$

(iii) Magnetic field ∝ No. of turns in the coil/solenoid [Board, 2023]

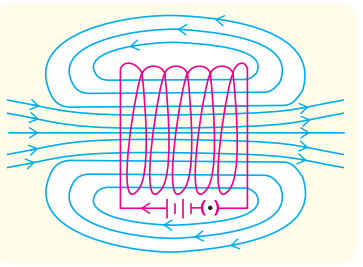


Fig. 12.4

• Magnetic field is additive in nature, i.e., magnetic field of one loop adds up to magnetic field to another loop. This is because the current in each circular turn has the same direction. [Board 2024, 2023, 2022]

• The strength of magnetic field produced by a current carrying circular coil can be increased by:

(a) Increasing the number of turns of the coil.

(b) Increasing the current flowing through the coil.

• **Magnetic field due to current in a solenoid:** Solenoid is a coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder. The end of the **solenoid** having clockwise current acts as south, while on the other hand, having anti-clockwise current acts as north pole. Thus, a solenoid acts as a normal magnet.

Solenoid: A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder, where insulation prevents electrical contact between wires.

Direction of magnetic field

(i) Outside the solenoid : North to South

(ii) Inside the solenoid : South to North

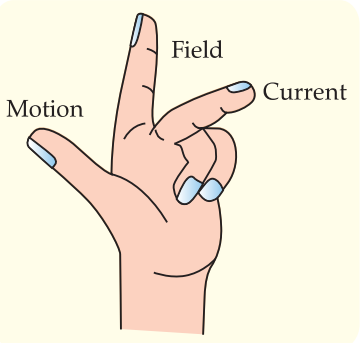
• When a material is placed inside a carrying current solenoid, it gets magnetised.

• **Force on a current carrying conductor in a magnetic field:** Ampere suggested that when a current I passes through a conductor of length l placed in a magnetic field B, then the force experienced is given by $F = IBl \sin\theta$, where θ is the angle between the length of the conductor and magnetic field.

Force is maximum when $\theta = 90°$ and minimum when $\theta = 0°$. [Board, 2023, 2022]

• **Fleming's left hand rule:** Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If forefinger points in the direction of magnetic field, middle finger in the direction of current, then the thumb points in the direction of force acting on the conductor and hence, the direction of its motion.

[Board 2024, 2023, 2022]



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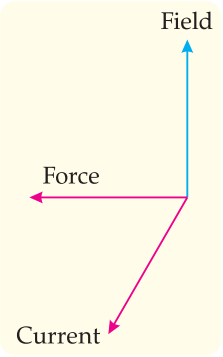

Fig. 12.5 : Fleming's left hand rule

2 AC, DC and Domestic Electric Circuits

Concepts covered: • Direct current • Alternating current • Frequency of AC • Advantage of AC over DC • Domestic electric circuits.

- **Direct Current (DC) and Alternating Current (AC)** [Board, 2020]

(i) Alternate current (AC): The current which reverses its direction periodically is called alternate current. In India, most of the power stations generate alternate current. The direction of current changes after every 1/50 second in India.

Frequency = 1/ Time period = 50 Hz

(ii) Direct current (DC): The current which does not reverse its direction and flows in one direction is called direct current. Sources of DC are solar cell, battery and storage cells.

- **Advantages of AC over DC:** [Board, 2023]

1. AC is easy to generate than DC.
2. AC generation is cheaper than DC generation.
3. The loss of energy during transmission is negligible for AC compared to DC.
4. AC to DC conversion is very easy.
5. AC amplitude can be easily stepped up or stepped down.

- **Domestic electric circuits:** We receive electric supply through mains supported through the poles. In our houses, we receive AC electric power of 220 V with a frequency of 50 Hz.

- An electric circuit consists of three main wiring components:

(i) Live wire with red or brown insulation cover.

(ii) Neutral wire with black or blue insulation cover.

(iii) Earth wire with green or yellow-green insulation cover. It protects us from electric shock in case of **leakage of current** especially in metallic body appliances. It provides a low resistance path for current in case of leakage of current.

Leakage of Current: The unintended flow of electric current from a circuit, often through insulation or faulty connections, posing safety risks.

- Faults and safety measures in domestic electric circuit. [Board, 2024]

(i) Short circuiting: It is caused by touching of live wire and neutral wire either directly or via conducting wire.

(ii) Overloading of an electric circuit: The overheating of electrical wire in any circuit due to flow of a large current beyond the specified limit through it is called overloading of the electrical circuit.

- **Safety measures are:**

(i) Electric fuse: It is a protective device which is used for protecting the circuit from short-circuiting and overloading. [Board, 2022]

It is a piece of thin wire of material having a low melting point and high resistance. Fuse is always connected in series to live wire and works on the principal of heating effect. [Board, 2023]

(ii) Earth wire: The metallic body of electric appliances is connected to the Earth by means of earth wire so that any leakage of electric current is transferred to the ground. This prevents any electric shock to the user.

MNEMONICS

Concept: Electrical wiring

Mnemonics: **R**are **L**awn **B**eautiful **N**ature **G**reen **E**arth

Interpretations:

Red: Live; **Black:** Neutral; **Green:** Earth

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UNIT-V : NATURAL RESOURCES

Chapter-13
Our Environment

1 Ecosystem and Food Chain

Concept covered: • Ecosystem and its components • Food chain, food web and trophic levels • Energy flow • 10 per cent law • Biological magnification.

• Everything that surrounds us is environment. It includes both living (biotic) and non-living (abiotic) components.

MNEMONICS

Concept: Major components of environment
Mnemonics: WASAP
Interpretations:
W: Water, **A:** Air, **S:** Soil, **A:** Animals, **P:** Plants

• Interaction between these biotic and abiotic components forms an ecosystem.

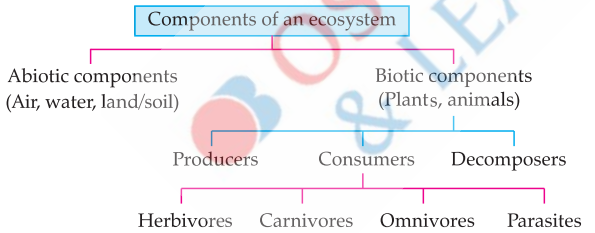
• **Ecosystem:** All the interacting organisms in an area, together with the non-living constituents of the environment, form an ecosystem. For example, forest, pond, etc.

• **Types of Ecosystem:** It is of two types:

(a) Natural Ecosystem: The ecosystem which exists in nature on its own. For example, forest, lake, ocean, etc.

(b) Artificial Ecosystem: Man-made ecosystem is called artificial ecosystem, e.g., crop field, aquarium, garden, etc.

• Biotic components include producers, consumers and decomposers.



• **Producers** include green plants and certain photosynthetic bacteria. They contain chlorophyll pigment which helps them to carry out the process of photosynthesis in the presence of light.

• Herbivores, carnivores, omnivores and parasites are the various types of consumers.

• **Consumers** are those organisms which depend upon the food produced, either directly from producers or indirectly by feeding on other consumers. They are also called **heterotrophs.**

• **Parasites** are those organisms that live outside (**ectoparasites**) or inside (**endoparasites**) the body of another organisms, i.e., host. For example, parasites of human include fleas and lice.

Etoparasites are parasites that live on the surface of their host, such as ticks, lice, and fleas.

Endoparasites are parasites that live inside the body of their host, such as tapeworms, roundworms and protozoa.

• **Decomposers** are microorganisms (bacteria and fungi) that obtain energy from the chemical breakdown of dead remains of animals and plants and waste products of organisms. Decomposers break down the complex organic substances into simple inorganic substances that go into the soil and are used up again by the plants. They help recycle nutrients, replenishing of natural resources.

Food chains, Trophic levels with Energy Flow
Scan Me!

• **Food chain** is the sequence of organisms through which food energy flows in an ecosystem. It is a succession of organisms that eat other organisms and may, in turn, be eaten themselves. **[Board, 2022]**

Example:

Grass (Producer) → Grasshopper (Herbivore) → Frog (Carnivore)

Decomposer (fungi and Bacteria) ← Eagle (Top Carnivore) → Snake (Carnivore)

• The first link in a food chain is always a producer (green plants) as they have the ability to trap solar energy with the help of chlorophyll. The last link is always a decomposer.

• **Trophic Levels** are the various steps or levels in the food chain where transfer of food or energy takes place. Producers are the first trophic level, herbivores are the second trophic level, carnivores or secondary consumers are the third trophic level and large carnivores or tertiary consumers are the fourth trophic level.

• **Food Web** is the network of various food chains which are interconnected at various trophic levels. Since an organism can occupy more than one position in different food chains in a food web it occupies more than one trophic level. It represents the feeding relationship within the community.

• The green plants in a terrestrial ecosystem capture about 1% of the energy of sunlight that falls on their leaves and convert it into food energy. A terrestrial ecosystem is a land-based ecosystem, such as forests, grasslands, deserts, or tundras, where organisms interact with each other and their environment.

• When green plants are eaten by primary consumers, a great deal of energy is lost as heat to the environment,

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some amount goes into digestion and in doing work, and the rest goes towards growth and reproduction. An average of 10% of the food eaten is turned into its own body and made available for the next level of consumers.

• Therefore, 10% can be taken as the average value for the amount of organic matter that is present at each step and reaches the next level of consumers.

• **10 Per cent Law:** It states that only 10% of total food energy is transferred from one trophic level to the next level. The remaining 90% energy is used in life processes (digestion, growth, reproduction, etc.) by the present trophic level.

• Due to this gradual decrease in energy, food chains contain 3–4 trophic levels.

• **The entire process of energy flow can be summarised in the following four steps:** [Board, 2022]

➢ The flow of energy in an ecosystem is always linear or unidirectional. The energy captured by the producers does not revert to the solar input. Also, the energy which passes to the herbivores does not come back to autotrophs.

➢ At every step in a food chain the energy received by the organism is also used for its own metabolism and maintenance. The left over is passed to next higher trophic level. Thus, energy flow decreases with successive trophic levels.

➢ The number of trophic levels in a food chain is restricted by 10% flow of energy. Very less amount of energy is available to the last trophic level.

Biological magnification

Scan Me!

• **Biological Magnification:** The concentration of harmful chemicals such as **pesticides** and unused chemical fertilisers goes on increasing with every next trophic level in a food chain. This is called as biological magnification. [Board, 2025, 2024]

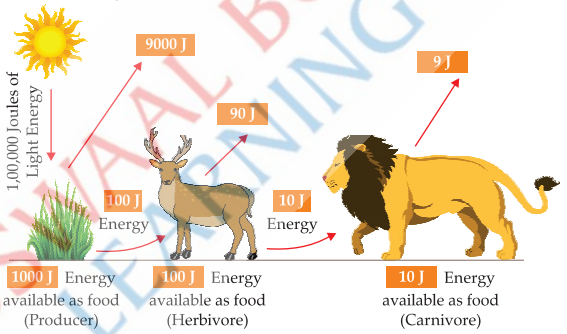
• The maximum concentration of such chemicals is found to accumulate in human body as human occupies the top level in any food chain.

Pesticides are chemical substances used to kill or control pests, such as insects, weeds, fungi, or rodents, to protect crops and public health.

Example 1

Suppose 100,000 J of solar energy falls on green plants. Only about 1% of this energy (i.e., 1,000 J) is captured by green plants through photosynthesis; the rest (99,000 J) is lost (reflected, absorbed by non-photosynthetic surfaces, or as heat).

Of the 1,000 J stored by plants, only 10% (i.e., 100 J) is transferred to herbivores when they consume the plants. The remaining 90% is used by plants for life processes or lost as heat. Similarly, only 10% of the 100 J (i.e., 10 J) is passed on to carnivores when they eat herbivores, with the rest lost.



2 Affect of Human Activities on Environment

Concept covered: • Methods of waste disposal • Environmental Problems caused by humans • Ozone depletion

• **Environmental problems caused by humans are:**
(i) Pollution due to mismanagement of waste disposal.
(ii) Depletion of the ozone layer due to the over-release of ozone-depleting substances (e.g., CFCs).

• **Waste Materials:** These are unwanted or unusable materials which are discarded after primary use, or are worthless, defective and of no use.

• **Garbage contains following type of materials:**
(a) Biodegradable Wastes: Substances that can be decomposed by the action of microorganisms are called biodegradable wastes. For example, fruit and vegetable peels, cotton, jute, cow-dung, paper, etc. [Board 2017]
(b) Non-biodegradable Wastes: Substances that cannot be decomposed by the action of microorganisms are called as non-biodegradable wastes. For example, plastic, polythene, metals, synthetic fibres, radioactive wastes, pesticides, etc. [Board 2024]

• **Methods of Waste Disposal:**

(a) Biogas Plant: Biodegradable wastes can be processed in biogas plant to produce biogas and manure.

(b) Sewage Treatment Plant: The drain water can be cleaned in sewage treatment plant before adding it to rivers to avoid pollution and eliminate harmful substances.

(c) Land Fillings: The wastes are buried in low lying areas and are compacted by rolling with bulldozers.

(d) Composting: Organic wastes are filled in a compost pit and covered with a layer of soil. After about three months, garbage changes to manure.

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(e) Reduce: The method for minimizing waste by using fewer resources and making mindful consumption choices. This can include using less plastic, saving electricity, and opting for sustainable products.

(f) Recycling: Non-biodegradable wastes are recycled to make new items.

(g) Reuse: It is a conventional technique of using an item again, e.g., newspaper for making envelopes.

(h) Biodegradable and non-biodegradable wastes should be discarded seperately in two different dustbins.

Reduce, Reuse and Recycle

Scan Me!

• **Formation of ozone in the atmosphere:** Ozone is continuously formed in the stratosphere through the action of ultraviolet (UV) radiation from the Sun on molecular oxygen. At the same time, it is also degraded back into molecular oxygen and atomic oxygen (O) by high-energy ultraviolet radiation.

$$O_3 \rightarrow O_2 + [O] + Heat$$

This oxygen atom then recombine with oxygen (O_2) molecule to form ozone molecule.

• **Ozone (O_3):** It is a molecule formed by three atoms of oxygen. Ozone performs an essential function of shielding the surface of the earth from ultraviolet radiation of the sun. [Board, 2024]

Ozone layer

Scan Me!

• Ozone layer is a layer of the earth's atmosphere in which most of the atmosphere's ozone is concentrated.

• **Ozone Depletion:** Ozone depletion is the reduction in concentration of ozone layer. [Board, 2022]

• **There are several reasons for depletion of the ozone layer.**

• The foremost is the use of chlorofluorocarbons (CFCs). The other factor responsible for ozone destruction is the pollutant nitrogen monoxide (NO).

• When the harmful chemicals like chlorofluorocarbons (CFCs) are released into the atmosphere, they accumulate in the stratosphere. These compounds break down ozone molecules, leading to the thinning of the ozone layer.

• Thus, the ozone layer in the atmosphere becomes thinner and gets depleted allowing more ultraviolet rays to pass through it. This phenomenon is referred as the ozone hole.

• **Ozone layer can be protected by:**

(a) Stopping the release of chlorofluorocarbons (CFCs).

(b) Removing the pollutant nitrogen monoxide.

(c) Limiting the use of appliances like air conditioners and refrigerators that use CFCs.

• The Montreal Protocol (an international treaty signed in Canada, in 1987) aims to control the emission of ozone depleting substances.

• Under this treaty, United Nations Environment Programme (UNEP) successfully negotiated an agreement to stop CFC production at 1986 levels by all the countries.

• It is now compulsory for all manufacturing companies worldwide to produce refrigerators that are free of CFCs.

Key Facts

Ozone is present in stratosphere a layer of the atmosphere located, 20–30 km above the Earth's surface.

The thinning of the ozone layer over Antarctica was first observed in 1985 and was termed as ozone hole.

●●

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